MINIMAL-PERIOD RIGIDITY FOR RIBBON-OUTER SKEW SCHUR COMPOSITIONS

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Abstract. We prove that minimal factorizations of Schur-equal connected skew diagrams have the same period (the content shift in the defining translation between inner copies) when the larger-period outer is a ribbon. At each diagram, the overlap's diagonal span is minimized first, then the inner diagram's. At every strictly smaller area, we assume that every connected Schur class containing diagrams unrelated by rotation admits a nontrivial composition, and that Schur equality preserves the existence of nontrivial compositions and, for minimal factorizations, the marked inner together with the outer Schur function, modulo the permitted identifications. The compositions satisfy Hypotheses I–IV of McNamara and van Willigenburg, with an arbitrary competing outer; we exclude classes containing a nontrivial composition whose inner is a row or column. Jacobi–Trudi determinant identities synchronize both diagram boundaries and yield a legal composition on the original diagram contradicting minimality; equal-period factor identification remains separate.

1. Introduction

Distinct connected skew diagrams, not related by rotation, can have the same skew Schur function [5, Ex. 7.8]. Marked composition explains many such equalities by assembling copies of an inner diagram according to an outer diagram and allowing suitable factor reversals. A natural inverse question is whether equality of the resulting functions determines the scale at which a source-minimal composition is assembled. We prove a period-selection theorem when the larger-period chart [7] has a ribbon outer, under a classification hypothesis at strictly smaller areas.

The composition of McNamara and van Willigenburg [3] records an inner diagram, two marked end copies, their displacement, and their attachment type. A proper chart $F = D \circ_\omega E$ has connected factors of at least two cells. If its period is p and the marked overlap has span w, then

$$N_F = pN_D + w, \qquad N_E = p + w. \tag{1.1}$$

The period is the content of the displacement between copies, not the full span of the inner. The chart is *source-minimal* when

$$\mathrm{score}(D, E, \omega) = (w, N_E) = (w, p + w) \tag{1.2}$$

is lexicographically least among the proper charts of the same source F. On the line-inner-free domain, these are exactly the least-period charts, by the least-root-period corollary of [7]. The forward composition and reversal identities under Hypotheses I–IV are available from Olson-Harris [4]. Our problem is to recover geometric information from an equality of their outputs.

Local common-factor gluing compares two charts already carried by the same diagram [7]. It gives equal fibers at equal periods, reassociation at divisible periods, and a legal common-divisor chart in the remaining compatible cases. These conclusions do not directly compare prescribed

Date: October 19, 2026.

2020 *Mathematics Subject Classification.* Primary 05E05; Secondary 05A17, 15A15.

Key words and phrases. skew Schur function, marked composition, source-minimal period, boundary synchronization, Jacobi–Trudi determinant, ribbon outer, composition necessity.

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This work is archived at doi:10.5281/zenodo.23482349.

charts on different Schur-equal parents: the second chart has not yet been realized on the first diagram.

Corner transport gives a further reduction [8]. If $D_1, \dots, D_t$ are the connected components of the outer interior D^-, the extreme Jacobi–Trudi cofactor of $D \circ_\omega E$ contains the transported child factors $s_{D_i \circ_\omega E}$, with exact multiplicities. A ribbon has empty interior. In the ribbon-outer case this particular descent has no outer-interior child to follow, although local cap and normalization factors remain. The present argument replaces that child-factor step by boundary reconstruction.

1.1. **Main theorem.** For a connected Schur class, *rootedness transfer* means that the existence of one proper chart implies that every class member has a proper chart. *Root existence* means that a class that is not a single rotation orbit has a rooted member. *Rooted-key invariance* means that rooted members have the same marked source-minimal key, with the ordinary gauge acting on both the marked inner and outer function. Let $\mathsf{J}_{<n}$ denote these three assertions for every connected Schur class of area strictly below n. Their full-fiber and factor-reversal consequences are recalled in Section 3. No current-area instance is included in $\mathsf{J}_{<n}$.

Theorem 1.1 (Ribbon-outer period rigidity)**.** *Assume* $\mathsf{J}_{<n}$*. Suppose a connected Schur class contains no member with a proper line-inner chart. Let*

$$F = D \circ_\omega E, \qquad H = C \circ_\eta U, \qquad s_F = s_H, \qquad |F| = n, \tag{1.3}$$

be proper source-minimal charts satisfying Hypotheses I–IV. If their periods satisfy $p \geq q$ *and* $D^- = \varnothing$*, then* $p = q$*.*

The competing outer C is arbitrary. Only the hypothesis on D initially allows $N_D = |D|$ in (1.1); N_C is not replaced by $|C|$ before ribbon inheritance is proved. Neither F nor H is assumed to be a ribbon. The conclusion compares the two prescribed source-minimal periods, not all periods of arbitrary proper presentations.

The hypothesis $\mathsf{J}_{<n}$ is the joint induction hypothesis, not a shorthand for period rigidity alone. In particular, root existence at smaller areas is needed when a smaller inner might be chartless. This induction step does not by itself establish the joint package at all areas.

1.2. **What diagonal heights forget.** Let $[\ell_k, h_k]$ be the rows of a diagram on content k, and put

$$\begin{gathered} A_k = \ell_{k-1} - \ell_k, \qquad B_k = h_{k-1} - h_k, \qquad H_k = h_k - \ell_k + 1, \\ H_k - H_{k-1} = A_k - B_k. \end{gathered} \tag{1.4}$$

Here $A_k, B_k \in \{0, 1\}$. The two choices $(0, 0)$ and $(1, 1)$ have the same difference. Thus a periodic height word need not have periodic boundary words. In beta coordinates, the forgotten choices are precisely the common entries of the row and column lists. Removing such an entry preserves every diagonal height, area and span, but changes both membership words.

The proof first reduces (1.3) to a positive nondividing comparison with distinct attachment phases and two original line outers (Theorem 4.8). Writing $g = \gcd(p, q)$ and $N = Lg + c$, $0 < c < g$, it obtains periodic structure for the *whole* cores and heights. The remaining task is to synchronize the paired records of the restored parents. This is stronger than identifying a core class or the determinant of a return block.

1.3. **The synchronization mechanism.** The two original composition annihilators act on the same normalized whole function. Their commutator, compared at the q and $p + q$ horizons, aligns the ordinary clocks and supplies a complementary cap response. Original p-periodicity identifies the two finite low-index decks; it does not identify their different high-index alphabets.

The constant coefficient in z of the cap response detects offending records at the extreme connected components of the weak return core. Its first coherent-unit coefficient has a pole with a residue that is a sum of particular weak-minor products. A simultaneous separated specialization gives these products one positive sign. Coincident end factors therefore add their contributions, rather than cancel them. Only the outward cap discrepancies that could defeat the next sign argument are removed.

In the original determinant, the same whole differential identity is a sum of full row and column replacements. Each surviving replacement is a feasible weak Jacobi–Trudi determinant. After the cap exclusions all nonzero weighted terms have one sign. Ordinary comparisons give boundary equality; mixed comparisons give a directed inclusion (Theorem 8.3). A one-way inequality cannot stay strict around a finite period cycle, so both conclusions yield gcd synchronization.

The synchronized words and the actual singleton diagonals then construct a legal period-g chart on the same source (Theorem 9.2). Only after that construction is its score compared with (1.2). This bypasses an exhaustive classification of the intermediate rational return responses.

1.4. **Organization and relation to companion preprints.** Section 2 fixes the common geometric and determinant conventions. Section 3 separates strict descent and actual reconstruction from minimality. Section 4 reduces the original arbitrary-C comparison to the line-outer branch. The aligned construction in Appendix A uses the cell-only cut theorem of Appendix B and applies weak-cap reconstruction [6] to the actual end windows; the exceptional constructions in Appendices C and D use the contact rigidity of Appendix E. Sections 5 and 6 derive periodic structure, actual cap systems and the two-horizon response. Sections 7 and 8 prove the pole filter and complete shifted-determinant noncancellation. Section 9 constructs the finer chart and completes Theorem 1.1. Appendix F gives an optional alternative geometric realization of the same scalar obstruction, using the shared structural, operator, end-pole and contact inputs.

Companion preprint [7] supplies same-parent geometry and source-minimal fiber domination. Preprint [8] supplies the cofactor transport and exact normalization. The erosion, zero-free class transport and inflation-of-legal-charts lemmas of [9] supply the actual zero-free class reduction; their exact transport domain does not assert that arbitrary inflation preserves Schur equality. The weak-cap and absolute-flag theorem of [6] supplies reconstruction from complete rational coframes, including units and disconnected proper blocks. The supplied-period covariance and terminal readout of [10] are used in the whole-source extraction below. After the periods agree, marked-factor and prescribed-fiber identification remain separate questions.

2. Marked charts, boundary words, and Jacobi–Trudi data

2.1. **Diagrams and full beta lists.** Diagrams are finite skew Young diagrams in English notation, identified up to translation. Connected means nonempty and edge-connected. We write $|S|$, $r(S)$, $c(S)$ and N_S for area, number of rows, number of columns and number of occupied diagonals. The content of (i,j) is $j-i$. For connected S, $N_S = r(S)+c(S)-1$. Rotation through 180 degrees is denoted by S^*; antitransposition is $\mathrm{AT}(i,j)=(-j,-i)$. The symbols Row_j and Col_j denote the j-cell row and column. The symmetric-function involution ι satisfies $\iota(h_j)=e_j$ and $\iota(s_S)=s_{\mathrm{AT}(S)}$, where e_j is the elementary symmetric function. Write ι_γ for the identity or ι according as $\gamma=\mathrm{id}$ or AT.

A row profile lists the row lengths, followed after a semicolon by the adjacent overlap lengths minus one; these latter quantities are called gaps.

Normalize the southwest content to zero. The full beta lists are strictly decreasing integer lists

$$X=(N,x_1,\dots,x_{r-1}),\qquad Y=(y_0,\dots,y_{r-2},0),\qquad s_S=\det(h_{x-y})_{x\in X,\,y\in Y}.$$

We always retain $h_0=1$ and $h_j=0$ for $j<0$. A row with endpoints (x,y) occupies contents $y,\dots,x-1$. Consequently

$$H_S(k)=\#\{y\in Y: y\le k\}-\#\{x\in X: x\le k\},\qquad 0\le k<N. \tag{2.1}$$

The two internal boundary words are $A_k=\mathbf{1}_{k\in Y}$ and $B_k=\mathbf{1}_{k\in X}$, with the diagonal-interval convention of (1.4). They are defined at $1\le k<N$; there are no artificial boundary steps at 0 or N.

A pair of binary boundary words and positive heights beginning and ending at one reconstruct a connected skew diagram. Indeed, starting from the southwest singleton, the diagonal interval endpoints change by zero or minus one at each step. Their intervals then form the two skew boundaries; every cell off the last diagonal has an eastern or northern neighbor on the next

diagonal. We will use this criterion for unions and shortening splices as well as for original diagrams.

A *common beta* is an element of $X \cap Y$. Deleting it from both lists preserves (2.1), hence the entire height word, area, span and connectedness. Repeating the deletion gives the *zero-free core* $\mathrm{Core}(S)$. Zero-free means absence of common betas; negative-index zero entries are not removed. Restoring a common beta changes the two membership words by $00 \leftrightarrow 11$.

Lemma 2.1. *If connected diagrams S, T have equal Schur functions, their zero-free cores have equal Schur functions. Their original row counts and numbers of common betas also agree.*

Proof. The original row count is the maximum ordinary degree, where $\deg h_j = 1$ for $j > 0$. If $u = |X \cap Y|$, selecting all u nonattacking unit entries in the determinant leaves, up to its actual Laplace sign, the homogeneous nonzero determinant of the zero-free core, of degree $r - u$. Every other selection has larger ordinary degree. Thus the least ordinary degree and its entire homogeneous part determine u and the core Schur polynomial up to sign. Actual skew Schur functions are positive at a sufficiently large positive alphabet, so two such associates have sign $+1$. □

2.2. **Legal marks and composition.** We use the conventions of [7], following [3]. A marked inner ω consists of a connected diagram E, connected end copies W_- and $W_+ = W_- + v$, and their attachment data. The lower copy contains the southwest cell and the upper copy the northeast cell. Put

$$O = E \setminus (W_- \cup W_+), \qquad w = N_W, \qquad p = N_E - w = c(v).$$

The cases $(a), (b), (c), (d)$ have lower/upper attachment types $(H,H), (V,V), (H,V), (V,H)$, respectively. Cases $(a), (b)$ are ordinary and $(c), (d)$ are mixed. Horizontal attachment uses the western or eastern neighbor of the relevant extreme middle cell; vertical attachment uses its southern or northern neighbor.

The mark is legal under the following conditions.

(I) No strictly larger skew diagram containing W_- on the same content diagonals has translated copies at both ends of E.
(II) There is an occupied diagonal between the two copies meeting neither: $0 \le w < p$.
(III) Removing either marked copy leaves a connected skew diagram.
(IV) In $\boldsymbol{E} = \bigcup_{k \in \mathbb{Z}} (E + kv)$ no hatted-middle copy is adjacent to a hatted-mark copy, where, for $e = (1,1)$,

$$\widehat{O}_1 = \{x \in O_1 : x + e \in O_1\}, \qquad \widehat{W}_1 = \{x \in \boldsymbol{E} : x + e \in W_1\} \cup \{x \in W_1 : x + e \in \boldsymbol{E}\}.$$

Adjacency means sharing an edge. No Hypothesis V is imposed. At $w = 0$ the mark is the horizontal edge reading, of displacement $v_H = \mathrm{ne}(E) - \mathrm{sw}(E) + (0,1)$. A vertical edge reading has displacement $v_H - e$ and is converted by the ordinary gauge, which antitransposes the outer. Figure 1 gives a row-inner example.

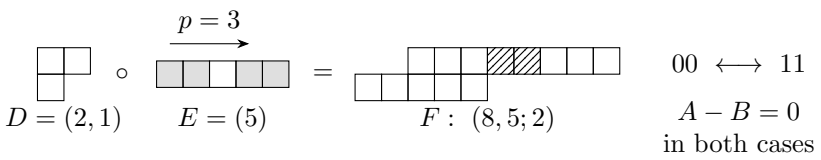


FIGURE 1. A legal row-inner chart, illustrating period and overlap rather than the hypotheses of the hard comparison. The marked inner has $p = 3$ and $w = 2$. The ribbon outer has three cells; its row action gives profile $(8, 5; 2)$ and span $3 \cdot 3 + 2 = 11$. The hatched cells are the overlap of the two copies in the upper row. The inset records the elementary ambiguity left by diagonal heights. No equality between two different source-minimal charts is asserted by this illustration.

For a cell $d = (i, i+j)$ of the outer D, let $[\ell_j^D, h_j^D]$ be its content-j diagonal interval. The copy origin is

$$L_\omega(d) = \begin{cases} jv + ie, & (a), \\ jv - ie, & (b), \\ jv + (i - \ell_j^D)e, & (c), \\ jv - (h_j^D - i)e, & (d). \end{cases}$$

In cases $(c), (d)$ the vertical boundary-ribbon pairs contribute the supplemental copies $W_+ + L_\omega(d) + e$ and $W_+ + L_\omega(d) - e$, respectively, with d the lower cell. These copies are part of the composition. A proper chart $D \circ_\omega E$ has $|D|, |E| \geq 2$.

2.3. **Local geometry and the coding table.** We record the precise parts of local gluing used below. Their proofs are the local geometry, coding and mark-window arguments of [7].

Proposition 2.2. *For a legal positive mark of period p and width w, the following hold.*

1. *$W_- = E \cap (E - v)$. The middle has at most one cell on each marked diagonal. The private diagonals w and $p-1$ are singletons.*
2. *The middle occupies $[s, p+t-1]$, where $1 \leq s \leq w$, $0 \leq t < w$; for mixed marks $t < s$.*
3. *If $N_D = M$, then $N_{D \circ_\omega E} = Mp + w$. Its first p diagonals and its last p diagonals are the corresponding two translated windows of E, and their nonempty private overlap reconstructs E.*
4. *The full boundary words are p-periodic away from the exceptional residues in* (2.2)*. The initial and terminal values are those in* (2.3)*.*

(2.2)

	A	B
(a)	t	s
(b)	s	t
(c)	$\varnothing$	t, s
(d)	t, s	$\varnothing$

	interior first slot	interior second slot
(a)	$A_{jp+t} = A_j^D$	$B_{jp+s} = B_j^D$
(b)	$B_{jp+t} = 1 - A_j^D$	$A_{jp+s} = 1 - B_j^D$
(c)	$B_{jp+t} = 1 - A_j^D$	$B_{jp+s} = B_j^D$
(d)	$A_{jp+t} = A_j^D$	$A_{jp+s} = 1 - B_j^D$

(2.3)

	initial	terminal
(a)	$A_t = 1,\ B_s = 0$	$A_{N-w+t} = 0,\ B_{N-w+s} = 1$
(b)	$B_t = 0,\ A_s = 1$	$B_{N-w+t} = 1,\ A_{N-w+s} = 0$
(c)	$B_t = B_s = 0$	$B_{N-w+t} = B_{N-w+s} = 1$
(d)	$A_t = A_s = 1$	$A_{N-w+t} = A_{N-w+s} = 0$

Endpoint steps at $0, N$ are omitted. In case (d), the private-interface values are $B_w = 1$, $B_p = 0$; case (c) has $A_w = 0$, $A_p = 1$. At empty width, each boundary has only the joining residue zero.

For later geometric constructions it is useful to spell out the height part of the coding. Set

$$M_j = h_j^D - \ell_j^D + 1, \quad S_j = h_{j-1}^D - \ell_j^D + 1, \quad S_0 = S_M = 1, \quad M_{-1} = M_M = 0.$$

The displayed marked inner is called the native. In case (d), if Ω_r is the marked height and ν_r the native private height, then

$$H_{D \circ_\omega E}(pj + r) = \begin{cases} \nu_r + M_j - 1, & w \leq r < p, \\ \Omega_r + M_{j-1}, & 0 \leq r < t, \\ \Omega_r + S_j - 1, & t \leq r < s, \\ \Omega_r + M_j, & s \leq r < w. \end{cases} \tag{2.4}$$

The marked lines include the two end bands, and the private line has $0 \leq j < M$. Formula (2.4) includes the supplemental copies. For an ordinary (a) mark whose interval at residue r is

$[a_r, b_r]$, the interior marked endpoints are

$$\ell(pj+r) = jv_1 + a_r + \begin{cases} \ell^D_{j-1} - 1, & r < t, \\ \ell^D_j, & r \geq t, \end{cases}$$

$$h(pj+r) = jv_1 + b_r + \begin{cases} h^D_{j-1}, & r < s, \\ h^D_j + 1, & r \geq s. \end{cases}$$

Antitransposition gives the remaining cases, acting on the full chart.

2.4. **Functions, cofactors, and normalizations.** Work in $\Lambda = \mathbb{Q}[h_1, h_2, \ldots]$. For $j > 0$, weighted degree assigns h_j degree j, and ordinary degree assigns degree one; the physical $h_0 = 1$ has degree zero. The ordinary degree of a polynomial is its highest such degree. In (A.3), $\mathrm{in}_{\mathrm{ord}}$ instead denotes the least ordinary-degree component, and $\mathrm{ord}_{\mathrm{ord}}$ its degree. The exceptional-phase notation Top_g in Equations (C.2) and (D.3) denotes the highest component for the grading $\deg h_j = \lfloor j/g \rfloor$. We write $\mathrm{wt}(P)$ for the weighted degree of a homogeneous polynomial. For nonconstant P write $\operatorname{sp} P = \max\{j : h_j \text{ occurs in } P\}$. Set $\operatorname{sp} P = 0$ for nonzero constants. A disconnected diagram has polynomial span equal to the maximum component span, not the separation of its extreme cells.

Write $S^- = \{x \in S : x + (1,1) \in S\}$ and

$$Q_S = (-1)^{r(S)-1} s_S, \qquad \kappa_S = \partial_{h_{N_S}} Q_S = s_{S^-}, \qquad f_S = \frac{Q_S}{\kappa_S} = \frac{s_S}{\partial_{h_{N_S}} s_S}.$$

We also write $\kappa(S) = \kappa_S$ and $\kappa(s_S) = \kappa_S$. The largest raw generator occurs only at the top-right matrix entry, so $f_S = h_{N_S}$ minus the proper Schur complement. The corner identity is the rim-removal cofactor calculation in [1, 5]. Connected skew Schur functions are irreducible; we use the corrected irreducibility theorem [1]. Rows, columns and all positive higher-row overlap multisets are Schur invariants [5, Cor. 8.11].

For any equal-length strict lists I, C, let $M(I, C) = \mathrm{JT}(I, C) = (h_{i-j})_{i \in I, j \in C}$. We call the lists weakly feasible when their decreasing entries satisfy $i_k \geq c_k$. Empty rows and disconnected pieces are allowed; the empty determinant is one and its inverse action is empty. Weakly feasible Jacobi–Trudi determinants are nonzero: the diagonal matching uniquely minimizes $\sum (i_k - c_{\pi(k)})^2$, so its moment monomial has coefficient one. Its signed leading-monomial refinement is proved in Lemma 7.1.

For a legal mark, including an empty mark, put $A = s_W$, $B = s_{\widehat{W}}$, and $C = s_{\widehat{O}}$. The modified-composition homomorphism is defined on the actual row probes by

$$\Phi_\omega(h_j) = A^{j-1} s_{\mathrm{Row}_j \circ_\omega E}, \qquad j \geq 1. \tag{2.5}$$

The complete composition theorem gives

$$\Phi_\omega(s_D) = \epsilon_\omega(D) A^{|D| - r(D)} B^{|D| - c(D)} C^{|D| - N_D} s_{D \circ_\omega E}, \tag{2.6}$$

with its prescribed ordinary gauge and mixed sign. When that convention uses an outer involution, it is carried on both sides. All local factors have polynomial span below the period; negative exponents in (2.6) are interpreted in the fraction field. This is the all-type formula of [4], with the finite-cap normalization of [8].

For a mark of period p and width w, define its finite caps by

$$K_- = E^- \cap \{0 \leq c(x) < w\}, \qquad K_+ = E^- \cap \{p \leq c(x) < p + w\}, \qquad \Theta_\omega = s_{K_-} s_{K_+}.$$

The row-amalgamation formula and Corner transport theorem of [8] give

(2.7)

$$\kappa(\mathrm{Row}_j \circ_\omega E) = \Theta_\omega C^j B^{j-1}, \qquad \kappa(E) = \Theta_\omega C,$$

$$\kappa(D \circ_\omega E) = \Theta_\omega A^{r(D) - 1 - \sum_i r(D_i)} B^{c(D) - 1 - \sum_i c(D_i)} C^{N_D - \sum_i N_{D_i}} \prod_i s_{D_i \circ_\omega E},$$

where the D_i are the connected components of D^-. All displayed exponents are nonnegative. The factors A, B, C, Θ_ω have span below p; more precisely their spans are at most $w, w, p - w -$

$2, w-1$ when those bounds are nonnegative. At empty width $A = B = \Theta_\omega = 1$ and $C = s_{E^-}$. Each displayed child is a connected prime of span $pN_{D_i} + w$. This identifies every factor of span at least p with its actual occurrence and multiplicity. A one-cell D_i contributes s_E, not a proper chart. When one mark is fixed, write Θ for Θ_ω in its corner formulas.

Lemma 2.3. *Suppose two actual connected outers D, D' are composed through the same legal marked inner. If the resulting parents are Schur-equal, then $s_D = s_{D'}$.*

Proof. At a finite cutoff the generators in (2.5) are triangular affine functions of the fresh pivots h_{jp+w}, with nonzero proper-cofactor coefficients. Adjoin the fraction field of every nonpivot variable. Successively solving for the pivots makes their images algebraically independent over this field, which contains all normalization factors. The equality of the parents and (2.6) therefore say that two polynomials in these independent images, with rational coefficients, differ by a coefficient-field scalar. Comparing one nonzero coefficient makes that scalar rational. Primitivity of actual skew Schur polynomials makes it ± 1, and positivity on an ordinary positive alphabet gives $+1$. The same argument applies after a fixed carried involution. The original normalizers were not assumed equal; equality of the outer functions now makes all their area, row, column and span exponents agree. □

3. Strict descent and actual realization

3.1. **The smaller-area package.** Let $\mathcal{F}(\omega, s_D)$ be the fiber of a legal marked inner and an outer function: it consists of the actual compositions with that marked inner or its transported rotation, with the ordinary gauge acting on the outer function as well. Let $K(F)$ be the key of a source-minimal chart, as in [7].

Definition 3.1. The package $\mathsf{J}_{<n}$ asserts, at every connected Schur class of area less than n:

(1) rootedness transfer: one rooted member implies all members are rooted;
(2) root existence: a class that is not a rotation orbit has a rooted member;
(3) rooted-key invariance: rooted members have the same key.

Lemma 3.2. *Under $\mathsf{J}_{<n}$, a connected diagram G of area less than n is either chartless with $\mathrm{Cl}(G) = \{G, G^*\}$, or every prescribed source-minimal chart $G = A \circ_\omega E$ has full fiber*

$$\mathrm{Cl}(G) = \mathcal{F}(\omega, s_A).$$

The corresponding factor-reversal exhaustion holds at those smaller areas.

Proof. If a chartless member had a rooted mate, rootedness transfer would be contradicted. If its wholly chartless class were nontrivial, root existence would be contradicted. This proves terminal rigidity. For a rooted class, rootedness transfer and key invariance put every member in the source-minimal fiber, by the component-key and domination theorems of [7]. Conversely the forward identity puts that fiber inside the class. A class with a line inner is already a full line fiber by the unconditional row/column transport theorem. Factor-reversal exhaustion then follows by induction on actual outer prefixes, reappending the same last marked factor; these prefixes have strictly smaller area. □

Proposition 3.3. *At one actual connected parent the following assertions hold.*

(1) *Equal-period proper charts have the same fiber.*
(2) *If a supplied proper period d divides another supplied period p, the period-p inner has a literal chart through the supplied period-d marked inner; the coarse chart reassociates through it for every connected outer.*
(3) *Nondividing periods with gcd greater than one yield a proper legal gcd chart, unless the attachment pairing is impossible. Coprime proper periods force line inners.*
(4) *Every proper-chart fiber through a source is contained in each of its prescribed source-minimal fibers.*

These are the equal, divisible, gcd and domination results of [7]. They apply to charts on one parent, not directly across (1.3).

The ribbon factor-reversal characterization stated in [2, Cor. 4.2], with the necessity direction supplied by [11], has the following consequence.

Lemma 3.4. *Let F be a connected ribbon with a source-minimal chart $F = D \circ_\omega E$. Then*

$$\mathrm{Cl}(F) = \mathcal{F}(\omega, s_D).$$

Proof. The row and column length multisets are Schur invariants, so area and $N_S = r(S) + c(S) - 1$ agree within a connected Schur class. Since $|F| = N_F$, every connected $H \in \mathrm{Cl}(F)$ also has one cell on each occupied diagonal, and hence is a ribbon.

Write R_α for the ribbon with top-to-bottom row lengths α. Denote the composition product by $\alpha \diamond \epsilon = \epsilon^{\odot\alpha_1} \cdot \cdots \cdot \epsilon^{\odot\alpha_{\ell(\alpha)}}$, where $\cdot$ is concatenation and $\odot$ merges the two meeting parts. If ω_ϵ is the horizontal empty mark on R_ϵ, then, up to translation,

$$R_\alpha \circ_{\omega_\epsilon} R_\epsilon = R_{\alpha\diamond\epsilon}.$$

Indeed, an east step of the outer joins successive copies by a horizontal edge and merges their meeting rows; a north step joins them by a vertical edge and keeps those rows separate. The empty mark is legal, and this chart is proper when $|\alpha|, |\epsilon| \geq 2$. Reversal of a row word rotates its ribbon. Reversing all row words commutes with $\diamond$, so the classification applies in this row convention as well.

For $H \in \mathrm{Cl}(F)$, that classification gives row words

$$\beta = \delta_1 \diamond \cdots \diamond \delta_s, \qquad \gamma = \epsilon_1 \diamond \cdots \diamond \epsilon_s, \qquad \epsilon_i \in \{\delta_i, \delta_i^*\},$$

for F and H, respectively. Delete all weight-one factors, which are identities for $\diamond$. If only one factor remains, then $H = F$ or F^*, and both belong to the prescribed fiber: rotation transports the whole chart and preserves its outer Schur function.

Otherwise split both products before their final factor. The two prefix ribbons have equal Schur functions by the forward direction of the classification, and their final ribbons agree up to rotation. Both prefixes and both final factors have at least two cells. The empty-mark identity therefore supplies proper charts placing F and H in one common-final-factor fiber. By Proposition 3.3(4), that fiber is contained in $\mathcal{F}(\omega, s_D)$. Thus $\mathrm{Cl}(F) \subseteq \mathcal{F}(\omega, s_D)$; the reverse inclusion is the forward composition and reversal identity, with the ordinary gauge carried on the outer function. □

This unconditional ribbon case needs no current-area induction call and is used separately from Lemma 3.2.

Lemma 3.5 (Changing a proper outer)**.** *Suppose $D \circ_\omega E$ is source-minimal and E is not a line. For every connected A with $|A| \geq 2$, the chart $A \circ_\omega E$ is source-minimal. Its period divides every proper period at that source. For every such A, the connected Schur class of $A \circ_\omega E$ has no member with a proper line-inner chart.*

Proof. The source-minimality and divisibility assertions are the outer-independent minimality lemma of the preprint [8]; its proof is included to fix the source at which the minimum is used. Let p, w be the period and width. A proper competing period not divisible by p would, by Proposition 3.3, either make E a line or give a real period-d chart with $d \mid p$ and $d < p$. Divisible reconstruction places that same marked d-factor on E. Its aligned reassociation through the original displacement of content p then places a proper d-chart on $D \circ_\omega E$. Its width is $w \bmod d \leq w$, with strict improvement of the inner span at a width tie, contradicting the original minimum. Thus every period at $A \circ_\omega E$ is a multiple of p. If r is such a multiple, the span law makes its width $bp + w$ for some $b \geq 0$; neither its width nor its inner span improves the displayed score.

Put $G = A \circ_\omega E$. If G had a proper chart of period r with a line inner T, the divisibility just proved would give $p \mid r$. At $r = p$, equal-period reconstruction from the complete end windows identifies the two inners in their permitted physical frames, making E a line. At $r > p$, the supplied-divisor construction of Proposition 3.3 gives an actual composition $T = V \circ_\omega E$, with V connected and $|V| \geq 2$. Its copy union contains a translate of E; supplemental copies only add cells. A row or a column cannot contain a nonline diagram. Both alternatives are impossible.

Rotation, transposition and the ordinary gauge preserve this distinction, so the argument covers both line orientations and their genuine edge readings.

Suppose now that another connected H with $s_H = s_G$ carried a proper line-inner chart. The unconditional full row/column transport of [7] places that same marked line chart on G, with the actual gauge transported as well. Its inner still has at least two cells, and its outer has the same transported Schur function as the outer at H, hence the same area, at least two. The chart on G is therefore proper, contradicting the preceding exclusion. No lower-area classification is used. □

Every invocation of Lemma 3.2 below is on an actual diagram.

Lemma 3.6. *If $F = D \circ_\omega E$ is proper, then $|E| < |F|$ and $|D| < |F|$. Every connected component of F^- also has area less than $|F|$.*

Proof. Put $\alpha = |W|$, $\beta = |\widehat{W}|$, and $\gamma = |\widehat{O}|$. The actual row union has $j|E| - (j-1)\alpha$ cells, so $\Phi_\omega(h_j)$ has weighted degree $j|E|$. The finite caps satisfy $K_- \cup (K_+ - v) = \widehat{W}$ and $K_- \cap (K_+ - v) = W^-$. Since $|W^-| = |W| - w$ and $E^- = K_- \sqcup \widehat{O} \sqcup K_+$,

$$|E| = N_E + |E^-| = p + \alpha + \beta + \gamma.$$

At empty width the same equality follows from $|E| = p + |E^-|$. Taking weighted degrees in the full normalization (2.6) therefore gives the exact area identity

$$|D \circ_\omega E| = p|D| + \alpha r(D) + \beta c(D) + \gamma N_D.$$

The formula includes mixed supplemental copies because they occur in that same normalization. In an opposite ordinary gauge the corresponding row and column coefficients are exchanged together. All coefficients are nonnegative, and every proper chart has $p \geq 2$. Thus $|F| \geq 2|D| > |D|$. Comparing with the cell outer gives

$$|F| - |E| = p(|D| - 1) + \alpha(r(D) - 1) + \beta(c(D) - 1) + \gamma(N_D - 1) > 0.$$

Finally $|F^-| = |F| - N_F < |F|$. Every connected component is strict in area, and any transported child whose function divides this proper product is strict by weighted degree. No span bound is used as an area bound. □

3.2. **Erosion, inflation, and zero-free classes.** Use the actual erosion $\mathcal{E}$ and one-step profile inflation $\mathcal{I} = I_1$ of [9]. Erosion keeps the cells with an eastern neighbor, in their original rows; inflation adds one to every row length and adjacent overlap-minus-one. The erosion lemma gives the exact area drop, and the zero-free class transport lemma gives inverse class bijections when the source is connected, zero-free and not a line. The inflation-of-legal-charts lemma transports supplied proper charts with nonrow outer and zero-free inflated parent, including the genuine empty-mark and maximal-width atom alternatives. These are independent transport results. The use of the smaller-area package remains in the following proposition.

Proposition 3.7 (Zero-free classes from strict descent)**.** *Assume the joint package below the area of a connected zero-free diagram S. Then its entire class is a rotation orbit or a single full common-factor component. In particular every prescribed source-minimal chart has full fiber.*

Proof. A line is immediate. Otherwise let $T = \mathcal{E}(S)$. The erosion and zero-free class transport lemmas of [9] make T connected, of area $|S| - r(S) < |S|$, and identify its entire class with that of S by inverse erosion and inflation. Apply Lemma 3.2 only at this strictly smaller diagram T. If T is chartless, its class, and hence the class of S, is its rotation orbit. If T is rooted, choose a fresh source-minimal chart $T = C \circ_\mu V$. The entire class has this marked fiber. If C is a row, its class is just that row, so inflation again leaves only the rotation orbit. Otherwise every mate outer is nonrow, since row count is a Schur invariant. Apply [9, inflation-of-legal-charts lemma] to each actual mate chart: its inflation belongs to the zero-free class of S. The resulting charts have one common marked factor. In the first two branches their outer functions are still equal.

In the maximal-width atom branch the actual outers become $\mathcal{I}(C), \mathcal{I}(C')$. The atom conclusion makes both zero-free, so neither C nor C' has a Jacobi–Trudi index -1. Since $s_C = s_{C'}$,

their row counts have a common value r. The no--1 raising identity in the proof of [9, zero-free class transport lemma] therefore gives

$$s_{\mathcal{I}(U)} = h_1^r s_U(h_j \mapsto h_{j+1}/h_1) \qquad (U = C, C'),$$

so these whole outer functions agree as well. This uses the actual no--1 domain, not a general equality-preserving property of inflation. Thus all class members share one common-factor component. Source-minimal domination and the component-key theorem, now at S, give the assertion for every prescribed minimum. No same-area class receives the lower package. □

3.3. **Marks under core deletion.**

Lemma 3.8 (Mark-sensitive core reduction)**.** *A legal marked inner of period p and width w induces a legal marked inner on its zero-free core, with the same period and width. At positive distinct phases the outer is retained as D or $\mathrm{AT}(D)$; at positive coincident phases and at empty width it is replaced by $\mathrm{Core}(D)$. These are literal composition identities on core diagrams, not transports of source minima.*

Proof. Replace each boundary pair by $((A-B)_+, (B-A)_+)$. At each distinct free slot of (2.2), the other bit is fixed. If it is zero, core deletion retains the free bit; if it is one, the free bit moves to the other boundary and is complemented. Thus both outer bits survive, with one of the four attachment types. If the resulting mixed order is reversed, the full end intersection adds the coincident thin tail on $[s, t-1]$, exchanges the phases and replaces D by $\mathrm{AT}(D)$; the identity $(A_{\mathrm{AT}(D)}, B_{\mathrm{AT}(D)}) = (1 - B_D, 1 - A_D)$ gives exactly the new slots. At a coincident slot the two outer bits are cored together, giving $\mathrm{Core}(D)$. A legal full-hat ordinary dual gives the (a) reading when needed; its connected full-span condition and proof are recorded in Lemma D.1.

The transformed lower and upper windows are obtained by deleting the appropriate northwest or southeast thin tails. Their corrected endpoint words agree and their heights are the original marked heights, with the extra full-intersection interval in the reversed mixed case. Hence they are congruent connected skew diagrams. The full intersection and singleton ends fix maximality at every displacement. Removing either mark leaves a thin tail joined through a private singleton to the unchanged positive private band, so both complements are connected skew. Finally, the hatted middle is confined to $[w+1, p-2]$, whereas all periodic hatted marks lie in marked bands. The content separation proves IV. Applying the complete coding table, including supplements, proves equality of both core boundary words and therefore the literal composition identities. □

3.4. **A class-valued divisible quotient.**

Proposition 3.9. *Suppose actual proper charts $F = D \circ_\omega E$ and $H = C \circ_\eta U$ are Schur-equal, their common class has full source-minimal fibers, and the joint package holds below their area. Let $p = kq$, $k \geq 2$, and write their widths as $w_s = bq + w_f$, $0 \leq b < k$. There is an actual connected J with $N_J = k + b$ such that*

$$s_E = s_{J \circ_\eta U}. \tag{3.1}$$

No source minimum of either displayed chart is required.

Proof. Induct on q. If U is a line, full line transport places the same marked q-chart on F and divisible gluing reconstructs (3.1). Otherwise choose a fresh minimum of period d on H. Local comparison with the displayed nonline U forces $d \mid q$: a smaller supplied divisor or a proper gcd chart would improve the fresh minimum, and coprimality would make U a line. This argument does not require the fresh inner to be nonline.

At $d = q$ the original fine chart is itself a minimum, so full transport and divisible gluing finish. At $d < q$, full transport gives the same actual marked d-factor W on both parents. Divisible gluing separately gives $\widetilde{E} = A \circ W$ and $U = B \circ W$, where any rotation of the first parent is carried through its full source chart. Aligned reassociation gives actual equal quotient parents, retaining outer functions up to the genuine gauges, of periods $p/d, q/d$. If the d-width is v, the new widths are $(w_s - v)/d$ and $(w_f - v)/d$, still differing by bq/d. The quotient parents

are strict-area outers of proper compositions through W; their displayed native spans are at least two. Lower full transport therefore supplies all recursive hypotheses.

The recursive conclusion is $s_A = s_{J_1 \circ B}$, a whole-function equality, not a literal chart on A. Forward it through the same marked W and undo the fine aligned gauge to obtain (3.1). This uses the already constructed all-outer reassociation for U, not a converse Mark lift. The span identity then gives $p + w_s = qN_J + w_f$, so $N_J = k + b \geq 2$. Every recursive step lowers both q and the actual parent area. □

The proposition is used on same-area zero-free core classes only after Proposition 3.7 has supplied their full fibers. Original minima are not assigned to those cores. The return from core functions to the entire original function is the aligned reconstruction in Appendix A.

4. Reduction of the original comparison

Fix the hypotheses of Theorem 1.1 and suppose $p > q$. Write

$$N = pM + w = qJ + u, \qquad M = N_D = |D| \geq 2, \quad J = N_C \geq 2.$$

At this stage J is a span, not a cell count. The displayed inner diagrams are nonline, since the class contains no proper line-inner chart. All gauges below act on complete marked charts.

4.1. **Actual reconstruction at divisible periods.** The distinction between constructing an actual reference and applying a minimum is useful throughout the reduction.

Proposition 4.1 (Aligned reconstruction)**.** *Suppose $p = kq$, $k \geq 2$, and write $w = bq + u$ with $0 \leq b < k$. Retain the original whole equality, the source ribbon D, the prescribed fine minimum through the nonline marked inner U, and $\mathsf{J}_{<n}$. The construction does not require the displayed period-p chart to be source-minimal. There is an actual connected diagram J_0 of span $k+b$ such that*

$$s_E = s_{J_0 \circ_\eta U}. \tag{4.1}$$

The construction uses only the original finite source words, of total outer weight at most M.

Proof. Appendix A proves the assertion in three steps. The terminal and penultimate source identities first give a reduced primitive polynomial P with the *entire* image $\Phi_\eta(P) = Ls_E$, where L uses only raw generators below q. If the fine mark has distinct phases, the mark-sensitive core map retains the entire coarse polynomial, and the actual class-valued quotient of Proposition 3.9 realizes P. At zero offset the fine outer interior either has a noncell component or consists only of cells. In the former case a full ordered conormal calculation supplies the actual end windows and the returned section; in the latter case the integral full-unit cut theorem constructs the actual remainder. In both cases P is the canonical polynomial of an actual J_0. Composing through the original η/U, connected irreducibility and the low-span normalizers give (4.1). The span law gives $N_{J_0} = k + b \geq 2$. The construction and its three branches are proved in Propositions A.1, A.4 and A.6 and Theorem B.1. □

Corollary 4.2. *The original source-minimal comparison cannot have $p = kq > q$.*

Proof. Put $G = J_0 \circ_\eta U$. It is proper, $s_G = s_E$, and $|G| = |E| < n$. Outer-independent minimality, using the originally prescribed fine minimum, makes the q-chart of G minimal. Lower full-fiber transport puts that chart on E, up to its genuine transports. Aligned reassociation with the original period-p mark constructs a period-q chart on F. Its score $(u, q+u)$ is strictly smaller than $(bq+u, p+bq+u)$. □

4.2. **Fresh factors and the arbitrary competing outer.** For the rest of this section, assume that neither period divides the other. The actual core class has full fibers by Proposition 3.7. A fresh minimum in that class can be stripped without assigning either original minimum to a core.

Lemma 4.3 (Fresh-factor descent)**.** *In an actual full-fiber class with two displayed nondividing proper periods, if both displayed natives are nonline, a fresh minimal period d divides both displayed periods. Stripping its same marked factor gives actual equal quotient parents of strictly smaller area and divides both periods by d. The procedure terminates at a line alternative. If one displayed native is a line, both displayed natives can instead be put in a common line orientation by actual line transport and same-parent gluing.*

Proof. Compare a fresh minimum with each displayed nonline chart at the same parent. A supplied smaller divisor or a proper gcd chart would improve its score, and coprimality would force the displayed native to be a line. Thus its period divides each displayed period, whether or not its own native is a line. Full transport supplies the same marked factor at both parents before divisible gluing is applied. The all-outer aligned identity then gives equal quotient functions with their full gauges and normalizers. The quotients are actual outers of proper compositions, hence have smaller area by Lemma 3.6. Their displayed native spans remain at least two. Every further full-fiber call is therefore supplied by $\mathsf{J}_{<n}$. The gcd decreases strictly at a nontrivial step.

If a line native appears, its full line fiber transfers that actual chart to the other parent. For coprime periods, the same-parent coprime theorem makes both natives lines in the same axis. Otherwise let $h > 1$ be their gcd. The actual gcd chart exists by local gluing, and divisible reconstruction of the supplied line native makes its period-h inner a line. Strip this supplied factor in its genuine line gauge. The quotient periods are coprime, so both quotient natives are lines in the same axis; one recomposed native is the supplied line, so its quotient axis is the line-preserving one. Reattachment through the same line factor therefore makes the other native a line in that axis as well. An incompatible attachment pair is excluded by local gluing, not retained as a line chart. These operations construct and strip one actual factor at a time; no composite mark is inferred by reassociating an unconstructed tower. □

At a positive distinct-phase mark the original outer survives core reduction as itself or its actual antitransposition. At a coincident or empty mark, it becomes the outer's common-beta core. This difference determines what can be recovered about the original C.

Proposition 4.4. *Suppose the original source mark is positive with distinct phases and $p = Ag$, $q = Bg$, where $A > B \geq 2$ and $\gcd(A, B) = 1$.*

1. *If D is a line and the fine mark is also positive with distinct phases, then the original C is a line.*
2. *If D is a line and the fine mark is coincident or empty, then the original C is a ribbon.*
3. *If D is a ribbon but not a line, then, in a genuine row or column orientation, every row or every column length of D is divisible by B. Consequently $D = D_1 \circ_{\varnothing} V_B$ literally, where V_B is a B-cell line and D_1 is a proper ribbon.*

Proof. Take the actual whole cores and apply Lemma 4.3. At the terminal line alternative read both natives as row-line actions, with all physical gauges carried. Their periods P, Q retain ratio A/B; denote the widths by v, t. If the transported source ribbon has row lengths r_i, its row-layer parent has lengths $Pr_i + v$ and every adjacent gap v. The fine parent has row lengths $Qs_j + t$ and gaps $Qh_j + t$.

If the source outer is a row, row-count invariance makes the fine parent a row, and the actual fine row action makes its outer a row. If the source outer is a column of length at least two, equality of the gap and row multisets gives

$$v = Qh + t, \qquad P + v = Qs_j + t,$$

so $Q \mid P$, contrary to the retained ratio. A line source must therefore be in the row alternative. Undo the full transports. A distinct fine mark retained C or $\mathrm{AT}(C)$, proving (i). A coincident or empty fine mark retained $\mathrm{Core}(C)$; a line core has area equal to span, and common-beta deletion preserves both, so C is a ribbon. This proves (ii).

For a nonline source ribbon there is at least one gap, and the same comparison gives

$$v = Qh + t, \qquad Pr_i + v = Qs_{\pi(i)} + t.$$

Thus $B \mid r_i$ for every transported source row. Row and column multisets are Schur invariants, so this divisibility holds for the actual original D in the corresponding axis. Divide those lengths by B and retain its zero gaps. This explicitly constructs the ribbon D_1. There are at least two rows in that orientation, whence $|D| \geq 2B$ and $|D_1| \geq 2$. The identity is the actual empty row action, or its transported column version. □

Lemma 4.5. *For the actual factorization in Proposition 4.4(iii), put $G = V_B \circ_\omega E$. There is a legal marked presentation*

$$F = D_1 \circ_{\omega_B} G \tag{4.2}$$

with width w and period Bp. The original period-p chart of G is proper and source-minimal, and $|G| < n$.

Proof. For V_B a row, use the two extreme original marked copies, whose displacement is Bv. Their full intersection, the two one-mark complements, and their hatted neighborhoods are read on the same repeated strip. The mark is maximal, the private singleton gap supplies separation, and the original thin tails give connected complements and hatted nonadjacency. The all-outer copy formula then gives (4.2), including mixed supplemental copies.

For a column, use its two complete period-p end windows. The true translation is $Bv-(B-1)e$ in case (a) and $Bv + (B-1)e$ in case (b), and Bv in the mixed reading. The same window tests prove the full intersection and complements. The hatted neighborhoods depend only on the retained mark windows; the appropriate original rank formula transports all supplemental copies. Equivalently this is the aligned identity of Proposition 3.3 applied after those windows have constructed the upper mark. Its content is Bp in every case.

The chart $G = V_B \circ_\omega E$ uses the original nonline E and a proper outer, so Lemma 3.5 gives its old period-p minimum. Since D_1 is proper in (4.2), G has strict area below F. The new period-Bp chart of F is not asserted minimal. □

Corollary 4.6. *A positive distinct-phase source with a nonline ribbon outer cannot occur in the original nondividing comparison.*

Proof. Use Lemma 4.5. Now $Bp = Aq$ is aligned with the unchanged original fine chart. Apply only Proposition 4.1 to (4.2); its source minimum is not required. It constructs $G_{\mathrm{ref}} = J_0 \circ_\eta U$ with $s_G = s_{G_{\mathrm{ref}}}$ and

$$N_{J_0} = A + \lfloor w/q \rfloor \geq A \geq 3.$$

Both are actual diagrams of area $|G| < n$. The old p-chart of G and the original-U q-chart of G_{ref} are minimal by Lemma 3.5. Lower rooted-key invariance forces their periods to agree, a contradiction. The words used by the regrouped construction have their original total weight: one Bp index is an original period-p index of weight B, and $B|D_1| = M$. □

4.3. **Coprime periods.**

Proposition 4.7. *The original comparison cannot have* $\gcd(p, q) = 1$.

Proof. In the full actual core class, a fresh minimum could not divide two coprime periods if both displayed natives were nonline. The line alternative therefore makes both native cores lines, hence rows because they are zero-free. Area and span are preserved, so the original E, U are ribbons. Positive coincidence is impossible on a ribbon: its two middle phases are the extreme phases in the appropriate genuine reading.

For a positive ribbon native, the mark-sensitive core transformation retains its outer without antitransposition. To check this choice, compose with the actual two-cell row. Its corner is one, so its core is a row. The transformed native is a row with its positive row mark, whereas an antitransposed two-cell outer would give a two-row parent. This fixes the choice in all original attachment cases.

If the source mark is empty, or if it is positive and D is a row, then F itself is a ribbon by the corner law. By Lemma 3.4, its prescribed source-minimal chart has full fiber. The component-key theorem of [7] then gives $K(F) = K(H)$, hence $p = q$. Otherwise D has at least two rows.

The row-layer comparison just used gives

$$w = qh + u, \qquad pr_i + w = qs_{\pi(i)} + u.$$

Coprimality makes every original row of D divisible by q. Thus $D = D_1 \circ_{\varnothing} \mathrm{Row}_q$ with $|D_1| \geq 2$. Regroup through the original mark and put $G = \mathrm{Row}_q \circ_{\omega} E$. This is a strict-area actual ribbon carrying its old minimal p-chart.

If the fine mark is positive it is distinct, and the distinct-phase branch of Proposition 4.1 supplies the proper actual q-reference to G. If it is empty, the same terminal readout supplies $\Phi_U(P) = Ls_G$. Here the empty ribbon map has unit fresh coefficients and $\kappa(G) = 1$; its top equation makes L a rational constant. The explicit ribbon-image projection of Lemma A.7 constructs the actual outer by retaining G's cuts at multiples of q. Either way lower full transport compares two strict-area old minima at p and q, a contradiction. □

4.4. **The complete reduction.** The empty-overlap, coincident-phase and zero-remainder exits are proved in Appendices C and D. Their different geometries supply different whole scalar identities. The ordered table below keeps each exit's original entry hypotheses.

Theorem 4.8 (Reduction to the distinct line case)**.** *Every strict-period input of Theorem 1.1 is either excluded by the ordered cases in Table 1 or has two positive distinct-phase original marks, two original line outers of at least two cells, and*

$$p = Ag, \quad q = Bg, \quad A > B \geq 2, \quad \gcd(A, B) = 1, \qquad N = Lg + c, \quad 0 < c < g. \tag{4.3}$$

The original parents, charts, and minima are unchanged at this last input.

Proof. Apply Table 1 in order. The divisible and coprime cases are excluded by Corollary 4.2 and Proposition 4.7. Then Propositions C.3 and C.4 exclude, respectively, two empty marks and exactly one empty mark, with C still arbitrary at entry. Propositions D.6 and D.7 exclude two positive coincidences and exactly one positive coincidence.

Both original marks are now positive and distinct. If D is not a line, Corollary 4.6 excludes the comparison; otherwise Proposition 4.4(i) proves that the original C is also a line. Only at this entry do we use the two-line zero-remainder case of Proposition C.4 to exclude $g \mid N$. The remaining periods and residue satisfy (4.3). A mixed positive mark always has distinct phases, a connected ribbon is either a line or a nonline ribbon, and a residue modulo g is either zero or positive. These alternatives exhaust the original inputs, retaining both original source minima throughout. □

Reduction stage	Conclusion or reference
$q \mid p$	Corollary 4.2
$\gcd(p, q) = 1$	Proposition 4.7
Both widths zero	Proposition C.3
Exactly one width zero	Proposition C.4
Both positive marks coincident	Proposition D.6
Exactly one positive mark coincident	Proposition D.7
Both distinct; D not a line	Corollary 4.6
Both distinct; D a line	C a line by Proposition 4.4
In the preceding line case, $g \mid N$	Proposition C.4
Remaining case	(4.3), treated below

TABLE 1. The original ribbon-outer comparison, with C arbitrary at entry and both original minima retained. Terminal exits are applied in order. The row deriving that C is a line is a transition: the following zero-remainder test is made within that line case. The remaining input is (4.3).

5. Periodic cores and actual boundary comparisons

Assume (4.3). Write the two original widths as $w = d_p g + c$ and $u = d_q g + c$. Both original line counts are at least two, so

$$N \geq 2\max(p,q) + c, \qquad a := N - q - c \geq q \geq 2g > c, \qquad m := N - q = a + c. \tag{5.1}$$

Here m is a shortened span; n continues to denote the original area.

5.1. **Propagation along actual factor chains.** For a connected diagram of span $N = LD+v$, $0 \leq v < D$, let $\mathcal{T}(D,v)$ denote the following two properties: both full boundary words are D-periodic on the *open* middle $v < k < LD$, and the diagonals $v + jD$ and $(j+1)D - 1$ are singletons for $0 \leq j < L$. Let $\mathcal{P}(D,v)$ denote the corresponding periodic-height and singleton properties. No boundary step is introduced at 0 or N.

Lemma 5.1. *If A has $\mathcal{T}(D_1, v_1)$ and $S = A \circ_\zeta W$ is a legal chart of period d and width v_0, then*

$$S \text{ has } \mathcal{T}(dD_1, dv_1 + v_0).$$

The property is invariant under whole rotation and antitransposition. Moreover, $\mathcal{T}(D,v)$ implies $\mathcal{T}(hD, N \bmod hD)$ whenever hD is the grouped period under consideration.

Proof. Set $D = dD_1$ and $v = dv_1 + v_0$. Away from the exceptional residues of the actual coding table, each boundary is literally d-periodic. An exceptional step $k = dj + r$, $0 \leq r \leq v_0$, lying in $v < k < LD$, reads an actual outer index with $v_1 < j < LD_1$. Translating k by D translates that index by D_1 without changing which outer bit or complement is read. The complete coding, including its mixed supplements, therefore proves the asserted open periodicity.

On a private residue $v_0 \leq r < d$, the exact height is

$$H_S(dj + r) = H_W(r) + H_A(j) - 1.$$

The native private endpoints $v_0, d-1$ are singletons. Substituting $j = v_1 + tD_1$ and $j = (t+1)D_1 - 1$ proves the two new singleton families. Rotation sends k to $N-k$ and interchanges the families; antitransposition sends (A_k, B_k) to $(1 - B_k, 1 - A_k)$ and preserves heights.

For grouping, if $N = LD + v = L'hD + c$, then $c = v + (L \bmod h)D$. The new open interval is contained in the old one, and its singleton samples are old samples. This proves the last assertion without asserting a new chart. □

Proposition 5.2 (Whole-parent structure)**.** *The original whole cores have $\mathcal{T}(g,c)$. Each positive connected component of their proper interiors has span less than g. On the original unit-bearing parents, $\mathcal{P}(g,c)$ holds and*

$$\kappa(S) \in \mathbb{Q}[h_1, \ldots, h_{g-1}] \tag{5.2}$$

for either original parent S.

Proof. Apply Lemma 4.3 to the two actual whole core parents. Throughout the descent their displayed outers remain lines. At its terminal line alternative read both natives as rows. A row outer gives a one-row parent; a column outer of length $M \geq 2$ gives M rows of length $P+v$ and gaps v. One row and a proper column outer cannot yield equal row counts. If both outers are columns, equality of row counts, gap multisets, and then row multisets forces equal terminal periods, contrary to the unchanged nondividing ratio. Thus both terminal *whole* parents are lines.

Apply Lemma 5.1 separately up the two actual factor chains. The product D_0 of their stripped periods divides g, and the accumulated width is the corresponding span remainder. This proves $\mathcal{T}(D_0, N \bmod D_0)$ on each whole core and then $\mathcal{T}(g,c)$ by grouping. The two chains need not have identical orientations or an accumulated common marked inner.

The corner estimate propagates at the same time. If every quotient component has span $< D_1$, a transported child at a lower period d and width $v_0 < d$ has span

$$dN_Z + v_0 \leq d(D_1 - 1) + v_0 < dD_1.$$

All remaining local corner factors have span $< d$. Starting from a line with empty proper interior gives the claimed bound $< D_0 \le g$. A one-cell component is a real factor in this calculation, not a proper chart to which transport is applied.

Restoring a common beta preserves the whole height and the proper-height support intervals. It can change the diagram and function of a component, but cannot move, join, or enlarge its content-support interval. Thus the component span bound and the height and singleton properties pass to the original parents. Their complete proper products give (5.2). The separate restored boundary words have not yet been synchronized. □

5.2. **Confinement of the original exceptional phases.**

Lemma 5.3. *For an original line chart of period p and positive width w, let t, s be its original phases. Then*

outer	*initial bound*	*terminal bound*
Row	$t \le c$	$w - s \le c$
Col	$s \le c$	$w - t \le c$.

Both full original boundary words are p-periodic on

$$\mathcal{I} = \{c+1, \dots, N-c-1\}. \tag{5.3}$$

Proof. Substitute the actual line bits $0,0$ or $1,1$ in the complete coding table. For a row outer there is only one possible initial height correction, at t, and one terminal deficit, at $Mp+s$. For a column outer the roles are s and $Mp+t$. The other free bit equals its interior line value; this includes the mixed supplemental copies.

If the initial row phase were $t > c$, its increment and its p-translated interior increment would both be in the periodic-height middle. Properness keeps the translated occurrence in range. Their heights require equal increments, but exactly one changed boundary bit changes the increment by one. This is impossible. Reading backward proves $w - s \le c$; interchanging the line selector gives the column bounds. Omitted endpoint steps remain omitted. All original cap deviations are now outside (5.3), so the actual coding gives full p-periodicity there. □

5.3. **Minimal cap-preserving splices.** For singleton contents $u < v$ in a connected diagram S, identify their singleton cells by keeping contents at most u and translating contents at least v. Write the resulting diagram as $\mathrm{Splice}(S; u, v)$. If $d = v - u$ and S has full flags X, Y, the complete new flags are

$$X' = (X \cap [1,u]) \cup ((X \cap (v,N]) - d),$$
$$Y' = (Y \cap [0,u]) \cup ((Y \cap (v,N)) - d).$$

The flag at u is retained, and the flag at v is discarded. The diagonal criterion proves actualness: the incoming step is the old step at u, the outgoing step is the old step after v, and all other positive diagonal intervals are inherited.

Define

$$S^-_{\mathrm{cap}} = \mathrm{Splice}(S; a-1, N-c-1), \qquad S^+_{\mathrm{cap}} = \mathrm{Splice}(S; c, q+c).$$

The cuts in Figure 2 are the private singletons of Proposition 5.2. Both diagrams have span $m = a+c$, preserve the complete first and last $c+1$ diagonals of S with the proper translations, and have strictly smaller area. Their height words agree because q is a multiple of g. They consequently have the same literal zero-free core, although their full Schur functions need not agree.

5.4. **The complete weak return frames.** Write the source's full lists as $X = (N, I)$ and $Y = (C, 0)$, so $I = X \setminus \{N\}$ and $C = Y \setminus \{0\}$, in their original decreasing orders. In the cap calculations these letters denote proper lists, not the competing outer. In the relative index

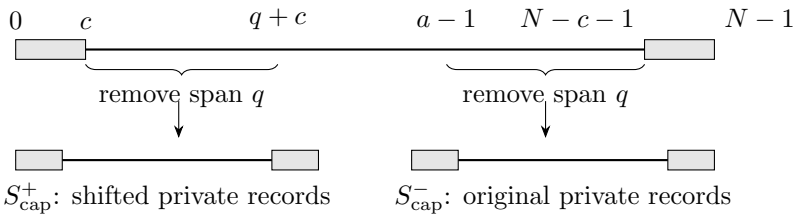


FIGURE 2. The two singleton splices, shown schematically and not to scale. Both preserve the full $(c+1)$-caps. Their private comparison is exactly $c < k < N-q-c$, so every pair $k, k+q \in \mathcal{I}$ is represented.

universe $K = \{0, \dots, c\}$ put

$$\begin{aligned} I_R &= (I \cap [a, m]) - a, & C_R &= (C \cap [a, m]) - a, \\ I_L &= (I \cap [q, q+c]) - q, & C_L &= (C \cap [q, q+c]) - q, \\ I_+ &= (I \cap [N-c, N)) - (N-c), & C_+ &= (C \cap [N-c, N)) - (N-c), \\ I_- &= I \cap [1, c], & C_- &= C \cap [1, c]. \end{aligned}$$

The square matrices

$$B_R = \mathrm{JT}(I_R, C_R), \quad B_L = \mathrm{JT}(I_L, C_L), \quad G_R = \mathrm{JT}(I_+, C_+), \quad G_L = \mathrm{JT}(I_-, C_-)$$

are actual weak proper blocks. Their determinants are nonzero products of connected component functions and unit pivots; their extreme inverse entries may vanish. Set $U_{\mathrm{cap}} = \{c\} \cup I_+$ and $V_{\mathrm{cap}} = \{0\} \cup C_-$. The normalized upper and lower cap vectors are

$$l(z) = \sum_{i \in U_{\mathrm{cap}}} \ell_i z^{c-i}, \quad \ell_c = 1, \qquad r(z) = \sum_{j \in V_{\mathrm{cap}}} r_j z^j, \quad r_0 = 1. \tag{5.4}$$

Their full residuals are

$$v_L(j) = \sum_{i \in U_{\mathrm{cap}}} \ell_i h_{i-j}, \qquad v_R(i) = \sum_{j \in V_{\mathrm{cap}}} h_{i-j} r_j.$$

Thus $v_L(j) = 0$ for $j \in C_+$, and $v_R(i) = 0$ for $i \in I_-$, by the actual upper-column and lower-row equations. Put $l_k = [z^k]l(z)$ and $r_k = [z^k]r(z)$, with coefficients outside $0, \dots, c$ equal to zero. For ordered lists J, K, the weighted diagonal matrices are $X_J = \mathrm{diag}(i)_{i \in J}$ and $Y_K = \mathrm{diag}(j)_{j \in K}$, with the entries in their original decreasing orders. Every cap-vector coefficient on its actual support is nonzero: deleting any row and the permanent bottom column, or the top row and any column, from a connected cap leaves a feasible weak Jacobi–Trudi minor. Its diagonal matching cannot cancel.

The original period-p copies provide reference frames. Write $B^*_R = \mathrm{JT}(I^*_R, C^*_R)$ and $B^*_L = \mathrm{JT}(I^*_L, C^*_L)$ for their two complete reference lists. In an actual ordinary-(a) row frame let $w = d_p g + c$, $\tau = t$ and $\sigma = s - d_p g$; then

$$B^*_R = \mathrm{JT}(U_{\mathrm{cap}} \setminus \{\sigma\}, C_+), \qquad B^*_L = \mathrm{JT}(I_-, V_{\mathrm{cap}} \setminus \{\tau\}). \tag{5.5}$$

For an ordinary-(a) column frame, set $\sigma = t - d_p g$, $\tau = s$. The corresponding references are

$$B^*_R = \mathrm{JT}(U_{\mathrm{cap}}, C_+ \cup \{\sigma\}), \qquad B^*_L = \mathrm{JT}(I_- \cup \{\tau\}, V_{\mathrm{cap}}).$$

The added entries are genuinely absent from the respective cap lists. In every reference formula σ is the upper active index and τ the lower active index. For an ordinary period-d chart, its clock is the fixed signed coefficient λ in the original annihilator $V_d - \lambda\delta_d$; the derivations are defined in Section 6. The column clock is therefore $-p(M-1) + s - t = -(N - p - c + \sigma - \tau)$. For a mixed-(d) frame, let z_0 be its initial corrected phase and o_0 its terminal deficient phase; put $\tau = z_0$, $\sigma = o_0 - d_p g$. Then

$$B^*_R = \mathrm{JT}(U_{\mathrm{cap}}, C_+ \cup \{\sigma\}), \qquad B^*_L = \mathrm{JT}(I_-, V_{\mathrm{cap}} \setminus \{\tau\}).$$

These identities follow by comparing the cap's complete flag word with its actual p-translate. Nonexceptional bits agree; the two exceptional bits are exactly the indicated deletion or addition. The singleton endpoints fix the $0, c$ records. Shared unit contacts are included.

Deleting all common betas from every return and reference leaves one strict disjoint weak core (I_0, C_0). Its height jumps determine its two lists, and the proper-height word is common to all internal g-bands. Write $A_0 = \mathrm{JT}(I_0, C_0)$ for its complete matrix. Consequently

$$(I_Z, C_Z) = (I_0 \cup Z, C_0 \cup Z), \qquad Z \subseteq \Omega := K \setminus (I_0 \cup C_0). \tag{5.6}$$

Write $Z_R = I_R \cap C_R$ and $Z_L = I_L \cap C_L$ for the restored common-beta sets of the right and left return frames. These are not independently chosen masks. The actual band sequence is A-periodic; the references occupy residues $0, d_p$ and the observed returns residues $B, d_p - B$ modulo A. Coincident residues carry the same complete frames. All zero-height buffers retain their actual triangular inverse actions.

6. The complete scalar and its second channel

6.1. **The whole source gradient.** Let S be one original parent. Put $A_S = \mathrm{JT}(I, C)$, $b = (h_i)_{i\in I}$, $w = (h_{N-j})_{j\in C}$, and $\kappa = \det A_S$. Moving the distinguished column to the front gives the canonical Schur complement

$$f_S = \frac{Q_S}{\kappa} = h_N - wA_S^{-1}b = \frac{s_S}{\partial_{h_N} s_S}. \tag{6.1}$$

Equal original Schur functions therefore give the same normalized whole function.

Temporarily replace every physical h_0 in every matrix by one variable ν, keep all negative entries zero, and put

$$\mathcal{H}(z) = \nu + \sum_{j>0} h_j z^j, \qquad \Theta_z = z\frac{d}{dz}.$$

This is a coherent homogenization, not an additional observation: $f_\nu(\boldsymbol{h}) = \nu f_1(\boldsymbol{h}/\nu)$. Every observer below is homogeneous of ordinary degree one, so a whole identity at $\nu = 1$ extends exactly. The operators fix ν. For a rational function of ν, write $[\nu^k]_\infty$ for its Laurent coefficient at infinity: substitute $\nu = t^{-1}$, expand at $t = 0$, and take the coefficient of t^{-k}.

Differentiating (6.1) gives the full gradient

$$\partial_{h_m} f_S = \mathbf{1}_{m=N} - w_m A_S^{-1}b - wA_S^{-1}b_m + wA_S^{-1}(A_S)_m A_S^{-1}b.$$

In particular the inverse-motion term is present. With the complete endpoint coframes

$$R_S(z) = 1 - \sum_{j\in C}(A_S^{-1}b)_j z^j, \qquad L_S(z) = 1 - \sum_{i\in I}(wA_S^{-1})_i z^{N-i},$$

all occurrences of each repeated raw label collect to

$$\partial_{h_m} f_S = [z^{N-m}] R_S L_S. \tag{6.2}$$

Define the derivations

$$\delta_s = \sum_{j\geq s} h_{j-s}\partial_{h_j}, \qquad V_s = \sum_{j\geq s}(j-s)h_{j-s}\partial_{h_j}, \qquad \mathscr{D}_s(\lambda) = V_s - \lambda\delta_s.$$

The whole proper factor is below g, so all operators at indices at least g fix it. The genuine normalized observers at q are therefore

$$O_H = [z^m]\mathcal{H}R_S L_S, \qquad O_\lambda = [z^m]\mathcal{H}(m - \lambda - \Theta_z)(R_S L_S). \tag{6.3}$$

The original mixed chart supplies $O_H = 0$; an original ordinary chart supplies $O_\lambda = 0$ at its own fixed signed clock. The term $h_q \mapsto h_0$ creates ν, not zero.

For an ordinary case-(a) line chart with J cells and phase offset $\theta = s_f - t_f$, the clock is

$$\lambda = \varepsilon q(J-1) + \theta, \qquad \varepsilon = 1 \text{ for a row}, \quad \varepsilon = -1 \text{ for a column}.$$

Indeed composition covariance [10, Composition covariance] acts on h_J or e_J; $V_1 h_J = (J-1)h_{J-1}$ and $V_1 e_J = -(J-1)e_{J-1}$, while $\delta_1 h_J = h_{J-1}$ and $\delta_1 e_J = e_{J-1}$. The full local normalizer is

fixed. Whole antitransposition gives the opposite ordinary reading with the rebuilt signed clock. A mixed chart instead supplies $\delta_q Q_S = 0$. Thus each zero above is an identity of the original whole function, not a presumed equality of the two splices.

6.2. **Both actual return corrections.** Let R^0, L^0 be the endpoint coframes of the actual first and last a-diagonal windows. Their entries use labels below a, and their heads through degree c are r, l of (5.4). Define, for $0 \le i, j \le c$,

$$P_i = [z^{a+i}]\mathcal{H}R^0, \qquad Q_j = [z^{m-j}]\mathcal{H}L^0, \qquad \rho = B_R^{-1}P_{I_R}, \quad \zeta = Q_{C_L}^t B_L^{-1}. \tag{6.4}$$

The full source systems give, through degree m,

$$R_S = R^0 - \sum_{j\in C_R} \rho_j z^{a+j}, \qquad L_S = L^0 - \sum_{i\in I_L} \zeta_i z^{m-i}. \tag{6.5}$$

Indeed the base equations are closed because every entry from a lower base row to a return column is a negative zero. The remaining equations are exactly the two complete square systems in (6.4). Their possible column m and row 0 are retained. The product of the two correction tails has degree at least $2a > m$, so contributes zero for a physical negative-slot reason.

Let

$$u_L^0(j) = \sum_{i\in U_{\mathrm{cap}}} (i-j)\ell_i h_{i-j}, \qquad u_R^0(i) = \sum_{j\in V_{\mathrm{cap}}} (i-j)h_{i-j}r_j,$$
$$x_R(z) = \sum_{i\in I_R} (v_L(C_R)^t B_R^{-1})_i z^{c-i}, \qquad y_L(z) = \sum_{j\in C_L} (B_L^{-1}v_R(I_L))_j z^j.$$

Write x_R^0, y_L^0 for the corresponding weighted projections obtained from u_L^0, u_R^0. Put

$$T_R = x_R/l, \qquad T_L = y_L/r \pmod{z^{c+1}}, \qquad w_c = rl.$$

The same $c+1$ original variables $h_a, \dots, h_{a+c}$ occur at both ends. In P_i their coefficients are r_{i-j}; in Q_j they are l_{c-j-i}. Thus the entire shared-high coefficient polynomials are

$$\begin{aligned} D(z) &= w_c - rx_R - ly_L = w_c(1 - T_R - T_L), \\ A(a,\lambda;z) &= (a+c-\lambda)w_c - \Theta_z w_c - r(x_R^0 - \lambda x_R) - l(y_L^0 - \lambda y_L), \end{aligned} \tag{6.6}$$

read modulo z^{c+1}. Each whole observer is equivalent to these $c+1$ coefficient equations and one full remainder. The denominators are low and independent of the high variables. The residual vectors P, Q and the individual inverse entries are not independent probes.

6.3. **The full cap-functional identity.** Write $f_E = f_{S_{\mathrm{cap}}^-}$ and $f_F = f_{S_{\mathrm{cap}}^+}$ for the normalized functions of the two actual cofaces. For $\langle T\rangle = [z^m]\mathcal{H}T$, their cross kernel K and balanced endpoint current J satisfy

$$K = \langle R^0 L^0\rangle, \qquad f_E = \langle R^0 l\rangle, \qquad f_F = \langle L^0 r\rangle, \tag{6.7}$$

$$J = \langle \Theta_z(R^0L^0) - R^0\Theta_z l - L^0\Theta_z r\rangle. \tag{6.8}$$

To prove (6.7), substitute the actual cap matrices G_R, G_L in (6.5); their cross corrections pair with $v_L(C_+) = 0$ and $v_R(I_-) = 0$. The upper self-complement is

$$P_c - h_{c-C_+}G_R^{-1}P_{I_+} = \langle R^0 l\rangle,$$

and the lower formula is dual. Differentiating the upper tail gives its index-weighted zero residual and the remaining $\langle R^0\Theta_z l\rangle$ term; the lower one gives $\langle L^0\Theta_z r\rangle$. This proves (6.8) without a contact/endpoint alias.

For either observer set $(\varepsilon_{\mathrm{obs}}, s, C_1, D_1) = (0, 1, x_R, y_L)$ in the Hall case, and

$$(\varepsilon_{\mathrm{obs}}, s, C_1, D_1) = (1, m-\lambda, x_R^0 - \lambda x_R + \Theta_z l, y_L^0 - \lambda y_L + \Theta_z r)$$

in the ordinary case. Direct elimination gives

$$O = -\varepsilon_{\mathrm{obs}} J + sK - \langle R^0 C_1 + L^0 D_1\rangle.$$

On the high equations, $rC_1 + lD_1 = srl \pmod{z^{c+1}}$. Since $l(0) = r(0) = 1$, define $\gamma = (C_1/l)_{\le c} = \sum \gamma_k z^k$ and

$$T_k = \langle R^0(z^k l)_{\le c} - L^0(z^k r)_{\le c} \rangle.$$

Then the exact whole identity is

$$O = -\varepsilon_{\mathrm{obs}} J + \gamma_0(K - f_E) + (s - \gamma_0)(K - f_F) - \sum_{k=1}^{c} \gamma_k T_k. \tag{6.9}$$

There is no reason to discard the last sum from the high equations alone. We will consume this same scalar by another exact representation, rather than require the T_k to be independent or $\gamma_{k>0}$ to vanish.

6.4. **Coherent cap limits.**

Lemma 6.1. *For the actual source reference indices σ, τ, the entire truncated cap responses have*

$$T_R \longrightarrow e_R \in \{0,1\}, \quad x_R^0 \longrightarrow (c - \sigma) l_\infty - \Theta_z l_\infty,$$
$$T_L \longrightarrow e_L \in \{0,1\}, \quad y_L^0 \longrightarrow \tau r_\infty - \Theta_z r_\infty$$

as $\nu \to \infty$. In particular an ordinary high zero satisfies

$$\lambda(1 - e_R - e_L) = a + \sigma - \tau > 0. \tag{6.10}$$

A Hall and an ordinary high zero cannot coexist on this same actual deck.

Proof. Delete paired units from the upper cap. Its remaining rectangular lists J_0, K_0 are disjoint with $|J_0| = |K_0| + 1$, $c \in J_0$. Its coframe limit ℓ is normalized by $\ell_c = 1$ and satisfies $\ell M(J_0, K_0) = 0$. Eliminating all other units leaves $\nu I + E_Z$, with E_Z independent of ν. Thus their coefficients are $O(\nu^{-1})$ and the normalized limit is finite and nonzero by genuine cap weak cofactors.

There are two reference-core possibilities. First, $I_0 = J_0 \setminus \{\sigma\}$, $C_0 = K_0$, with σ a vacancy. The residual on core columns is exactly zero; its only possible order-ν vacancy entry is $\nu \ell_\sigma$ at σ. If this vacancy is absent, the projected row tends to zero. If present, the ν^{-1} leading inverse and core back-substitution reconstruct ℓ, giving $x_R \to l_\infty$.

Second, $I_0 = J_0$, $C_0 = K_0 \cup \{\sigma\}$, with σ outside both rectangular lists. If σ was a shared cap unit its own full residual is zero, and the projection tends to zero. If it was absent from the cap columns, write $\ell A_0 = f e_\sigma^t$, $f \neq 0$; the projected row tends to ℓ. This proves the Boolean limit without replacing a zero own-unit residual by absent forcing.

In the weighted projection an equal-slot coefficient is $i - j = 0$, so no order-ν vacancy forcing survives. In the first case the limiting row has entries $(i - \sigma)\ell_i$; in the second,

$$u_L^0(C_0) A_0^{-1} \longrightarrow \ell X_{J_0} - f e_\sigma^t Y_{C_0} A_0^{-1} = \ell(X_{J_0} - \sigma I).$$

Both give the upper weighted formula. Transpose and reverse the finite index interval for the lower formula. Empty cores and endpoint indices are included. Dividing the ordinary equation in (6.6) by rl and taking these limits cancels both logarithmic derivatives and gives (6.10). A Hall zero would instead require $e_R + e_L = 1$. □

6.5. **The sum horizon and clock rigidity.**

Lemma 6.2. *The block calculation* (6.5)–(6.6) *applies at $p+q$. Its complete low cap and return data agree with those at q, while its high variables are a different actual band.*

Proof. Put $a' = N - p - q - c$. By (5.1),

$$a' \ge |p - q| \ge g > c.$$

The pins give the required singleton cuts, both correction tails begin at a', their product is beyond $a' + c$, and the base labels are below a'. These are exactly the block proof's requirements. No period-$(p+q)$ chart or stronger coface bound $N - p - q \ge 2g$ is used.

The two left bands are $[q, q+c]$ and $[p+q, p+q+c]$; the right bands are $[a'+p, a'+p+c]$ and $[a', a'+c]$. All positions lie in the original open periodic interval. Original p-periodicity equates

the full low frames. The end caps are unchanged. Each horizon separately has one shared high vector, but the two high vectors are not identified. □

The actual clocks are fixed integers. Direct calculation on every generator, including the equality case that creates ν, gives

$$[\mathscr{D}_p(\alpha), \mathscr{D}_q(\beta)] = (p-q)V_{p+q} + (q\beta - p\alpha)\delta_{p+q},$$
$$[\delta_p, \mathscr{D}_q(\beta)] = p\delta_{p+q}.$$

For h_j with $j = p+q+k$, the first coefficient is

$$(k+p-\beta)(k-\alpha) - (k+q-\alpha)(k-\beta) = (p-q)k + q\beta - p\alpha.$$

Below the sum both compositions vanish. These annihilator consequences use the two original whole identities on one f_S, not merely their high coefficients.

Proposition 6.3 (Type and clock compatibility)**.** *Mixed/ordinary comparisons do not occur in* (4.3)*. In an ordinary/ordinary comparison the clocks have a common sign ε and a common cap defect $\Delta \in [0, 2c]$:*

$$\alpha = \varepsilon(N-p-\Delta), \qquad \beta = \varepsilon(N-q-\Delta), \qquad D(z) = \varepsilon r(z) l(z).$$

After lawful whole positive-clock normalization,

$$\lambda = a + \sigma - \tau, \qquad T_R + T_L = 0. \tag{6.11}$$

For mixed/mixed charts the genuine relation is $T_R + T_L = 1$.

Proof. A mixed source and ordinary mate have $\delta_p f = 0$ and $\mathscr{D}_q(\beta) f = 0$. Their commutator gives $\delta_{p+q} f = 0$. By Lemma 6.2 the same q-deck has both a Hall and an ordinary high zero, contrary to Lemma 6.1. Interchange parents for the reverse ordered case.

For ordinary charts, use σ for the upper active index and τ for the lower active index in both reference types. In a physical source case-(a) frame write $\alpha = \varepsilon d_p$ with $d_p = N-p-c+\sigma-\tau > 0$. The weighted cap limit gives $\beta = \eta d_q$, $d_q = N-q-c+\sigma-\tau > 0$, and $D/(rl) \to \eta$. The whole commutator gives a sum-horizon clock

$$\gamma = \frac{p\alpha - q\beta}{p-q}.$$

Subtract its high equation from the q equation, whose low data are equal by Lemma 6.2. Since $a' = a - p$,

$$0 = -p\, rl - (\gamma - \beta)D, \qquad (\alpha - \beta)D = (q-p)rl.$$

The clocks cannot be equal. At coherent unit infinity the second identity gives $\alpha = \eta d_p$, hence $\eta = \varepsilon$. The full identity then gives $D = \varepsilon rl$. Take $\Delta = c - \sigma + \tau$.

For a negative clock, antitranspose the entire original comparison. On full flags this sends (X, Y) to $([0,N] \setminus Y, [0,N] \setminus X)$, and on a return band (I, C) to $(K \setminus C, K \setminus I)$ for $K = \{0, \dots, c\}$. Consequently a genuine completion reference becomes a deletion reference at an actual transformed cap index. Rebuild all frames and rehomogenize. The physical involution reverses the clock sign, proving (6.11) with actual deletion references (5.5). The mixed relation is its original Hall high equation. No whole Hall zero has been added to the ordinary case. □

7. Endpoint poles and outward boundary defects

The complementary response will be used only to remove the defects that could spoil a sign argument. The complete scalar remains to be consumed afterward. Throughout this section every true unit is the same ν.

For the comparison statements here and in Section 8, retain (4.3), the actual cap-preserving splices, the full cap and reference lists of Section 5, and the common weak core and coupled returns of (5.6). In the ordinary case use the whole positive-clock normalization of Proposition 6.3. The return sets remain observed bands of the same periodic source, not independently chosen masks. The input to Theorem 7.4 is the complementary response: (6.11) in the ordinary case and $T_R + T_L = 1$ in the mixed case. Theorem 8.3 additionally uses the original complete ordinary or Hall zero in (6.3); no independent scalar zero is assigned to either coface.

7.1. **Weak Jacobi–Trudi signs.**

Lemma 7.1 (Simultaneous negative-lex sign)**.** *For this sign specialization fix a positive real value of $h_0 = \nu$, including the physical value $\nu = 1$. Let $X_1 > \cdots > X_m$, $Y_1 > \cdots > Y_m$ satisfy $X_i \geq Y_i$. The weak determinant is nonzero. In lexicographic order $h_M \gg h_{M-1} \gg \cdots \gg h_1$, its leading monomial, evaluated with all positive generators negative, has sign*

$$(-1)^{\eta(X,Y)}, \qquad \eta(X,Y) = \#\{(i,j) : i \geq j,\ X_i > Y_j\}.$$

For every finite collection of such determinants, one specialization realizes all these signs simultaneously, with all denominators nonzero.

Proof. For nonvanishing, the diagonal matching is supported. Its squared-index cost $\sum_i (X_i - Y_i)^2$ is uniquely smallest among permutations by strict rearrangement. A determinant monomial records its positive indices and, at fixed matrix size, the number of zeros. No other supported permutation therefore has that diagonal monomial.

Induct on m for the sign. If $X_k = Y_k$, the actual unit pivot and its surrounding triangular zeros split the determinant into the weak upper-left and lower-right blocks. The exponent η adds. If $X_{k+1} < Y_k$, a negative-zero lower-left block gives the same factorization. Otherwise the positive block is connected. Its largest label $X_1 - Y_m$ occurs only at the top-right entry and has a nonzero feasible proper cofactor. The lex-leading term uses that entry. Its sorting sign is $(-1)^{m-1}$ and the negative generator adds one minus sign, giving $(-1)^m$. The full exponent is m plus the proper-cofactor exponent, completing the induction. Empty determinants have sign one.

For simultaneity, let B exceed twice the largest matrix size plus two and set $h_j = -T^{B^j}$. The exponent of T realizes lex order for all their bounded-degree monomials. For one sufficiently large positive T, every leading term dominates and all asserted signs hold. Apply this to the finite union of all numerators and denominators required in the argument. □

7.2. **Actual masks and cap-height placement.** Let $A_0 = \mathrm{JT}(I_0, C_0)$ be the common disjoint weak return core in (5.6). It may be empty or disconnected. Its height is

$$h_D(u) = \#\{i \in I_0 : i > u\} - \#\{j \in C_0 : j > u\} \geq 0.$$

The actual reference equations imply

$$\mathbf{1}_{I_R^*} - \mathbf{1}_{C_R^*} = \mathbf{1}_{U_{\mathrm{cap}}} - \mathbf{1}_{C_+} - \mathbf{1}_\sigma, \qquad \mathbf{1}_{C_L^*} - \mathbf{1}_{I_L^*} = \mathbf{1}_{V_{\mathrm{cap}}} - \mathbf{1}_{I_-} - \mathbf{1}_\tau. \tag{7.1}$$

Thus σ is either a vacancy or a core column, and τ a vacancy or a core row. Away from the active index, the two cap memberships at a vacancy agree. Denote their paired sets by

$$S_R = (U_{\mathrm{cap}} \cap C_+) \cap \Omega \setminus \{\sigma\}, \qquad S_L = (I_- \cap V_{\mathrm{cap}}) \cap \Omega \setminus \{\tau\}.$$

The Boolean limits have the following exact meaning. If σ is a vacancy, the cap contains its row but not its column, and $e_R = \mathbf{1}_{\sigma \in Z_R}$. If $\sigma \in C_0$, the cap contains both row and column when $e_R = 0$, and neither when $e_R = 1$. The lower alternatives are dual. These exhaust the binary possibilities in (7.1).

Define

$$A_R = \begin{cases} (Z_R \setminus S_R) \setminus \{\sigma\}, & e_R = 0, \\ S_R \setminus Z_R, & e_R = 1, \end{cases}$$

$$A_L = \begin{cases} (Z_L \setminus S_L) \setminus \{\tau\}, & e_L = 0, \\ S_L \setminus Z_L, & e_L = 1. \end{cases}$$

They are added units when $e = 0$ and missing cap units when $e = 1$, never the active anchor itself. The actual cap heights are

$$H_{\mathrm{upper}}(u) = 1 + h_D(u) - \mathbf{1}_{u \geq \sigma}, \qquad H_{\mathrm{lower}}(u) = 1 + h_D(u) - \mathbf{1}_{u < \tau}. \tag{7.2}$$

For the upper cap the distinguished bottom column is at -1; its top is c. The lower cap runs from 0 to the top at $c + 1$. Hence connectedness gives

$$h_D(u) \geq 1 \quad (\sigma \leq u < c), \qquad h_D(u) \geq 1 \quad (0 \leq u < \tau). \tag{7.3}$$

If $\sigma < c$, the highest positive core component has support $[b, c)$ with $b \le \sigma$, and every vacancy above σ lies in this one component. If $\tau > 0$, the lowest component has support $[0, d)$ with $\tau \le d$. At $\sigma = c$ or $\tau = 0$ the corresponding outward interval is empty. The two components may coincide or have equal functions; no connected full-span return core is assumed.

7.3. **The first complete projection defect.** Let ℓ^∞, r^∞ be the cap limits. For an absent nonactive vacancy define the limiting off-diagonal forcing

$$G_u = \sum_i \ell_i^\infty h_{i-u}, \qquad F_u = \sum_j h_{u-j} r_j^\infty.$$

These are not the full residuals at an own inserted unit. Such a unit satisfies

$$0 = \nu \ell_u + \sum_{i \ne u} \ell_i h_{i-u}, \qquad \ell_u = -\nu^{-1} G_u + O(\nu^{-2}), \tag{7.4}$$

and dually $r_u = -\nu^{-1} F_u + O(\nu^{-2})$. This derives its coefficient independently of its zero full residual.

Set

$$\phi_u(z) = z^{c-u} - M(u, C_0) A_0^{-1} (z^{c-i})_{i \in I_0},$$
$$\psi_u(z) = z^u - (z^j)_{j \in C_0} A_0^{-1} M(I_0, u).$$

Here $M(R, C) = (h_{r-c})$ on the indicated actual lists. Empty matrices have their empty action.

Lemma 7.2. *Coefficientwise through degree c,*

$$T_R = e_R + \nu^{-1} \sum_{u \in A_R} \frac{G_u \phi_u}{l_\infty} + O(\nu^{-2}), \qquad T_L = e_L + \nu^{-1} \sum_{u \in A_L} \frac{F_u \psi_u}{r_\infty} + O(\nu^{-2}). \tag{7.5}$$

Proof. For $e_R = 0$, all core columns are actual cap columns, so their full residual is exactly zero. Eliminate the whole disjoint core in the actual return matrix. Its vacancy complement is

$$\nu I_Z + E[Z, Z], \qquad E_{uv} = h^+_{u-v} - M(u, C_0) A_0^{-1} M(I_0, v), \tag{7.6}$$

where $h_j^+ = h_j$ for $j > 0$ and zero otherwise. The inverse begins with $\nu^{-1} I$. Cap units have zero full residual; added nonactive vacancies have leading forcing G_u. Core back-substitution gives ϕ_u, proving the upper addition formula.

For $e_R = 1$, the active row is included in the return, or absent from the cap as well. The only missing cap rows are $S_R \setminus Z_R$. Exact full row representation gives

$$x_R - l = - \sum_{u \in S_R \setminus Z_R} \ell_u \phi_u^{B_R},$$

where the superscript means the complete actual return inverse. A missing vacancy has no equal core/cross index, so $\phi_u^{B_R} \to \phi_u$. Substitute (7.4); the two minus signs give the same plus sign in (7.5). The lower proof is the actual column dual. All feedback in (7.6) is retained before taking this coefficient. □

7.4. **Local connected components and their endpoint products.** For the upper extreme component with absolute support $[b, c)$, the local indices corresponding to the band indices z, σ are $z - b, \sigma - b$, and its span is $s = c - b$. Translating every row and column label by $-b$ preserves every difference and hence every matrix entry. The lower extreme component already has lower endpoint zero. The formulas in this subsection use these local indices; the endpoint filter below returns to the original band coordinates.

Normalize an actual connected disjoint component to full flags $(s, I)/(C, 0)$, and write $D = \mathrm{JT}((s, I), (C, 0))$ for its whole matrix. Put

$$\begin{aligned} &K = \mathrm{JT}(I, C), \quad b = h_I, \quad a_0 = h_{s-C}, \\ &F_D = h_s - a_0 K^{-1} b = h_s - \omega_D, \\ &f_z = h_z - M(z, C) K^{-1} b, \qquad g_z = h_{s-z} - a_0 K^{-1} M(I, z), \quad P_z = f_z g_z. \end{aligned} \tag{7.7}$$

All f_z, g_z, ω_D use indices below s. Solving the proper columns first and then the bottom column gives the exact top-row projection

$$M(z, \text{full columns})D^{-1}e_{\text{top}} = f_z/F_D.$$

This is a complete inverse identity, not replacement of the frame by its determinant.

Lemma 7.3. *For the upper cap restricted to this component,*

$$G_z = g_z - F_D U_z, \qquad U_z \in \mathbb{Q}(h_1, \dots, h_{s-1}). \tag{7.8}$$

Dually $F_z = f_z - F_D V_z$ with V_z in the same lower field. At one simultaneous negative-lex specialization, every P_z at an actual interior vacancy is strictly positive, including products from different connected components.

Proof. Define $k(u,z) = h_{u-z} - M(u,C)K^{-1}M(I,z)$. Every upper anchor case has an explicit correction:

$$U_z = \begin{cases} 0, & \sigma = 0 \text{ is the bottom core column,} \\ k(\sigma,z)/f_\sigma, & \sigma \text{ is a vacancy,} \\ \dfrac{e_\sigma^t K^{-1} M(I,z)}{e_\sigma^t K^{-1} b}, & \sigma \text{ is a proper core column.} \end{cases}$$

Start with the endpoint-left vector $(1, -a_0K^{-1})$. At a fresh row σ its correction coefficient is $-F_D/f_\sigma$, giving the middle formula. At a shared core-column unit its own coefficient is $O(\nu^{-1})$, and the remaining limiting vector is corrected by $-F_D e_\sigma^t K^{-1}/(e_\sigma^t K^{-1} b)$. Removing that column gives the same limiting equations. The bottom-column case is the endpoint vector. This proves (7.8). The displayed denominators are nonzero actual weak Cramer coefficients and omit h_s.

For the lower side, $V_z = 0$ at the top core row, $V_z = k(z,\tau)/g_\tau$ at a fresh vacancy, and

$$V_z = \frac{M(z,C)K^{-1}e_\tau}{a_0 K^{-1} e_\tau}$$

at a proper core-row anchor. The same complete solve proves these formulas and their nonzero denominators.

The numerators of f_z, g_z are, with their actual Cramer sorting signs, the weak boards

$$(I \cup \{z\})/(C \cup \{0\}), \qquad (I \cup \{s\})/(C \cup \{z\}).$$

Their heights are the component height lowered by one on $[z,s)$ and $[0,z)$, respectively. They are feasible and nonzero. To compute the product sign, put $n_I(x) = \#\{i \in I : i > x\}$ and $n_C(x) = \#\{j \in C : j < x\}$. Counting positive entries and the two insertion permutations in Lemma 7.1 gives the normalized product sign

$$(-1)^{n_I(0)+n_I(s)+n_C(0)+n_C(s)} = (-1)^{|I|+|C|} = +1.$$

The denominator is $\det K^2$. This parity is positive across components, not just a common unspecified sign within one contact pair. Apply the simultaneous specialization to the finite union of all these weak numerators, denominators, cap normalizers, and shifted source matrices used below. □

7.5. **The endpoint filter, including equal poles.** Let $t_R = [z^0]T_R$ and $t_L = [z^0]T_L$. If the highest component has support $[b,c)$ and span $s_R = c - b$, then $A_0^{-1}e_c$ is supported on that component. Rows below b are negative against its columns. The upper cap equations on those high columns are closed. In band coordinates, $P_u^{D_R}$ denotes the local product P_{u-b} computed after subtracting b from the upper component's row and column labels; $P_u^{D_L}$ uses the unchanged lower coordinate u. By Lemmas 7.2 and 7.3,

$$\begin{aligned} [\nu^{-1}]_\infty t_R &= -\frac{S_R^{\text{end}}}{F_{D_R}} + R_R, \quad S_R^{\text{end}} = \sum_{u \in A_R \cap (b,c)} P_u^{D_R}, \\ [\nu^{-1}]_\infty t_L &= -\frac{S_L^{\text{end}}}{F_{D_L}} + R_L, \quad S_L^{\text{end}} = \sum_{u \in A_L \cap (0,d)} P_u^{D_L}. \end{aligned} \tag{7.9}$$

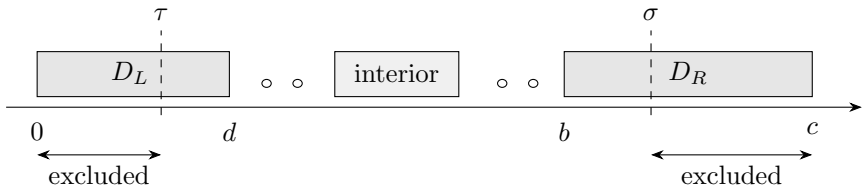


FIGURE 3. The cap heights place outward discrepancies inside the two extreme positive components. Unit buffers and intermediate feedback remain. When the two end factors agree, their specific endpoint residues add with the same sign.

With superscripts specifying the normalized components, the regular terms in the same local coordinates are

$$R_R = \sum_{u\in A_R\cap(b,c)} f^{D_R}_{u-b}U^{D_R}_{u-b}, \qquad R_L = \sum_{u\in A_L\cap(0,d)} g^{D_L}_{u}V^{D_L}_{u}.$$

Indeed the full endpoint summands are

$$-\frac{G_z f_z}{F_D} = -\frac{f_z g_z}{F_D} + U_z f_z, \qquad -\frac{F_z g_z}{F_D} = -\frac{f_z g_z}{F_D} + V_z g_z.$$

The upper expression uses $z = u - b$ and the component's top-row normalization; the lower one uses $z = u$ and its bottom-column normalization. Thus each regular sum has the same indexing set as its pole numerator. Each regular term uses only labels strictly below the span of its own end component. The two formulas include all intermediate weak components and unit buffers; they disappear from these particular coefficients only by the proved triangular support. If an outward endpoint interval is empty, that end contribution is zero.

Theorem 7.4 (End-pole filter)**.** *Under the ordinary or mixed complementary response,*

$$A_R \cap (\sigma, c] = \varnothing, \qquad A_L \cap [0, \tau) = \varnothing. \tag{7.10}$$

In fact each discrepancy set misses the interior of its corresponding active extreme positive component just defined: the upper component when $\sigma < c$, and the lower component when $\tau > 0$.

Proof. By Proposition 6.3, $t_R + t_L$ is the constant zero or one, so its ν^{-1} coefficient is zero. Choose the larger end-component span s. Regard h_s as independent over $\mathbb{Q}(h_1, \dots, h_{s-1})$. Terms from a smaller-span end and the regular terms at span s are independent of h_s. Each span-s denominator is the monic linear factor $h_s - \omega_D$.

If the two factors differ, separate residues make each corresponding S^{end} zero. If they coincide, the residue is $-(S^{\mathrm{end}}_R + S^{\mathrm{end}}_L)$. By Lemma 7.3 every nonempty summand has a simultaneous strictly positive specialization, so this sum cannot vanish unless both indexing sets are empty. Once a set is empty, its entire contribution in (7.9), including the regular part, disappears because it was a sum over that set. Treat the smaller end in the remaining exact equation. This proves the stronger end-component exclusion; (7.3) implies (7.10).

The sign test on a residue numerator uses the lower-index restriction of the one chosen specialization. It does not evaluate the original scalar at a pole. Equal factors do not identify inverse frames or erase their occurrence multiplicities. □

Figure 3 shows the two extreme components and the excluded outward intervals.

Example 7.5. For $I_0 = \{3\}$, $C_0 = \{0\}$ and vacancies $1, 2$, the return frames are

$$[h_3], \quad \begin{pmatrix} h_2 & h_3 \\ \nu & h_1 \end{pmatrix}, \quad \begin{pmatrix} h_1 & h_3 \\ \nu & h_2 \end{pmatrix}, \quad \begin{pmatrix} h_1 & h_2 & h_3 \\ \nu & h_1 & h_2 \\ 0 & \nu & h_1 \end{pmatrix}.$$

The middle determinants are both $h_1h_2 - \nu h_3$, but their labeled inverses differ. The component has $F_D = h_3$ and both endpoint products equal h_1h_2. Its endpoint pole is therefore

$$-\big(|A_R| + |A_L|\big)h_1h_2/h_3.$$

It cannot cancel. The full observer, not this coefficient alone, must still determine the private records. This is a symbolic interface illustration, not a supplied original pair with two minimum charts.

8. Whole-determinant noncancellation

Return to the physical value $\nu = 1$. Every term of (6.9), including its nonconstant low-coefficient cap-mode sum, remains present. The pole filter has not classified all rational response germs. The same original scalar has a representation that no longer requires such a classification.

For $x \in X$, let $\mathcal{R}_x$ be the determinant in the *original row order* after replacing the row label x by $x - q$. For $y \in Y$, let $\mathcal{C}_y$ be the determinant in the original column order after replacing y by $y + q$. For the original source, duplicate rows or columns and negative or out-of-horizon replacements give zero. Every remaining entry is the actual shared raw moment; no individual term is observed separately.

Lemma 8.1. *For any equal-length strict integer lists X, Y and positive integer q, let $s_S = \det \mathrm{JT}(X, Y)$ and take $\mathcal{R}_x, \mathcal{C}_y$ to be the literal replacement determinants in the original orders. Then the following Hall and weighted identities hold without connectedness or the original comparison hypotheses:*

$$\delta_q s_S = \sum_{x \in X} \mathcal{R}_x = \sum_{y \in Y} \mathcal{C}_y, \qquad V_q s_S = \sum_{x \in X} (x - q)\mathcal{R}_x - \sum_{y \in Y} y\mathcal{C}_y. \tag{8.1}$$

In the original comparison, the genuine positive ordinary clock $\lambda = r_* - \tau$, $r_* = a + \sigma$, gives the specialization

$$\mathscr{D}_q(\lambda) s_S = \sum_{x \in X} (x - q - r_*)\mathcal{R}_x + \sum_{y \in Y} (\tau - y)\mathcal{C}_y. \tag{8.2}$$

Proof. Differentiate the determinant by every entry occurrence. Replacing the whole row x gives the derivative contribution with entry h_{x-y-q}; summing the rows is the Hall derivative. The same argument by columns gives its second expansion. For the weighted derivative, $x - y - q = (x - q) - y$, which gives the two terms in (8.1). At $x - y = q$, the two weighted contributions cancel while the Hall contribution retains $h_0 = 1$. For the empty matrix, the determinant is one and both derivatives and all replacement sums are zero. In the original clock setting, split the single coefficient $\lambda = r_* - \tau$ between the two equal Hall expansions to obtain (8.2). This splitting does not assert a Hall zero in the ordinary case. □

For the original connected source, $\kappa(S)$ is fixed and nonzero. Thus the zero of (8.2) is precisely the original ordinary zero (6.3), up to its fixed nonzero normalization. The mixed zero is either full Hall sum in (8.1). Thus no cap-mode remainder has been omitted in changing representations.

8.1. Feasibility and signs in the original orders.

Lemma 8.2. *For the original connected source, every nonduplicate in-horizon shift above is a nonzero weak Jacobi–Trudi determinant. Relative to the reference sign $(-1)^{\eta(X,Y)}$ at one simultaneous specialization of Lemma 7.1, with $w_0 = x - q$ and $v_0 = y + q$,*

$$\begin{aligned} \frac{\operatorname{sign} \mathcal{R}_x}{(-1)^{\eta(X,Y)}} &= (-1)^{\#(X \cap (w_0, x)) + \#(Y \cap [w_0, x))} \\ &= (-1)^{H_S(w_0) - H_S(x-1) + \mathbf{1}_{w_0 \in Y}}, \\ \frac{\operatorname{sign} \mathcal{C}_y}{(-1)^{\eta(X,Y)}} &= (-1)^{\#(Y \cap (y, v_0)) + \#(X \cap (y, v_0])} \\ &= (-1)^{H_S(y) - H_S(v_0 - 1) + \mathbf{1}_{v_0 \in X}}. \end{aligned} \tag{8.3}$$

Proof. Lowering a row from x to $w_0 \geq 0$ subtracts one from the positive source height on $[w_0, x)$. Raising a column from y to $v_0 \leq N$ subtracts one on $[y, v_0)$. Thus the new sorted lists are weakly

feasible, even if the resulting diagram is disconnected or has diagonal units. Lemma 7.1 proves nonvanishing.

Sorting the changed row contributes $\#(X \cap (w_0, x))$ swaps. Its positive-entry count changes by $-\#(Y \cap [w_0, x))$. For a feasible r-row weak matrix, every entry strictly above the diagonal is positive, so η is the positive-entry count minus $r(r-1)/2$. This proves the first row formula; the column proof is the same. Cumulative counts give the displayed height forms modulo two. Apply Lemma 7.1 to the finite collection of all surviving shifted matrices at once. □

For $c < k < a$, k and $k+q$ lie in the original open middle. Their adjacent heights agree by $\mathcal{P}(g, c)$, using the actual pins at the first boundary. A surviving row shift $k+q \to k$ is a unit at $k+q$ and a vacancy at k; a surviving column shift is the reverse discrepancy. Both have relative sign $+1$. Their ordinary coefficients are

$$k - r_* < 0, \qquad \tau - k < 0. \tag{8.4}$$

The full cap ledger in Table 2 follows by substituting (7.2) in (8.3). The variable v in the table is a nonactive relative cap vacancy. For example, a missing nonactive upper cap unit at v has source row $x = N - c + v$ and returned position $x - q = a + v$, absent from both returned lists. Since v is a core vacancy, $h_D(v-1) = h_D(v)$. The actual heights give

$$H_S(a+v) = 1 + h_D(v), \qquad H_S(N-c+v-1) = 1 + h_D(v) - \mathbf{1}_{v>\sigma}.$$

Also $\mathbf{1}_{a+v \in Y} = 0$. The row formula in (8.3) gives relative sign $(-1)^{\mathbf{1}_{v>\sigma}}$; its ordinary coefficient is $v - \sigma$. Their product is negative on either side of σ.

Shift	Actual discrepancy	Relative sign	Ordinary coefficient
Row, $x - q = v$	added lower unit	$-$ if $v < \tau$; $+$ if $v > \tau$	$v - r_*$
Row, $x - q = a + v$	missing upper unit	$+$ if $v < \sigma$; $-$ if $v > \sigma$	$v - \sigma$
Column, $y = v$	missing lower unit	$-$ if $v < \tau$; $+$ if $v > \tau$	$\tau - v$
Column, $y = a + v$	added upper unit	$+$ if $v < \sigma$; $-$ if $v > \sigma$	$\tau - a - v$

TABLE 2. Signs in the original matrix orders, relative to the same reference sign. The only potentially wrong-sign weighted additions are exactly the outward records removed by Theorem 7.4.

The active positions follow from the binary membership alternatives in (7.1). In the upper rows below, the cap and return pairs are $(\mathbf{1}_{U_{\mathrm{cap}}}(\sigma), \mathbf{1}_{C_+}(\sigma))$ and $(\mathbf{1}_{I_R}(\sigma), \mathbf{1}_{C_R}(\sigma))$; the shifts are $\mathcal{R}_{N-c+\sigma}, \mathcal{C}_{a+\sigma}$. In the lower rows the pairs are $(\mathbf{1}_{I_-}(\tau), \mathbf{1}_{V_{\mathrm{cap}}}(\tau))$ and $(\mathbf{1}_{I_L}(\tau), \mathbf{1}_{C_L}(\tau))$; the shifts are $\mathcal{R}_{q+\tau}, \mathcal{C}_\tau$.

Anchor	Boolean value	Cap pair	Return pair	Row term	Column term
Upper vacancy	$e_R = 0$	$(1,0)$	$(0,0)$	live, weight 0	absent
Upper core column	$e_R = 0$	$(1,1)$	$(0,1)$	live, weight 0	duplicate
Upper vacancy	$e_R = 1$	$(1,0)$	$(1,1)$	duplicate	live
Upper core column	$e_R = 1$	$(0,0)$	$(0,1)$	absent	live
Lower vacancy	$e_L = 0$	$(0,1)$	$(0,0)$	absent	live, weight 0
Lower core row	$e_L = 0$	$(1,1)$	$(1,0)$	duplicate	live, weight 0
Lower vacancy	$e_L = 1$	$(0,1)$	$(1,1)$	live	duplicate
Lower core row	$e_L = 1$	$(0,0)$	$(1,0)$	live	absent

For ordinary state 00, every live active term has zero weight. For mixed state 10, the row Hall expansion has only absent or duplicate active row terms. For mixed state 01, the column Hall expansion has only absent or duplicate active column terms. The complementary live terms, of ordinary coefficient $\tau - a - \sigma$, are not discarded from the other expansion; that expansion is not used. At $\sigma = c$, the upper cap pair is $(1,0)$, with actual top row N. At $\tau = 0$, the lower pair is $(0,1)$, with actual bottom column 0. These are the vacancy cases in the ledger. A surviving horizon replacement has a genuine unit pivot, rather than being out of range.

For a coface T with normalized full beta lists X_T, Y_T, write $Z_T = X_T \cap Y_T$ for its common-beta set.

Theorem 8.3 (Whole-scalar synchronization)**.** *In the original distinct line comparison, the ordinary identity implies literal equality $S^-_{\mathrm{cap}} = S^+_{\mathrm{cap}}$ and equality of the returned frames with their positive-clock references. In the mixed case it gives one of the unit-set inclusions*

$$Z_{S^+_{\mathrm{cap}}} \subseteq Z_{S^-_{\mathrm{cap}}} \quad \text{or} \quad Z_{S^-_{\mathrm{cap}}} \subseteq Z_{S^+_{\mathrm{cap}}}. \tag{8.5}$$

These conclusions supply every private comparison $c < k < a$.

Proof. In the ordinary positive-clock frame, Proposition 6.3 gives $e_R = e_L = 0$, so the offending sets are additions. The end-pole filter excludes added upper return units above σ and added lower return units below τ. Consider the *entire* sum (8.2).

Private terms have positive relative sign and negative weights by (8.4). For missing upper cap units, the sign reverses at σ and the weight is $v - \sigma$, making their product negative on both sides. For missing lower cap units the same holds with weight $\tau - v$. The only wrong-sign additions are precisely the two excluded outward sets. Every remaining addition has positive relative sign and negative weight. Active terms have zero weight or are absent; duplicates are zero. At the one simultaneous specialization, every nonzero weighted summand therefore has the same strict sign. Its observed zero permits no such summand. All nonactive cap differences and all private differences disappear, giving the ordinary conclusions.

For mixed $(e_R, e_L) = (1, 0)$, the offending upper records are missing cap units and the offending lower records are additions. The filter removes upper missing units above σ and lower additions below τ. Use the complete *row* Hall sum. Every remaining term in the table has positive relative sign, as do all private terms. Active row terms are absent or duplicate. The zero sum has no nonzero row shift. In particular,

$$\mathbf{1}_{k+q\in X} \le \mathbf{1}_{k\in X} \qquad (c < k < a).$$

Since the private core words agree, this is the first inclusion in (8.5). The cap conclusions are exactly $U_{\mathrm{cap}} \subseteq I_R$ and $I_L \subseteq I_-$.

For $(e_R, e_L) = (0, 1)$ use the complete *column* Hall sum. The filter removes upper additions above σ and lower missing units below τ. All remaining column terms have positive relative sign and active terms are absent or duplicate. Hence

$$\mathbf{1}_{k\in Y} \le \mathbf{1}_{k+q\in Y} \qquad (c < k < a),$$

which is the reverse unit inclusion, with cap conclusions $C_R \subseteq C_+$ and $V_{\mathrm{cap}} \subseteq C_L$. No equal unit counts or prior Schur equality of the cofaces were used. □

The proof has used a restricted response coefficient to make the signs compatible, followed by the complete scalar to exclude all remaining discrepancies. Central nonconstant feedback was present throughout. Appendix F gives the alternative geometric realization of this same obstruction by cap-modified contact systems.

9. Finite synchronization and the legal gcd chart

9.1. From an inclusion to a full period.

Lemma 9.1. *Let a binary word be p-periodic on an interval $\mathcal{I}$ of at least $p + q$ consecutive integers. If $u(k + q) \le u(k)$ for every $k, k + q \in \mathcal{I}$, or all these inequalities have the reverse direction, then the word is $\gcd(p, q)$-periodic on $\mathcal{I}$.*

Proof. Choose p consecutive representatives whose q-translates still lie in $\mathcal{I}$. Reduce their comparisons using the original p-period. Addition by q modulo p is a permutation. Around each finite cycle, a chain of inequalities returns to its starting value, so every inequality is equality. Its cycles are exactly the residue classes modulo $\gcd(p, q)$. Original p-periodicity transfers these equalities to every position of the interval. □

In the original problem,

$$|\mathcal{I}| = N - 2c - 1 \geq 2p - c - 1 \geq p + q, \tag{9.1}$$

since $p - q \geq g \geq c + 1$. The conclusion of Theorem 8.3 applies to every $c < k < N - q - c$, precisely all $k, k+q \in \mathcal{I}$. Lemma 5.3 supplies its original p-period. Thus the complete common-beta membership word of S is g-periodic on $\mathcal{I}$. The core words already have this period by Proposition 5.2; restoring the synchronized paired records gives *both full original boundary words* with this period. All singleton pins are unchanged.

9.2. **A fold from one original line chart.** The geometric reconstruction uses only one supplied original chart and the synchronized full words, not a second chart presumed to be carried by S.

Theorem 9.2 (One-chart reconstruction)**.** *Let $S = L_M \circ_\omega E$ be an actual proper positive I–IV chart with line outer L_M, $M \geq 2$, period $p = Ag$, and width $w = bg + c$, where $A \geq 2$, $0 \leq b < A$, $0 < c < g$. Put $N = Lg + c$, $L = MA + b$. Suppose both full boundary words are g-periodic on $c < k < Lg$, and all diagonals $c + jg, (j+1)g - 1$ are singletons. Then an actual connected G of span $g + c$, a legal width-c mark ζ, and an actual L-cell line D_L satisfy*

$$S = D_L \circ_\zeta G \tag{9.2}$$

literally. Its score is $(c, g + c)$.

Proof. Choose generic bits A_r^0, B_r^0 from the actual internal steps $g + r$, $0 \leq r < g$. They lie strictly inside the synchronized middle because $L \geq 4$. Set

$$C_r = A_r^0 - B_r^0, \qquad \Delta = \sum_{r=0}^{g-1} C_r, \qquad H_r^0 = 1 + \sum_{i=1}^{r} C_i.$$

The singleton samples one period apart give $\Delta = 0$. In particular the four diagonals $c, g-1, N-g, N-c-1$ are actual singletons. Translate the last g-diagonal window so that its singleton at $N - g$ coincides with the first window's singleton at c. On the resulting overlap $[c, g-1]$, all internal boundary steps agree by synchronization. Their union G, depicted in Figure 4, is connected skew of span $g + c$ with singleton ends. Its area is strictly below that of S: the original end windows are disjoint, an occupied middle lies between them, and the new union identifies a nonempty overlap.

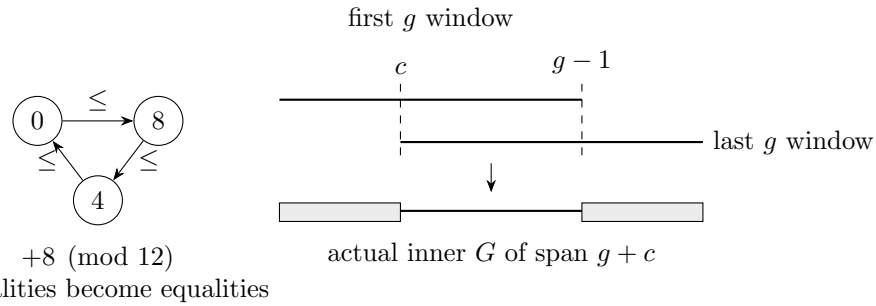


FIGURE 4. Left: one gcd cycle in the arithmetic illustration $p = 12, q = 8$. Right: the retained end windows agree on their singleton-aligned overlap and form the actual folded inner. Shaded end bands contain the explicitly constructed width-c marks; their cells are selected in the proof. Neither panel asserts an example of a contradictory original minimal pair.

Use a whole antitransposition, when necessary, to put the original chart in case (a) or (d); it will be undone on the final construction. Let $\rho = 0$ for a row source outer and $\rho = 1$ for a column source outer.

Ordinary source. The actual interior coding gives $A_S(jp+t) = B_S(jp+s) = \rho$. Write $\tau = t \bmod g$, $\sigma = s \bmod g$; then

$$A^0_\tau = B^0_\sigma = \rho. \tag{9.3}$$

The only initial and terminal deviations from the generic words yield

$$\begin{array}{l|ll} \rho = 0 & t = \tau \in [0,c] & s = bg+\sigma, \quad \sigma \in [0,c] \\ \rho = 1 & s = \sigma \in [1,c] & t = bg+\tau, \quad \tau \in [0,c-1]. \end{array}$$

These bounds follow by comparing each exceptional cap bit with its existing period-p interior occurrence. Every other cap step is generic. If $i = \mathbf{1}_{\tau>0}$, $j = \mathbf{1}_{\sigma<c}$, the four pins give

$$C_0 = i(1-\rho)+\rho, \quad -C_c = \rho + j(1-\rho), \quad \sum_{r=1}^{c-1} C_r + (i-j)(1-\rho) = 0. \tag{9.4}$$

The initial correction is i for $\rho = 0$ and one for $\rho = 1$; zero drift and $H_{g-1} = 1$ identify it with C_0, and $H_c = 1$ gives the partial sum. The terminal calculation gives the second relation.

Choose the new case (a) when $\rho = 1$, or when $\rho = 0$ and $\tau < \sigma$. Choose the new case (b) otherwise. In the latter case $\tau > 0$ and $\sigma < c$: the forbidden pair $(0,0)$ would leave the original positive mark's diagonal at bg empty (or violate $s \geq 1$ when $b = 0$), and (c,c) would leave the initial diagonal $c-1$ empty (or violate $t < w$ when $b = 0$).

The retained marked copies on G are defined cell by cell:

new case	delete in lower band r	delete in upper band $g+r$
(a)	southeast cell for $r \geq \sigma$	northwest cell for $r < \tau$
(b)	northwest cell for $r \geq \tau$	southeast cell for $r < \sigma$.

Retain every other cell. In case (a), their common internal words are the generic words with the internal A-slot τ and B-slot σ set to one; in case (b), those slots are set to zero. Equations (9.3) and (9.4) give binary steps and singleton ends.

Their heights are positive. For $\rho = 1$ the new lower mark has exactly the first c diagonal heights of the original positive mark. For $\rho = 0$ the original mark has height

$$H^0_{r \bmod g} + \mathbf{1}_{t>0,\ r\geq t} - \mathbf{1}_{r\geq s}. \tag{9.5}$$

When $\tau < \sigma$, the new height is $H^0_r + \mathbf{1}_{\tau>0, r\geq\tau} - \mathbf{1}_{r\geq\sigma}$. Before σ it is an initial-parent height; afterward it is (9.5) at $bg+r$. In the new case (b) it is H^0_r, positive from the initial parent for $r < \tau$ and from the old mark for $r \geq \sigma$. These ranges cover every marked residue. Thus the retained copies are congruent connected skew diagrams, with a translation of content g.

For new case (a), the middle consists of a southeast singleton lower tail, the private band, and a northwest singleton upper tail. Their joins are actual singleton diagonals; inherited endpoint steps give both connected one-mark complements. The actual bits $B_\sigma = 0$ and $A_{g+\tau} = 0$ give the horizontal attachments. Separation is $c < g$. Write the displacement as $(h, h+g)$. Any containing mark on the same c content diagonals sends the southwest singleton into content g, forcing $h' \leq h$. At relative content $\tau < c$ the upper tail has ended; the northwest retained cell forces $h' \geq h$. At this displacement the mark is exactly $G \cap (G-(h,h+g))$, proving maximality against every possible displacement. The hatted middle is confined to $[c+1, g-2]$; all periodic hatted-mark cells have residues $0, \ldots, c-1$. Their content distance is at least two, proving the entire bi-infinite condition IV. Whole antitransposition gives the corresponding checks for new case (b).

Define the new outer bits by

$$(A^{D_L}_j, B^{D_L}_j) = \begin{cases} (A_S(jg+\tau), B_S(jg+\sigma)), & (a), \\ (1-B_S(jg+\sigma), 1-A_S(jg+\tau)), & (b). \end{cases}$$

They form a constant equal pair by (9.3), hence an actual row or column of L cells. The complete coding reconstructs every source step and both defining windows of G, so its composition is literally S.

Mixed source. Normalize to case (d). Its fixed word gives $B_0^0 = 0$, $B_c^0 = 1$ and its original slots give $A_S(jp+t) = \rho$, $A_S(jp+s) = 1-\rho$. The two projected slots are distinct, because otherwise one generic bit would have to equal zero and one. Let z_0 be the original phase with generic bit zero and o_0 the phase with generic bit one. Synchronization confines the initial correction and terminal deficit to $0 \le z_0 \le c$ and $bg \le o_0 \le bg + c$. Put

$$\tau = z_0, \quad \sigma = o_0 - bg, \quad \alpha = \min(\tau, \sigma), \quad \beta = \max(\tau, \sigma).$$

Then $0 \le \alpha < \beta \le c$ and

$$C_0 = \mathbf{1}_{\tau>0}, \qquad -C_c = \mathbf{1}_{\sigma<c}, \qquad \sum_{r=1}^{c} C_r = -C_0.$$

A wide mixed column source is automatically excluded by $t < s$ and these bounds, but the narrow column source remains in the construction.

The same window union G has southeast displacement $\lambda = (-h, g-h)$, where $h = \sum_{r=0}^{g-1} B_r^0$. Delete the northwest cell from lower marked residues $r \ge \beta$ and upper marked residues $g + r$ with $r < \alpha$. Before deletion their heights are

$$L_r = H_r^0 + \mathbf{1}_{\tau>0, r\ge\tau}, \qquad U_r = H_r^0 + \mathbf{1}_{\tau>0} - \mathbf{1}_{\sigma<c, r\ge\sigma}.$$

After deletion both are

$$\begin{cases} H_r^0 + \mathbf{1}_{\tau>0, r\ge\tau} - \mathbf{1}_{r\ge\sigma}, & \tau < \sigma, \\ H_r^0, & \tau > \sigma. \end{cases}$$

Positivity follows from the initial parent before σ and the original mark at $bg + r$ afterward in the first case. In the second, if $b = 0$ it is the old mark; otherwise the initial range $r < \tau$ and old-mark range $r \ge \sigma$ cover every residue. Their southeast word is B^0 and their northwest word is A^0 with internal slots α, β set to one and zero. The budget gives singleton ends, so the retained copies are congruent, connected and southeast-justified, with the stated displacement.

The middle has two northwest singleton tails and the private band. Its inherited steps and singleton joins prove both complements connected. The bits $A_\beta = 1$ and $A_{g+\alpha} = 0$ give the vertical/horizontal attachments, including $\alpha = 0$ through the actual private endpoint. For maximality, the image of the southwest singleton gives $\lambda_1' \le \lambda_1$; the inverse image of the northeast singleton in the lower content-$c-1$ interval gives the reverse inequality. At that displacement the deleted-tail ranges are disjoint, so the mark is the full intersection. The same content-distance argument proves IV, including neighbors outside the retained mark. Finally define

$$A_j^{D_L} = A_S(jg + \alpha), \qquad B_j^{D_L} = 1 - A_S(jg + \beta).$$

The generic slot bits sum to one, so these are a constant equal pair. The full mixed coding, with every supplemental copy, reproduces the two original words and both windows. This proves (9.2); whole antitransposition gives case (c).

The formulas include $c = 1$ directly. Since $g \ge 2$, G is proper, and $L \ge 4$ makes the outer proper. Undo all full transports on the constructed chart. No minimum on G or on a coface has been used. □

9.3. **Completion of period rigidity.**

Proof of Theorem 1.1. Suppose $p > q$. Theorem 4.8 excludes every original input except (4.3). In that case Proposition 5.2 and Lemma 5.3 provide the whole-parent structure, actual caps and original p-period. Proposition 6.3 excludes opposite types and aligns the ordinary clocks. Theorems 7.4 and 8.3 give all private q-shift equalities or a single consistent direction of inclusion. The budget (9.1) and Lemma 9.1 give both full original g-periodic boundary words, with the original singleton pins retained.

Theorem 9.2 now constructs the actual same-source chart before its score is compared with the prescribed original score:

$$(c, g + c) < (bg + c, (A + b)g + c).$$

If $b > 0$ the width improves; if $b = 0$ the inner span improves because $g < p$. This contradicts the source's own minimum. Hence $p = q$. Every lower-class call occurred at a stated strict area, except for the same-area zero-free class theorem proved independently from strict erosion. □

Corollary 9.3. *Let $F = D \circ_\omega E$ and $H = C \circ_\eta U$ be actual proper legal I–IV charts with $s_F = s_H$ and common period p. If D is a ribbon, their overlap widths agree, $N_D = N_C$, and C is a ribbon.*

Proof. Write w, u for the overlap widths and N for the common span. The span law gives $N = pN_D + w = pN_C + u$. Since $0 \le w, u < p$, it follows that $w = u$ and $N_D = N_C$. The complete proper corners agree. By (2.7), the ribbon-source corner contains only factors of span less than p. A positive component T of C^- would contribute the connected prime $s_{T\circ_\eta U}$ of span $pN_T + u \ge p$ to the equal complete corner, which is impossible. Thus $C^- = \varnothing$. □

Concluding remarks

The period comparison separates algebraic synchronization from geometric reconstruction. Core reduction and diagonal heights recover the expected gcd structure but leave paired boundary choices undetermined. Extreme component residues exclude the choices that could defeat noncancellation; the entire differential identity then forces the remaining comparisons. For mixed charts a directed inclusion suffices, because finite period cycles turn it into equality. The synchronized words construct the legal finer chart on the original source, and that chart makes source minimality decisive.

This proves the ribbon-outer period step under the strict lower-area joint package. Equal-period marked-factor and prescribed-fiber identification is a different assertion. Root existence and the positive-interior induction steps are also separate. The method here is the passage from complete determinant identities to literal boundary constraints without an exhaustive classification of their intermediate rational response representations.

Appendix A. Aligned reconstruction before minimality

This appendix proves the reconstruction assertion used in Proposition 4.1. The two original parents, the source ribbon, the prescribed fine minimum and $\mathsf{J}_{<n}$ remain fixed. The displayed coarse chart is not required to be minimal. Write

$$p = kq, \qquad w = bq + u, \qquad 0 \le b < k, \quad 0 \le u < q, \qquad R = k + b, \quad N_E = Rq + u.$$

The letter P below denotes a reduced coarse polynomial, not an assumed skew Schur function.

A.1. **The actual composition operators.** For a fixed marked inner of period d, let A, B, C be its mark, hatted-mark and hatted-middle functions. All these factors use generators below d. Use the row-generator map of (2.5), with the ordinary involution carried whenever the chart is changed. Its fresh coefficient at h_{jd+v}, where v is the width, is nonzero, primitive, and uses only generators below d. This follows by expanding the actual row-probe determinant at its unique largest entry and applying the full corner transport formula.

Use composition covariance and terminal readout from [10], with the actual row map (2.5) and full normalizer (2.6). In the ordinary case (a), write $\xi = \delta_d s_E$ and $\theta = V_d s_E/\xi = s - t$. For case (b) the read map is $\Phi_\omega \circ \iota$ and the read outer is $\iota(s_D)$; the actual $\xi = \delta_d s_E$ and $\theta = V_d s_E/\xi$ carry the physical signs. In either ordinary reading $|\theta| < d$, and

$$\delta_{rd}\Phi = \xi^r \Phi \delta_r, \qquad V_{rd}\Phi = \xi^r \Phi(dV_r + \theta\delta_r).$$

The genuine empty edges have $\theta = 0$. For a mixed chart rotate the whole chart to case (d), put $\rho = s - t \in (0, d)$ and $\chi = (-1)^{-v_1} ABC$, and use

$$\delta_{rd}\Phi = 0, \qquad V_{rd}\Phi = \chi^r \Phi(dV_r + \rho\delta_r).$$

Here $\Phi(1) = 1$ is the algebraic unit. An exhausted row probe instead has geometric remainder $b_0 = A$; the cited identities retain that local factor. Whole antitransposition obeys $\delta_j \iota = (-1)^{j-1} \iota \delta_j$ and $V_j \iota = (-1)^j \iota V_j$, and transports the map, outer involution, and entire normalizer together.

For the terminal application take the original ribbon source $F = D \circ_\omega E$ of period p, with $M = |D| \geq 2$. Terminal readout [10, Terminal readout] gives

$$\frac{\delta_p^{M-1} s_F}{\delta_p^M s_F} = \frac{s_E}{\xi_s} \qquad \text{in the ordinary case,}$$

$$\frac{V_p^{M-1} s_F}{V_p^M s_F} = \frac{s_E}{\rho_s \chi_s} \qquad \text{in the positive-phase mixed case.}$$

Here $\xi_s = \delta_p s_E$ in the ordinary source; in the rotated mixed source, $\rho_s = s_s - t_s$ and $\chi_s = (-1)^{-(v_s)_1} A_s B_s C_s$ use that source's own phases, displacement, and three local factors. Both denominators are nonzero, and the full source normalizers cancel. The elementary-generator identities used for the line observers are

$$\begin{aligned} (dV_r + \rho\delta_r)h_j &= (d(j-r)+\rho)h_{j-r}, \\ (dV_r + \rho\delta_r)e_j &= (-1)^{r-1}(\rho - d(j-r))e_{j-r} \end{aligned} \qquad (j \geq r).$$

For $j < r$ both outputs vanish. At exhaustion the scalar output is not assigned to a proper one-cell or empty chart.

Proposition A.1. *Let $F = D \circ_\omega E$ and $H = C \circ_\eta U$ be actual proper legal I–IV charts with equal whole Schur functions, of periods p, q and overlap widths w, u, respectively. Suppose D is a ribbon with $M = |D| = N_D \geq 2$, and*

$$p = kq, \qquad k \geq 2, \qquad w = bq + u, \qquad 0 \leq b < k, \quad 0 \leq u < q.$$

Put $R = k + b$, so $N_E = Rq + u$. Use the complete fine map Φ_η of (2.5), carrying the outer involution and the full local normalizer whenever a physical frame is changed. Then either the two original operator types are incompatible, or one obtains a primitive irreducible weighted-homogeneous polynomial $P \in \mathbb{Z}[h_1, h_2, \ldots]$, linear in its largest generator h_R, and a nonzero primitive low polynomial L such that

$$\Phi_\eta(P) = L s_E. \tag{A.1}$$

Here $\mathrm{sp}(P) = R$, and L uses only the original generators below q. This construction does not assume a chart on E through U. Neither source-minimality nor a lower-area classification hypothesis is required for this extraction.

Proof. Apply composition covariance to the two original source words in terminal readout [10, Composition covariance and terminal readout]. If an ordinary source is paired with a mixed fine chart, its nonzero terminal Hall word is incompatible with the fine Hall zero. Otherwise let T_1, T_0 be the two complete coarse outputs. The fine normalizer cancels in their ratio, giving

$$\Phi_\eta(T_1/T_0) = R_{<q}\, s_E / X_s, \tag{A.2}$$

where $X_s = \xi_s$ or $\rho_s\chi_s$, and $R_{<q}$ is a nonzero fine low factor. Every connected factor of X_s has span below p, but X_s is not assumed to have span below q. Since $N_E = p + w \geq p$, it is coprime to the actual connected polynomial s_E.

Reduce $T_1/T_0 = P/G$ before interpreting the numerator, with P, G coprime and weighted homogeneous, and make P primitive integral. Put $x_j = h_{jq+u}$. At a finite cutoff write the actual fine generators as $\Phi_\eta(h_j) = c_j x_j + d_j$, where c_j is a nonzero primitive low polynomial and d_j uses earlier raw labels. Adjoin the fraction field K of all nonpivot raw variables. Triangular substitution is an isomorphism over K, so $\Phi(P)$ and $\Phi(G)$ are coprime there. Equation (A.2) and the connected prime s_E, which depends on the fresh pivot x_R, give $\Phi(P) = c_K s_E$ for $c_K \in K^\times$.

Write $c_K = a/b$ in coprime nonpivot polynomials. Polynomiality would make b divide s_E. Since s_E is irreducible and depends on x_R, b is constant. Thus c_K is a nonpivot polynomial. Order the selected pivots from largest to smallest. The leading selected-monomial coefficient of $\Phi(P)$ is a nonzero rational coefficient of P times a product of the c_j, hence is low. Every purely nonpivot factor consequently divides a low polynomial and is low up to a rational unit. This proves (A.1).

The same primitive pivot argument modulo each integer prime shows that $\Phi(P)$ is primitive. Gauss's lemma makes L primitive after absorbing the original rational scalar. The unique largest pivot of s_E then shows that P has span R and is linear in h_R. Any factorization of P would, under the triangular isomorphism, factor the one connected high prime; a nonconstant coarse factor cannot become a coefficient-field unit. Hence P is irreducible. Only the original words of lengths $M-1$ and M were used. □

Lemma A.2. *The polynomial $\partial_{h_R}P$ is, with its canonical sign, a primitive positive product Γ of connected component functions from the actual fine outer interior. Multiplicities are retained. A cell component contributes h_1; a noncell component provides a proper fine reference.*

Proof. Differentiate (A.1) at its unique largest pivot:

$$c_R\Phi_\eta(\partial_{h_R}P) = L\,\partial_{h_{N_E}}s_E.$$

The complete source proper corner divides the proper corner of F: for a ribbon outer the latter is $\Theta_s A_s^{r(D)-1}B_s^{c(D)-1}C_s^M$, whereas $\kappa(E) = \Theta_s C_s$. Every high connected factor of that same corner, read on H, is the function of an actual component Z of C^- composed through the original η/U; every other fine factor is below q. Triangular prime transport and coefficient-field independence identify all coarse factors of $\partial_{h_R}P$ with these actual s_Z, including their exponents. Primitive pivot coefficients retain integer content. The actual positive proper coefficient fixes the canonical sign. A noncell Z has at least two cells and makes $Z\circ_\eta U$ proper; a cell makes only the function s_U and is not assigned such a chart. □

A.2. **Distinct phases retain the entire coarse polynomial.**

Lemma A.3. *Let the fine positive mark have distinct phases. Its actual marked core map has a fixed $\gamma\in\{\mathrm{id},\mathrm{AT}\}$ and a legal mark η_0 on $U_0=\mathrm{Core}(U)$. For every nonzero weighted-homogeneous coarse polynomial f of weight d,*

$$\mathrm{in}_{\mathrm{ord}}\,\Phi_\eta(f) = \epsilon^d\Phi_{\eta_0}(\iota_\gamma f), \qquad \mathrm{ord}_{\mathrm{ord}}\,\Phi_\eta(f) = d\,r(U_0), \tag{A.3}$$

where ϵ is the actual core phase and ι_γ is the identity or the carried involution. In particular the input on the right is the whole f, not its initial ordinary-degree part.

Proof. At distinct phases the transformed mark itself is zero-free: each tail operation either removes one boundary bit or adds a bit where the other bit is zero. Thus its function A_0 is ordinary homogeneous. The actual row-probe core has $r(\mathrm{Row}_j\circ U_0) = jr(U_0)-(j-1)r(W_0)$, so $\Phi_{\eta_0}(h_j)$ is homogeneous of ordinary degree $jr(U_0)$. The elementary-probe normalization gives the analogous assertion for its other mark factor B_0.

Put $\Psi=\Phi_{\eta_0}$, and let A^*,B^* be the positive ordinary cores of the original mark and hatted-mark products. Their actual initial signs are included in the phases below. The geometric core formula gives

$$G_j := \mathrm{in}_{\mathrm{ord}}\,\Phi_\eta(h_j) = \epsilon_j\vartheta^{j-1}\Psi(\iota_\gamma h_j), \qquad G_1=\epsilon_1\Psi(h_1).$$

The two actual two-cell probes give the following two normalizations:

γ	G_2	$H_2 := \mathrm{in}_{\mathrm{ord}}\,\Phi_\eta(e_2)$
id	$\epsilon_2(A^*/A_0)\Psi(h_2)$	$\epsilon_2'(B^*/B_0)\Psi(e_2)$
AT	$\epsilon_2(A^*/B_0)\Psi(e_2)$	$\epsilon_2'(B^*/A_0)\Psi(h_2)$.

Call the first ratio ϑ and the second ϑ'. They are positive primitive low Schur-product ratios. If d_A,d_B are their ordinary degrees, the competing degrees in $\Phi_\eta(e_2)=\Phi_\eta(h_1)^2-\Phi_\eta(h_2)$ are

$$\deg_{\mathrm{ord}}G_1^2 = 2r(U_0), \qquad \deg_{\mathrm{ord}}G_2 = 2r(U_0)+d_A, \qquad \deg_{\mathrm{ord}}H_2 = 2r(U_0)+d_B.$$

If $d_A>0$, the initial difference is the square of s_{U_0}, whereas the actual column probe has one connected two-copy prime. Their low normalization factors cannot reconcile these high primes. If $d_A<0$, the initial difference is $-G_2$, identifying the row and column two-copy primes up to a low factor, contrary to Lemma 2.3. Hence $d_A=0$.

Put $Y_j = \Psi(\iota_\gamma h_j)$. In the independent fine pivot coordinates, Y_1^2, Y_2 are distinct monomials, so their difference cannot lose its degree. Both terms survive, and

$$Y_1^2 - \epsilon_2 \vartheta Y_2 = \epsilon_2' \vartheta' (Y_1^2 - Y_2).$$

Comparison of the two coefficients gives

$$\epsilon_2' \vartheta' = 1, \qquad \epsilon_2 \vartheta = 1.$$

Positivity gives $\vartheta = \vartheta' = 1$ and $\epsilon_2 = \epsilon_2' = 1$. For $j \geq 3$, use the actual hook $(j-1, 1)$, whose function is $h_1 h_{j-1} - h_j$. Its normalization is $A^{j-2}B$. The two calibrated ratios make its core normalization agree with that of the carried hook, also when the roles of A_0, B_0 are exchanged. For its actual core phase η_j, the initial identity is

$$\epsilon_1 \epsilon_{j-1} Y_1 Y_{j-1} - \epsilon_j Y_j = \eta_j (Y_1 Y_{j-1} - Y_j).$$

Comparing the two independent coefficients gives $\epsilon_j = \eta_j = \epsilon_1 \epsilon_{j-1}$. Together with $\epsilon_2 = \epsilon_1^2 = 1$, this yields $\epsilon_j = \epsilon_1^j$. Triangular independence prevents cancellation for any nonzero weighted-homogeneous input and proves (A.3). These are identities of known fine probe compositions, not new observations on the source. □

Proposition A.4. *If the original fine mark is positive and distinct, the polynomial P of Proposition A.1 is the canonically phased function of an actual connected diagram of span R.*

Proof. Apply the actual marked-core construction to the original equal parents. The zero-free class theorem and Proposition 3.9 supply an actual J_0 of span R with $s_{E_0} = s_{J_0 \circ_{\eta_0} U_0}$. Apply (A.3) to the entire image (A.1). Its initial right side is the actual core function times a low factor. The class-valued quotient gives that same function as the normalized image of s_{J_0}. Therefore the images of $\iota_\gamma P$ and s_{J_0} differ by a low rational scalar. In the independent fine pivot coordinates this scalar is rational constant. Primitivity and the actual top proper phase make it the canonical sign. Apply the physical identity or antitransposition to J_0 prescribed by γ. The resulting actual diagram realizes the entire P. □

A.3. **Whole coframes recover actual end windows.** The zero-offset reconstruction uses the weak-cap and absolute-flag theorem of [6]. Its domain is arbitrary strictly ordered positive lists X, Y with $X_i \geq Y_i$, including unit and disconnected proper blocks. Its joint-polynomial numerator irreducibility and reconstruction of the absolute flags from the whole rational coframe are both retained; connected-coframe coprimality alone would not supply this interface. The application below is needed when the fine outer interior contains a noncell, even if that factor comes from a wide source mark rather than from E^-.

A permitted transport here is the original marked U, or its simultaneous half-turn with both full end marks and their attachments, up to translation. Write $(U^\dagger, \eta^\dagger)$ for the chosen representative. An ordinary gauge is accompanied by its outer involution and is undone before the row maps are compared; in the original ordinary reading, $\Phi_{\eta^\dagger} = \Phi_\eta$. The first and last q windows use this same representative, with the lower window southwest-normalized and the upper window read in its reflected end frame. The transport is supplied by the original minimum's smaller-area fiber, not by a new factorization of U.

Lemma A.5. *In the original aligned comparison, suppose the actual C^- has a noncell component. If the fine mark has zero offset, the first and last q windows of the original E are the complete windows of one permitted transport of the original marked U. Every cap unit is retained.*

Proof. Choose a noncell $Z \subset C^-$. Its actual reference $Z \circ_\eta U$ is proper, and its connected function occurs as a high factor of the common whole proper corner. Each corresponding actual component of F^- has area below n. By the original fine minimum, Lemmas 3.2 and 3.5 give it that same fine marked factor. We use the *whole parent* F for this argument, so the component may come from a source mark or a hatted source piece, not only from E^-.

At a finite cutoff put $y_j = \Phi_\eta(h_j) = c_j h_{jq+u} + d_j$, and retain all nonpivot raw variables as spectators. Over the fraction field of the raw variables below q, triangular inversion is polynomial

in y_j and the high spectators. Put $K = N_C$, so $N_F = Kq + u$, and use the current carried fine frame. If A_η, B_η, C_η are its local mark products, the complete low normalization is

$$\mathcal{N}_\eta = \epsilon_\eta(C) A_\eta^{|C|-r(C)} B_\eta^{|C|-c(C)} C_\eta^{|C|-K}.$$

The original whole equality and its top derivative give

$$\Phi_\eta(s_C) = \mathcal{N}_\eta s_F, \qquad c_K \Phi_\eta(\partial_{h_K} s_C) = \mathcal{N}_\eta \partial_{h_{N_F}} s_F.$$

Any prescribed outer involution acts on both displayed coarse polynomials. The nonzero factor $\mathcal{N}_\eta$ uses only raw labels below q; negative normalization exponents are read in that low fraction field. Define the partial compensator

$$\Delta_s^{<j} = \partial_{h_s} + \sum_{i<j} a_{si} \partial_{h_{iq+u}}.$$

Differentiation in one spectator, holding all y_j fixed, is

$$\Delta_s = \partial_{h_s} + \sum_j a_{sj} \partial_{h_{jq+u}}, \qquad a_{sj} = -c_j^{-1}\big((\Delta_s^{<j} c_j) h_{jq+u} + \Delta_s^{<j} d_j\big).$$

For a high spectator $s \geq q$, induction on j gives $a_{sj} = 0$ whenever $jq + u < s$: the entire jth pivot polynomial uses only raw labels at most $jq + u$, and all earlier compensators are already zero. Since s is not a selected pivot, the first possibly moving pivot is strictly above s. Every raw label below s is fixed. The low coefficients c_j are fixed as well, but the lower terms d_j retain their full earlier-pivot feedback in the displayed recursion. These two identities express s_F and its complete proper product as polynomials in the y_j over the low fraction field, independent of the high spectators. Degree additivity in each spectator makes every connected factor of the proper product independent of it as well.

Fix one high component prime and retain every one of its e actual occurrences $G_1, \dots, G_e$, in their original order. Eliminate all regular components and unit records by their full block inverses. The resulting proper matrix is upper block triangular, with the unchanged diagonal blocks G_i and complete effective bridges $E_{i,i+1}$. Modulo the selected prime, each G_i has rank one less than its size. Its proper corner is a nonzero coefficient-field unit: differentiate its actual fine image at h_{vq+u}, obtaining a nonzero multiple of $\Phi(\partial_{h_v} s_Z)$, which is not divisible by the prime. Normalize the rank-one adjugates using these corners. Their vectors remain polynomial in the spectators.

The only maximal-pole path visits all e occurrences. The complete source residue is a nonzero coefficient-field scalar times

$$(u_{\mathrm{eff}} a_1) \prod_{i=1}^{e-1} (b_i^t E_{i,i+1} a_{i+1}) (b_e^t g_{\mathrm{eff}}). \tag{A.4}$$

The final factor contains the actual creation column. The source is a connected prime of larger span, so its residue is nonzero. Polynomial degree additivity in each spectator therefore makes every complete link in (A.4) spectator-flat.

Each link has an actual connected numerator. For consecutive occurrences, retain all intervening regular records, delete the last column of G_i and the first row of G_{i+1}, and nothing else. Between these deletions an old row pairs with the next old column; positive rows and the old weak diagonals prove connectedness, including all intervening unit records. If there are ℓ regular records, its raw determinant is

$$(-1)^\ell \det(H_{\mathrm{reg}}) e_{\mathrm{last}}^t \operatorname{adj}(G_i) E_{i,i+1} \operatorname{adj}(G_{i+1}) e_{\mathrm{first}}.$$

The sign follows by moving ℓ regular rows past $r(G_i)$ rows and the matching columns past $r(G_i) - 1$ columns. At the left end keep the original first row and delete the first row of the first component; at the right end delete the last component column and keep the original column 0. The endpoint signs are the same total-regular-rank signs. Their proper corners are products of regular determinants and nonzero component corners, hence nonzero in the residue field.

If such a numerator had high nonpivot span d, then Δ_d would act on it simply as ∂_{h_d}, since every compensating pivot is higher than d. Its derivative is its nonzero proper corner,

contradicting spectator-flatness. Thus every endpoint and consecutive span is $jq+u$. The left endpoint gives $\min Y(G_1) \equiv 0 \pmod q$; consecutive spans propagate this congruence, and every component maximum is congruent to u. The right endpoint gives the same final phase. Repeat the argument for every high prime in the complete proper product, including the image of a one-cell component. Thus all high component occurrences have aligned absolute endpoints before any last-band bound is used.

The last-band bound follows from these actual aligned occurrences. Every high connected factor of the proper corner of F, of span $N_F = Kq+u$, has smaller aligned span $vq+u$; hence $v \le K-1$. All remaining factors have span below q. For an actual endpoint or consecutive numerator of span $jq+u$, its complete proper coefficient is the product of the intervening regular component determinants and one or two repeated-component proper corners. A regular high component lies strictly inside the numerator endpoints. Its aligned endpoints give span $vq+u < jq+u$, hence at most $(j-1)q+u$. For a repeated component of span $vq+u$, the full fine corner formula bounds its high proper factors by $(v-1)q+u$; its local factors have span below q. This also covers a one-cell occurrence, whose proper factors are all low. The proper coefficient of F is fixed on $(K-1)q+u < s < Kq+u$; that of each actual link numerator is fixed on its corresponding band $(j-1)q+u < s < jq+u$.

Write $X_F = (N_F, I_F)$ and $Y_F = (C_F, 0)$ for the whole parent's full beta lists. Let $g_F = R_F$ be the complete lower endpoint coframe and $g_{F^*} = L_F$ the reflected upper endpoint coframe in the actual beta frame of F, as in (6.2). Their normalizations are at the actual bottom column and top row. Let $g_{U,-}, g_{U,+}$ be the corresponding whole weak-cap coframes obtained by solving the known first and last q-window matrices of the supplied marked U, including all their unit records. These native window coframes have degree less than q. At this point the source coframes are only read modulo z^q; their full matrices have not been truncated, and physical boundedness of the source caps has not been assumed.

Differentiate the whole-parent image above at a raw label in its last open q-band. Earlier selected images use smaller labels, and later pivots exceed N_F, so only y_K contributes. Divide the summed derivative by the nonzero top cofactor, which is fixed on this band. The intrinsic full gradient and the actual Kth native row-probe give, for $0 \le t < q$,

$$[z^t] g_F g_{F^*} = \frac{\partial_{h_{N_F - t}} s_F}{\partial_{h_{N_F}} s_F} = \frac{\partial_{h_{Kq+u-t}} y_K}{c_K} = [z^t] g_{U,-} g_{U,+}.$$

The case $t = 0$ is the common normalization one. Thus

$$g_F(z) g_{F^*}(z) = g_{U,-}(z) g_{U,+}(z) \pmod{z^q} \tag{A.5}$$

Decompose each endpoint proper system at its actual zero-height cuts and unit pivots, retaining every triangular bridge. Suppose a block meeting a column of exponent below q has a largest outer label $d \ge q$. Choose d maximal across both endpoint systems, and then the earliest column exponent b at which it occurs. Since the constant coframe term is one, $0 < b < q$.

Each block attaining (d, b) is connected, of span $s = d - b > 0$. Its matrix and all lower blocks omit h_d; that variable occurs only in the first entry of its effective creation border, with coefficient one. The inverse corner therefore gives

$$\partial_{h_d} [z^b](g_F g_{F^*}) = -\sum_{T \in \mathcal{E}_{d,b}} \frac{\kappa(T)}{Q_T},$$

where $\mathcal{E}_{d,b}$ consists of the one or two endpoint blocks attaining the same earliest border. Earlier colors are independent of h_d, so no product of two moving terms contributes. This is ordinary differentiation of the selected rational coefficient, not extraction of an h_d-coefficient or a residue. For every such block,

$$\frac{Q_T}{\kappa(T)} = h_s - \omega_T, \qquad \omega_T \in \mathbb{Q}(h_1, \dots, h_{s-1}).$$

Regard the sum as a rational function of h_s over $\mathbb{Q}(h_i : 1 \le i < d,\ i \ne s)$. Distinct monic factors have separate residues -1; coincident factors contribute their negative multiplicity. The

sum is nonzero. But the right side of (A.5) uses only raw labels below q, so its h_d-derivative is zero. This contradiction proves that both endpoint systems are physically bounded below q.

Their completed low blocks have equal row and column counts:

$$\#(I_F \cap [1,q)) = \#(C_F \cap [1,q)).$$

Thus (2.1) gives $H_F(q-1) = 1$. The reflected upper count gives $H_F(N_F - q) = 1$. These are singleton cuts in the full source lists, including every common beta.

Apply the same last-q gradient to each actual link numerator above. The proved bound on its complete proper coefficient makes that coefficient fixed on the entire last open band. Its conormal equation has only the raw derivative there and the last-pivot compensation; all lower compensators vanish and all higher pivots exceed its span. Division by its actual nonzero proper coefficient gives (A.5) modulo the selected high prime, with all occurrences and their original normalization retained. After taking the physically bounded caps, every coefficient belongs to the raw low field. The high prime has an independent fresh pivot, so reduction modulo that prime is injective on this field. The identity is therefore an exact truncated low identity. For each selected noncell occurrence G_i, let V_i be the permitted transport of the original marked U supplied by its strict-area full fiber. Its first and last q-windows survive deletion of the opposite extreme flag, since $N_{G_i} \geq 2q + u$. Write $g_{F,-}, g_{F,+}$ for the bounded lower and reflected upper source caps. All these coframes have constant term one and degree below q.

The left endpoint numerator retains the upper source window and the lower window of G_1. Its normalized last-band identity is

$$g_{F,+}g_{V_1,-} \equiv g_{U,+}g_{U,-} = g_{V_1,+}g_{V_1,-} \pmod{z^q}.$$

The last equality carries the permitted rotation or ordinary gauge: rotation exchanges the two endpoint coframes, while the ordinary gauge leaves the inner diagram unchanged. Cancellation takes place in $\mathbb{Q}(h_1, \ldots, h_{q-1})[z]/(z^q)$, where $g_{V_1,-}$ is a unit, and gives

$$g_{F,+} \equiv g_{V_1,+} \pmod{z^q}.$$

Both polynomials have degree below q, so they are equal as whole coframes. Their actual weak cap lists are strictly ordered and positive, with $X_i \geq Y_i$; no unit or disconnected block has been removed. The weak-cap and absolute-flag theorem [6] therefore identifies their complete absolute flags.

The numerator between G_i and G_{i+1} retains the upper window of G_i and the lower window of G_{i+1}. Its complete normalized identity is

$$g_{V_i,+}g_{V_{i+1},-} \equiv g_{U,+}g_{U,-} = g_{V_i,+}g_{V_i,-} \pmod{z^q}.$$

Cancel $g_{V_i,+}$ and use the degree bound to obtain $g_{V_{i+1},-} = g_{V_i,-}$. Their native two-cap products then give $g_{V_{i+1},+} = g_{V_i,+}$. The same pair of actual windows therefore propagates through every occurrence. The right endpoint numerator gives

$$g_{V_e,+}g_{F,-} \equiv g_{U,+}g_{U,-} = g_{V_e,+}g_{V_e,-} \pmod{z^q},$$

and hence $g_{F,-} = g_{V_e,-} = g_{V_1,-}$. Whole-coframe recognition identifies the opposite source window. A cell occurrence is handled by its actual equality with s_U and the same intrinsic two-cap product; it is never given a proper one-cell outer.

Finally $q < p$, and the first and last p windows of F are the actual windows of E. The reconstructed q windows of F are therefore exactly the corresponding windows of E. This proves the assertion regardless of which source proper factor supplied the noncell reference. □

A.4. **A returned section realizes the whole polynomial.**

Proposition A.6. *If the actual fine outer interior contains a noncell component, the zero-offset branch of Proposition A.1 has an actual connected canonical realization of its entire P.*

Proof. We first align the capped core quotient. Use Lemma A.5 to fix one transported fine marked frame in which the complete first and last q windows of E and U match. Core deletion is local on the two boundary words, so the corresponding windows of E_0 and U_0 still match.

We strengthen the induction of Proposition 3.9 as follows. At every stage retain actual proper Schur-equal charts with $p = kq$, $k \geq 2$, the stated width relation, an ordinary supplied fine chart, full source-minimal fibers for the current parent class, the joint package below its area, and these two matched windows of the displayed inners. The conclusion retained is a literal quotient through the supplied fine mark. Neither displayed chart is assumed source-minimal. The full fibers are supplied by Proposition 3.7 initially and by Lemma 3.2 at the strictly smaller quotient parents. The geometric induction allows ordinary fine marks generally; it does not assume that a recursive offset remains zero.

If U_0 is a line, unconditional full line transport places its same marked q-chart on the first parent. Supplied-divisor gluing with that parent's actual p-chart gives a literal q-chart on E_0, with the genuine ordinary gauge carried.

Otherwise choose a fresh source-minimal chart on the second parent, of period d and width v. Its full fiber is supplied by the current induction hypotheses, not by inheriting an original minimum. Compare it locally with the supplied chart through the nonline U_0. If d, q are nondividing and have gcd $h > 1$, Proposition 3.3 constructs an actual proper h-chart. Its width is $v \bmod h \leq v$; at a width tie its inner span is smaller than $d + v$. It therefore improves the fresh minimum. If the periods are coprime, the same local comparison forces U_0 to be a line, contrary to this branch. If d is a larger multiple of q, the already supplied q-chart has width $u = v \bmod q \leq v$ and again improves the score, strictly in inner span when the widths agree. Consequently $d \mid q$.

At $d = q$, the displayed fine chart has the same width and inner span as the fresh minimum, so it too is source-minimal. Its full fiber supplies the fine chart on the first parent. The matched q windows identify its transported inner with U_0, since $N_{U_0} = q + u < 2q$. If the two actual markings form the ordinary double, use their genuine gauge with its outer involution. Supplied-divisor gluing then finishes as in the line branch.

Suppose $d < q$. The fresh divisor factor is ordinary: a mixed period-d factor would annihilate δ_q, whereas the supplied proper ordinary q-chart has a nonzero Hall response [10, Mixed-vanishing lemma and factor recovery at a prescribed period]. Full transport and supplied-divisor gluing place its marked inner on both parents and on their displayed inners, initially allowing its full rotation. The first and last d windows lie in the matched q windows. Since this inner has span $d + v < 2d$, the two windows identify the transported inners literally. If their actual marks form the ordinary double, carry the genuine outer involution so that one common marked factor ζ, on W, is used on both sides. Write

$$E_0 = E_1 \circ_\zeta W, \qquad U_0 = U_1 \circ_\zeta W.$$

The descended fine mark on U_1 is the one already supplied by aligned reassociation.

Put $q' = q/d$. The quotient fine chart is also ordinary. Otherwise its $\delta_{q'}$ response would vanish, and the ordinary Hall covariance through ζ would make the original δ_q response vanish. The local normalization factors have raw labels below d, so they and their inverses are δ_q-constant; the prescribed involutions are carried on both sides. This uses the full covariance of [10, Composition covariance].

The matched windows descend at the level of both boundary endpoints. Let a be the first coordinate of the displacement of ζ, and let $[\lambda_c, \mu_c]$ be its native interval at a private residue $v \leq c < d$. A quotient interval $[\ell_j, h_j]$ gives the source interval at $dj + c$

$$\begin{cases} [ja + \lambda_c + \ell_j,\ ja + \mu_c + h_j], & \text{case } (a), \\ [ja + \lambda_c - h_j,\ ja + \mu_c - \ell_j], & \text{case } (b). \end{cases}$$

Each formula recovers both quotient endpoints. Thus no ordinary dual is imposed on a mark for which it has not been supplied. For a quotient inner of span m, the first $q = dq'$ window contains the samples $dj + v$ for $0 \leq j < q'$, and the last contains those for $m - q' \leq j < m$: its initial content is $d(m - q') + v$. The complete first and last q' windows of E_1 and U_1 therefore match, including their common betas.

Strip the common factor and apply Lemma 2.3. The quotient parents are actual outers of proper compositions, hence have strictly smaller area by Lemma 3.6; the lower package supplies

their full fibers. Their periods are $p/d, q/d$, and their widths are $(w-v)/d, (u-v)/d$, so the relation $w = bq + u$ is retained. At each recursive call use the line branch when its fine inner is a line, and otherwise choose a fresh minimum. Since $d \geq 2$, the fine period strictly decreases.

Induction gives the literal quotient through the supplied descended fine mark. Reattach by the already supplied fine aligned identity and undo its prescribed outer involution. Denote the resulting actual outer by J_0. Then

$$E_0 = J_0 \circ_{\eta_0} U_0 \tag{A.6}$$

literally, in the matched frame. If a class-valued quotient was chosen earlier, faithfulness identifies its function with s_{J_0}, not its diagram or lists. No arbitrary coarse mark is lifted through a newly chosen fine factor.

Continue in the matched marked frame of (A.6). Write X_E, Y_E for the original full source lists. Set $\rho = 0$ in the horizontal empty reading. For a positive coincident mark use its genuine ordinary-(a) reading and set $\rho = \tau$, where $0 < \tau < u$ is the phase of the current transported mark.

The literal composition identity, the complete coding table (2.2), and its caps (2.3) give

$$X_{E_0} \cap (\rho + q\mathbb{Z}) = qX_{J_0} + \rho, \qquad Y_{E_0} \cap (\rho + q\mathbb{Z}) = qY_{J_0} + \rho.$$

At empty width this includes the actual top row and bottom column. At positive width the coincident return is absent from both fixed native lists, and the initial and terminal cap values supply the selected endpoints $Rq + \tau$ and τ.

The phase is not held fixed through a rotation. Since $N_E = Rq + u$, reflection sends $qy + \rho$ to $q(R-y) + (u-\rho)$; it therefore sends ρ to $u - \rho$ and rotates the quotient. A genuine ordinary gauge carries its outer involution before these lists are read. Define

$$X_J = \{(x-\rho)/q : x \in X_E,\ x \equiv \rho \pmod q\},$$
$$Y_J = \{(y-\rho)/q : y \in Y_E,\ y \equiv \rho \pmod q\}.$$

The selected core top row is $Rq+\rho$, and the selected core bottom column is ρ. Put $Z_E = X_E \cap Y_E$ and

$$Z_\rho = \{(z-\rho)/q : z \in Z_E,\ z \equiv \rho \pmod q\}.$$

A restored common beta cannot occur at either selected endpoint, which is already an unmatched core flag. Since $q > u$, there is no additional selected position below ρ or above $Rq+\rho$ within the source span. Thus $Z_\rho \subseteq \{1, \ldots, R-1\}$, and

$$X_J = X_{J_0} \dot\cup Z_\rho, \qquad Y_J = Y_{J_0} \dot\cup Z_\rho.$$

The lists have equal cardinality and retain every selected unit. For $0 \leq j < R$,

$$\#\{y \in Y_J : y \leq j\} - \#\{x \in X_J : x \leq j\} = H_{J_0}(j) > 0.$$

The endpoint heights are one. The sorted lists are weakly feasible and reconstruct a connected skew diagram J of span R. Unselected source records are not deleted. In the positive reading, the physical top row $Rq + u$, bottom column 0, and remaining records in the terminal band $[Rq, Rq + u]$ stay as separate native cap records; the selected contact row does not replace the physical top. We now identify the polynomial of J, rather than assume it.

For the empty case let $g(z)$ be the actual lower native cap coframe and $l(z) = [\mathcal{H}_{<q}(z)g(z)]_{<q}$, both with constant term one. Initially let $Y(u) = 1 + \sum_{j\geq 1} Y_j u^j$ and $S(u)$ be independent series with constant term one. Set

$$G^\dagger = g - 1 + S(z^q), \qquad H^\dagger = \frac{l - 1 + Y(z^q)S(z^q)}{G^\dagger}. \tag{A.7}$$

The low raw moments are unchanged. On every actual native row-probe, the complete coframe is $G^\dagger/S$, and

$$H^\dagger G^\dagger/S = (l-1)/S + Y(z^q).$$

The fixed native rows vanish and the selected rows have the prescribed independent residuals. The actual generator calibration is $\Phi(h_j) = \xi^j Y_j$, after using the fixed signed empty normalization. The coefficient of the top pivot in the jth empty row image is ξ^j, by its row-corner

formula and actual row phase. Hence the same scale cancels in the source function divided by its top derivative. Every finite vector is truncated only beyond its actual matrix horizon.

Choose now S to be the complete coframe of the actual J. Its denominator is the actual proper coefficient $\kappa(J)$ and omits Y_R. Use $G^\dagger$, not the native row-probe vector $G^\dagger/S$, as the source vector. Its entire row residual series is

$$H^\dagger G^\dagger = l - 1 + Y(z^q)S(z^q).$$

The support in (A.7) lies on the actual source's lower-cap columns and selected columns. Set unused inverse coordinates to zero without deleting their rows or units. Below q the native cap equations apply; above q off the selected residue the numerator is zero; on the selected residue they are the whole J coframe equations. Thus the entire source inverse is solved. Its normalized scalar is $Q_J/\kappa(J)$, while (A.1) makes it P/Γ in the same independent coordinates. Both denominators omit the largest coarse generator. Cross multiplication, irreducibility and primitive canonical normalization give $P = Q_J$.

For a positive coincident mark use the true phase τ, not the empty selected residue. Let g be its actual lower native q-cap coframe, and put $l = [\mathcal{H}_{<q}g]_{<q}$. Write $g_\tau = [z^\tau]g$ and $l_\tau = [z^\tau]l$. Both are nonzero actual contact-cofactor ratios. Take Y, S independent again, and keep them independent throughout the native calibration. Define

$$\begin{aligned} G^\dagger &= g + g_\tau z^\tau (S(z^q) - 1), \\ L^\dagger &= l + l_\tau z^\tau (Y(z^q)S(z^q) - 1), \qquad H^\dagger = L^\dagger/G^\dagger. \end{aligned} \tag{A.8}$$

Then

$$H^\dagger G^\dagger/S = (l - l_\tau z^\tau)/S + l_\tau z^\tau Y(z^q).$$

All fixed native rows vanish, and row $jq+\tau$ has residual $l_\tau Y_j$. The selected returned coefficient is g_τ; column 0 and its actual positive-copy columns remain. At the terminal cap the copy-column range is strictly below u, while the fixed rows may include u. These are exactly the bounds in the actual ordinary coding table. They ensure that this is the full coframe, with no omitted live tail.

For the actual row probe put $r_j = r(\mathrm{Row}_j \circ_\eta U)$ and $c_j = \partial_{h_{jq+u}}\Phi_\eta(h_j)$. The modified row image (2.5) and its complete proper corner (2.7) give

$$c_j = (-1)^{r_j - 1} A^{j-1} \Theta C^j B^{j-1}.$$

Successive copies have the constant row increment $r_2 - r_1$. Its sign is the actual one-copy phase in the local product ξ, so

$$\xi = (-1)^{r_2 - r_1} ABC, \qquad c_1 = (-1)^{r(U)-1}\Theta C, \qquad c_j = c_1 \xi^{j-1}.$$

All these factors use raw labels below q, whose values the positive section preserves.

Let g_+ be the complete upper native q-cap coframe, reflected and normalized at the actual top row. With S still independent of Y, differentiating (A.8) gives

$$\partial_{Y_j} H^\dagger = \frac{l_\tau z^{jq+\tau} S(z^q)}{G^\dagger} = \frac{l_\tau z^{jq+\tau}}{g(z)} \pmod{z^{jq+u+1}}.$$

Thus no raw label below $jq + \tau$ moves. On the remaining band the complete native gradient is $\partial_{h_{jq+u-a}}\Phi_\eta(h_j) = c_j[z^a](gg_+)$, for $0 \le a \le u - \tau$, by (6.2). The chain rule therefore yields the full coefficient

$$\partial_{Y_j}\Phi_\eta(h_j) = c_j l_\tau [z^{u-\tau}]\frac{gg_+}{g} = c_j l_\tau [z^{u-\tau}]g_+.$$

The last coefficient is the actual upper contact-cofactor ratio and is nonzero, as is l_τ. Both are low and independent of j, S, Y. At $Y_j = 0$, the native vector $G^\dagger/S$ annihilates the entire row-probe matrix: its only possible remaining row residual was $l_\tau Y_j$ at $jq + \tau$. Its column-0 coefficient is one. The full determinant, and hence its modified image, therefore vanishes. Consequently

$$\Phi_\eta(h_j) = \mu \xi^j Y_j, \qquad \mu = \frac{c_1 l_\tau [z^{u-\tau}]g_+}{\xi} \neq 0,$$

with one low μ independent of j, S, Y. Calibrate this constant using the known actual two-column fine composition: put $S(z^q) = 1 - Y_1 z^q$ and $Y_2 = Y_1^2$. The vector $G^\dagger$ is then supported on that probe's actual columns and solves its full rows: the coefficients of $Y(z^q)S(z^q)$ at z^q and z^{2q} vanish. Thus the probe determinant vanishes. The identity $\Phi(e_2) = \Phi(h_1)^2 - \Phi(h_2)$ gives $\mu\xi^2(\mu - 1)Y_1^2 = 0$. Hence $\mu = 1$. This uses a known fine identity, not an extra source observation.

Now choose the actual J coframe for S, and use $G^\dagger$ as the source vector. Its row residual series is

$$H^\dagger G^\dagger = l + l_\tau z^\tau \big(Y(z^q)S(z^q) - 1\big).$$

Every source row is solved except the true contact row $Rq + \tau$, whose residual is $l_\tau Q_J/\kappa(J)$. The vector's column-0 coefficient is one. Adjugate multiplication shows that Q_J divides the substituted source determinant after clearing only low factors and $\kappa(J)$, never Q_J itself. The whole image is a nonzero low multiple of $\xi^{\mathrm{wt}(P)}P(Y)$. Connected irreducibility of Q_J and irreducibility of P make them associates; independent coefficient comparison, primitivity and the actual top phase give $P = Q_J$. This also retains every regular term and every selected unit. It completes both zero-offset sections. □

Lemma A.7 (Ribbon image projection)**.** *For an actual empty marked ribbon U of span q, if the whole function of an actual ribbon G lies in its composition image, its coarse polynomial is the canonical function of the ribbon obtained by retaining G's cuts at multiples of q and dividing those positions by q.*

Proof. Use the canonical ribbon coarsening expansion. Set raw generators whose indices are not multiples of q to zero. Every actual row-probe image then reduces to its unique no-internal-cut term h_{jq}, with its fixed canonical phase, since $\kappa(U) = 1$. On G the surviving coarsenings are exactly those whose partial sums are multiples of q; they are the full coarsening expansion of the indicated actual divided-cut ribbon. The induced substitution on the coarse polynomial is injective. Therefore its entire polynomial, not just its projection, is the stated canonical ribbon function. The physical gauge is undone on the actual ribbon when necessary. □

The remaining aligned branch has only cells in the actual C^-. The next appendix proves the full-unit cut theorem used there. Once it realizes P, composing that actual diagram through the original η/U compares two connected span-N_E primes against low normalizers. They are associates, and primitivity and positivity give $s_{J\circ_\eta U} = s_E$. This finishes the reconstruction part of Proposition 4.1; its separate minimality corollary is Corollary 4.2.

Appendix B. Integral cuts with a cell-only proper interior

The aligned construction must also handle a fine outer whose proper interior consists only of cells. No component in that interior supplies a proper fine chart. The following cut theorem treats this case on the full original determinant, including every original common beta.

Theorem B.1 (Full-unit integral cut theorem)**.** *Let C be an actual connected diagram of span $S = L + R$, with $L \geq 2$, $R \geq 2$, and $\kappa(C) = h_1^c$. Suppose $0 \leq a \leq c$, put $r = c - a$, and suppose the two whole currents satisfy*

$$V_L Q_C = mQ_0, \qquad \delta_L Q_C = \alpha Q_0, \qquad m \in \mathbb{Z} \setminus \{0\}, \quad \alpha \in \mathbb{Z},$$

where Q_0 is primitive integral, has span R, and satisfies

$$\partial_{h_R} Q_0 = h_1^c, \qquad v_{h_1}(Q_0) = r. \tag{B.1}$$

Then there is an actual connected diagram J of span R, with $\kappa(J) = h_1^a$, such that

$$Q_0 = h_1^r Q_J.$$

It has area $R + a < S + c = |C|$. Neither the end-cap nonoverlap inequality $L \geq R - 1$ nor the inventory inequality $c \geq 2a$ is required.

B.1. **Physical cuts and the minimum child.** *Proof of Theorem B.1.* Every positive component of C^- is a cell. Indeed an actual connected noncell polynomial has a largest-generator term different from a power of h_1, and irreducibility precludes such a factor in h_1^c. Thus the original adjacent gaps g_i are zero or one, and there are exactly c gap-one edges. The actual interior row inequalities are $r_i > g_{i-1} + g_i$; hence every entry farther below the subdiagonal of the full Jacobi–Trudi matrix is negative zero. The matrix is upper Hessenberg. For a contiguous partition $\mathcal{P}$ of the original rows set

$$q_B = \sum_{i\in B} r_i - \sum_{\text{edges in } B} g_i, \qquad e(\mathcal{P}) = \sum_{\text{edges inside blocks}} g_i.$$

The complete interval-cycle expansion is

$$Q_C = \sum_{\mathcal{P}} (-1)^{|\mathcal{P}|-1} h_1^{e(\mathcal{P})} \prod_{B\in\mathcal{P}} h_{q_B}. \tag{B.2}$$

The phase is the interval-cycle sign multiplied by the one original canonical row phase. All repeated monomials are collected in this sum.

For $1 \le t \le R$, define the actual physical cut N_t by marking each block $B = [i, j]$ in (B.2), replacing its factor by h_{q_B-L}, and multiplying the occurrence by $\mathbf{1}_{y_j<t\le x_i-L}$. Equivalently it is the complete marked-entry cofactor sum with that same indicator. Then

$$V_L Q_C = \sum_{t=1}^{R} N_t. \tag{B.3}$$

The unweighted sum over every marked occurrence is $\delta_L Q_C$. In particular $h_L \mapsto 1$ remains in the latter identity.

Let $Z = X \cap Y$ be the original common betas, and let s be the original row count. Index the full decreasing lists here as $X = (x_1, \dots, x_s)$, $Y = (y_1, \dots, y_s)$. For $T \subseteq Z$, put $I_T = \{i : x_i \in T\}$, $J_T = \{j : y_j \in T\}$. Identify T also with the actual nonattacking unit positions (i, j) for which $x_i = y_j \in T$. Delete those rows and columns and subtract one from each retained outer beta. Call the resulting actual weak diagram B_T. Deleting a common beta preserves cumulative-count differences; after the shift its sorted diagonal indices are nonnegative. Thus B_T is an actual weak board, possibly with empty rows or several components, of area $|C| - (s - |T|)$. Retain all $d = s - |T|$ slots in their original decreasing orders and put

$$Q_{B_T}^{[d]} := (-1)^{d-1} \det(H_{x_i-1-y_j})_{i\notin I_T,\, j\notin J_T}, \qquad H_0 = 1, \quad H_j = 0 \quad (j < 0).$$

The matrix is $\mathrm{JT}(X_{\overline{I_T}} - 1, Y_{\overline{J_T}})$ in the H-variables. Its phase counts retained slots, including empty ones, not occupied rows of B_T. For a fixed ordinary degree d, define the injective map

$$\psi_d(h_{i_1} \cdots h_{i_d}) = \prod_{i_j>1} H_{i_j-1}.$$

The missing number of h_1 factors is recovered from d. Expansion in the actual nonattacking unit positions gives

$$q_d := \psi_d((Q_C)_d) = \sum_{|T|=s-d} \epsilon_T Q_{B_T}^{[d]}, \qquad \epsilon_T = (-1)^{\sum_{(i,j)\in T}(i-j+1)}. \tag{B.4}$$

To verify the identity directly, substitute $h_i = tH_{i-1}$ for $i > 0$, keep $h_0 = 1$, and take the coefficient of t^d. Each chosen unit deletes its row and column; every unchosen old unit becomes a negative zero in B_T. The chosen equal-beta positions preserve their relative order, so their Laplace sign is $(-1)^{\sum_{i\in I_T} i + \sum_{j\in J_T} j}$. Combining it with the original phase $(-1)^{s-1}$ and dividing by the retained phase $(-1)^{d-1}$ gives

$$(-1)^{s-d+\sum_{i\in I_T} i + \sum_{j\in J_T} j} = (-1)^{\sum_{(i,j)\in T}(i-j+1)} = \epsilon_T.$$

Deleting empty slots before assigning this phase would instead use the occupied-row convention and need not give the same polynomial.

Deleting all units gives the unique minimum child B_Z, a connected ribbon β_0 of span $S-1 = L+d_0$, where $d_0 = R-1$. It has $c+1$ rows. For $c=0$ it is a row. There are no empty slots in this child, so $Q_{B_Z}^{[c+1]} = Q_{\beta_0}$. Every original unit is subdiagonal, at $(j+1, j)$, and hence $\epsilon_Z = 1$. Consequently $q_{c+1} = Q_{\beta_0}$, with its existing canonical phase. The whole contraction identities for the two currents are

$$\begin{aligned} \psi_d((V_L Q_C)_d) &= (V_L^H + \delta_L^H) q_d, \\ \psi_d((\delta_L Q_C)_d) &= \delta_L^H q_d + \partial_{H_{L-1}} q_{d+1}. \end{aligned} \tag{B.5}$$

The first formula includes the boundary lost by shifting each outer beta by one. More explicitly, the contracted cut is $N_t(B_T) + O_{t-1}(B_T)$, with both cut sums formed from $Q_{B_T}^{[d]}$; O_{t-1} is its whole outer L-strip remainder at that endpoint. Empty slots retain their assigned phase. The second formula includes the actual Hall creation from an original factor h_L. These exhaust the marked occurrences, so neither is a termwise assumption.

The top term in (B.1) makes the weighted current attain ordinary degree $c+1$. Its proportional Hall current has no degree-c term. At the minimum layer (B.5) therefore gives

$$\partial_{H_{L-1}} Q_{\beta_0} = 0, \qquad W := (V_L^H + \delta_L^H) Q_{\beta_0}, \qquad W/m \in \mathbb{Z}[H]. \tag{B.6}$$

The minimum child therefore uses the canonical phase, while all higher layers retain their original slot counts.

B.2. **Creation-gated vertex saturation.** Let P_0, Q_0' be the proper prefix and suffix sums of the ribbon β_0, and put

$$T_0 = \{d_0 - v : v \in Q_0',\ 0 < v < d_0\}, \qquad f = \mathbf{1}_{d_0 \in P_0}, \quad g = R - \mathbf{1}_{d_0 \in Q_0'}.$$

For $0 \le t \le d_0$ form the actual ribbon of area d_0 with cut set

$$(P_0 \cap [1, d_0 - t]) \ \cup \ (T_0 \cap [d_0 - t, d_0 - 1]). \tag{B.7}$$

The endpoint $t = 0$ is zero when $d_0 \in P_0$, and $t = d_0$ is zero when $d_0 \in Q_0'$. Every other vertex contributes that actual canonical ribbon. Their sum is W, and

$$m = g - f = R - \mathbf{1}_{d_0 \in P_0} - \mathbf{1}_{d_0 \in Q_0'} > 0. \tag{B.8}$$

Here is the full coarsening proof, including overlapping end windows. Mark a factor of length $L+e$, $e \ge 0$, in a coarsening of β_0. If its outside prefix and suffix have weights u, v, then $e = d_0 - u - v$, and its weighted-plus-Hall charge is $e+1$, the number of integer vertices in $[v, d_0 - u]$. For two different marked long blocks in one coarsening these vertex intervals are disjoint: if the earlier block has length $b \ge L$ and the intervening weight is $b' \ge 0$, the later prefix increases by $b + b'$, leaving a gap of at least $L + b'$ between the two vertex intervals. Thus a fixed vertex receives at most one mark from that coarsening, even when $L < d_0$. Collecting all coarsenings at that vertex gives (B.7). At the switching location two apparent single-cut terms are corrected by the $H_0 = 1$ central term: their coefficient is $-1 - 1 + 1 = -1$. The same recurrence cancels an endpoint exactly in the two cases stated. Each live ribbon has top coefficient one, proving (B.8); the original nonzero weighted current excludes $m = 0$.

For any ribbon, $\partial_{H_j} Q = 0$ exactly when there is no contiguous row interval of weight j. Indeed all coarsening occurrences contributing to one output monomial have sign determined by its number of factors, so cannot cancel. The creation zero in (B.6) therefore forbids every original prefix/suffix pair of sum $d_0 + 1$. Moving the switch in (B.7) by one can delete a cut and add the neighboring cut. If both happened, that forbidden prefix/suffix pair would exist. Each changed live adjacency consequently makes just one strict split or merge of positive row lengths.

Suppose the masks were not equal. The only possibly invisible endpoint differences are also forbidden by the same sum-$d_0 + 1$ condition; this includes the one-live-vertex case. Choose a globally least squared-row-cost monomial among the m live ribbon diagonals. A nontrivial coarsening has strictly larger squared cost. All diagonal aliases at this least cost have the same sign, determined by their common factor count. Because a live adjacency changes its squared

cost strictly, this monomial occurs on between one and $m-1$ vertices. Its coefficient is not divisible by m, contradicting (B.6). Thus

$$P_0 \cap [1, d_0 - 1] = T_0, \qquad W = mQ_\beta \tag{B.9}$$

for one actual ribbon β of area d_0.

B.3. **The exact row count from the whole unit jets.** It remains to determine the row count of this ribbon without discarding higher original unit layers. Let $A_1, \dots, A_{c+1}$ be the maximal original zero-gap runs. Their spans R_i are at least two when $c > 0$; singleton rows inside a run remain. For an interval I of t consecutive runs, join neighboring endpoint rows by

$$A \star B = (a_1, \dots, a_{r-1}, a_r + b_1 - 1, b_2, \dots, b_s),$$

and write A_I for the actual joined ribbon. Its span is $\sigma(I) = \sum_{i \in I} R_i - (t-1)$. The full determinant expansion is

$$Q_C = \sum_{E \subseteq [c]} (-1)^{c-|E|} h_1^{|E|} \prod_{A \in \mathcal{A}(E)} Q_A,$$

where $\mathcal{A}(E)$ is the ordered list after joining the selected gap-one edges. Its phase follows by counting the final coarsening blocks. Every internal singleton contribution is still inside its ribbon factor.

Put $P_j^{\mathrm{jet}} = [h_1^j] Q_C$ and let V_L^0, δ_L^0 denote the induced operators after setting $h_1 = 0$. The only positive image creating a new h_1 comes from h_{L+1}, with coefficient one for both operators. Consequently

$$\begin{aligned} [h_1^j] V_L Q_C &= V_L^0 P_j^{\mathrm{jet}} + \partial_{h_{L+1}} P_{j-1}^{\mathrm{jet}}, \\ [h_1^j] \delta_L Q_C &= \delta_L^0 P_j^{\mathrm{jet}} + \partial_{h_{L+1}} P_{j-1}^{\mathrm{jet}}, \qquad P_{-1}^{\mathrm{jet}} = 0. \end{aligned} \tag{B.10}$$

For $a \geq 1$ and $r \geq 1$, the simultaneous vanishing of the jets below r is equivalent to

$$\begin{cases} \sigma(I) \leq L - 1, & |I| < r, \\ \sigma(I) \leq L + 1, \quad \sigma(I) \neq L, & |I| = r. \end{cases} \tag{B.11}$$

Subject to these bounds, exact weighted valuation r means that some length-r interval has span $L+1$, or some length-$r+1$ interval has span at least $L+2$.

For completeness, these equivalences follow from the whole path, not from separate jet terms. Project each actual ribbon to $h_1 = 0$ and set all remaining h_j, $j \geq 2$, to -1. If $\kappa_i = -\overline{Q}_{A_i}(-1)$, every $\kappa_i > 0$ by the coarsening signs. At jet j, under the preceding barriers, a nonzero weighted group uses all j mergers and is one interval G of $j+1$ runs. A creation uses all $j-1$ mergers and is one interval H of j runs with span $L+1$. Every term extracting an extra internal h_1 has smaller weight and is killed. With $\nu_G = -(V_L^0 \overline{Q}_{A_G})(-1) \geq 0$, the exact normalized jet is

$$\frac{(V_L Q_C)_j(-1)}{P_0^{\mathrm{jet}}(-1)} = B_j - C_j, \quad B_j = \sum_{|G|=j+1} \frac{\nu_G}{\prod_{i \in G} \kappa_i}, \quad C_j = \sum_{\substack{|H|=j \\ \sigma(H)=L+1}} \frac{1}{\prod_{i \in H} \kappa_i}. \tag{B.12}$$

The minus sign is the actual change of bridge-count phase. If H contributes to C_j and G extends it by the adjacent run A_b, then $\nu_G \geq 2\kappa_b$. Fully merge A_H with the first or last block of every surviving coarsening of A_b; that block has weight at least two, so each resulting weighted charge is at least two. Each extension has at most two such high preimages. One preimage gives strict surplus; two give at least their sum. An internal high interval has two extensions. If all high intervals are at the endpoints, the inequality $c+1 \geq j+2$ ensures that a two-preimage extension would introduce an internal high interval. Thus a nonempty C_j always gives $B_j > C_j$. The weighted zero excludes those high intervals and then every remaining active length-$j+1$ interval. The Hall zero excludes length-$j+1$ span L. This proves (B.11) by induction from the zeroth jet; the same calculation at jet r proves the exact-valuation assertion. The converse follows directly by substituting the bounds in (B.10).

The minimum ribbon is $\beta_0 = (R_1 - 1, \dots, R_{c+1} - 1)$. A contributing block above L must merge at least $r+1$ of its rows; a block equal to L must merge at least r rows and its Hall removal loses one factor. Hence $\deg_{\mathrm{ord}} W \le c + 1 - r = a + 1$. At that degree evaluate every H_j at -1. Its entire coefficient is

$$\#\{I : |I| = r,\ \sigma(I) = L + 1\} - \sum_{\substack{|I|=r+1 \\ \sigma(I)\ge L+2}} (\sigma(I) - L).$$

The same extension argument, now with unit requirements, makes this strictly negative whenever exact valuation r holds. Therefore $\deg_{\mathrm{ord}} W = a+1$, and (B.9) makes β have exactly $a+1$ rows.

At $r = 0$, $a = c \ge 1$, the nonzero projection of $V_L Q_C$ at $h_1 = 0$ forces some actual zero-gap run to have span at least $L + 2$. Acting on its unmerged row in β_0 preserves all $c + 1$ factors. These top-degree terms have one sign and cannot be canceled by a coarsening with fewer factors. Thus again β has $a+1$ rows. At $a = 0$, Q_0/h_1^c has weight R and its unique ordinary-degree-one term is h_R. The entire minimum-layer identity is therefore $W = m h_{R-1}$, so β is a single row directly. No strict-surplus argument is used in this boundary case. At $R = 2$ the vertex has area one; it excludes $a \ge 1$, and the $a = 0$ argument applies.

B.4. **The original unit records and simultaneous cut signs.** Write $X_0 = X \backslash Z$, $Y_0 = Y \backslash Z$, and set $\mathcal{S} = Y_0 \cap (0, d_0)$, with $|\mathcal{S}| = a$. Vertex saturation gives the actual proper-cell matching

$$\mathbf{1}_{Y_0}(e) = \mathbf{1}_{Y_0}(L + e), \qquad 0 < e < d_0, \tag{B.13}$$

and, for $f < e < g$, matching of both unmatched beta lists at distance L. The original units, however, have not yet been synchronized.

For a live physical cut $f < t \le g$, put $Z_t = Z \cap [t, L + t)$, $c_t = |Z_t|$, and $D_t = s - c_t$. Every marked interval cycle contributing to N_t must traverse precisely these forced zero-gap sites. Formula (B.2) therefore gives $\deg_{\mathrm{ord}} N_t \le D_t$. At degree D_t, delete exactly these sites and shave the retained rows. The resulting actual board B_t has one central ribbon covering $[t-1, L+t-1]$; its other ribbon components and empty slots are spectators. The exact top cut is its corrected central vertex $N_t(B_t) + O_{t-1}(B_t)$ times the whole spectators. Here $B_t = B_{Z_t}$, and the cut uses $Q_{B_t}^{[D_t]}$, retaining all D_t slots, including the empty spectator slots.

That vertex is nonzero. The only possible blind at its lower endpoint would require $t-1 \in Z$ and $L+t-1 \in Y_0$. At a strict interior endpoint the latter and (B.13) contradict the former. At $t - 1 = f = 0$, the original bottom endpoint cannot be a common beta. At $t - 1 = f = 1$, the blind would give $d_0 - 1 \in P_0$ and $1 \in Q_0'$ beside $d_0 \in P_0$, contrary to the forbidden prefix/suffix sum $d_0 + 1$. The upper blind similarly requires $L + t \in Z$ and $t \in X_0$. Interior matching excludes it. At $t = g = R$, the top endpoint cannot be a unit; at $t = g = R - 1 = d_0$, the corresponding prefix/suffix contradiction is the reflected one. Thus the bound is attained at every live cut.

Let the central ribbon have lower beta endpoint b_0 and upper endpoint $L + A_0$, with $0 \le b_0 \le t - 1 \le A_0 \le d_0$. Neither b_0 nor A_0 belongs to $\mathcal{S}$: at a lower boundary this would identify a unit with an unmatched inner beta; at an upper boundary (B.13) would identify its proper-cell outer beta with the required unit. The central ribbon row count is exactly

$$c - |\{e \in \mathcal{S} : e \le b_0\}| - |\{e \in \mathcal{S} : e \ge A_0\}| + 1.$$

Its corrected vertex has $1 + |\mathcal{S} \cap (b_0, A_0)|$ slots, including one slot when its width is zero. Hence the loss of slots is

$$c - a = r. \tag{B.14}$$

This count does not subtract a possibly negative "middle inventory". When $L < d_0$, the end sets $\mathcal{S}$ and $L + \mathcal{S}$ may overlap, but the two regions actually removed from this central component are disjoint. Each original proper cell is counted once. The same argument handles $L = d_0$ and its coincident boundary point.

If the central ribbon has s_t rows, its complete top cut product has phase $(-1)^{D_t + s_t - 2}$ times its normalized vertex ribbon and its raw spectator determinants. If a resulting diagonal has ℓ_t

positive parts and e_t empty slots, its coefficient is $(-1)^{\ell_t+e_t-1}$. By (B.14), the total output slot count is $D_t - r$, so this coefficient is precisely

$$(-1)^{D_t-r-1}. \tag{B.15}$$

The squared-row-cost minimum of each such product is unique, with unit coefficient of that sign; every nontrivial ribbon coarsening increases that cost. The same ordering holds before the fixed-degree shave, since the cost changes by one common degree-dependent constant.

If the sliding unit count c_t varied, the maximal degree $D_* = s - \min c_t$ would occur on fewer than $m = g - f$ live cuts. Choose a globally least-cost diagonal at that degree. All its aliases have the same sign by (B.15); no off-diagonal term from any cut can have that cost. Its coefficient has absolute value between one and $m - 1$, contradicting the original integrality of $V_L Q_C/m = Q_0$. Thus c_t is constant, which is equivalent to

$$\mathbf{1}_Z(e) = \mathbf{1}_Z(L + e), \qquad f < e < g. \tag{B.16}$$

All original units have been used in this argument. The minimum shaved ribbon was not chosen as a convenient replacement for their full output.

B.5. **Two actual strips and the connected predecessor.** Extend the original physical cuts by $N_0 = N_{R+1} = 0$. These values agree with their defining indicators, since $0 \le y$ and $x - L \le S - L = R$. Let $\mathcal{R}_x, \mathcal{C}_y$ be the actual raw row and column replacements of Lemma 8.1 with $q = L$, in the original matrix orders. Set a replacement to zero when its indicated source label is absent.

In the full marked-entry formula for N_t, increasing t from e to $e+1$ adds the boundary $y = e$ and removes the boundary $x = L + e$. The corresponding complete cofactor sums are precisely $\mathcal{C}_e$ and $\mathcal{R}_{L+e}$. With the one original canonical phase this gives

$$N_{e+1} - N_e = (-1)^{s-1}\big(\mathcal{C}_e - \mathcal{R}_{L+e}\big), \qquad 0 \le e \le R.$$

When $x - y = L$, both boundaries occur together and their created $h_0 = 1$ contributions cancel in this difference. For $x - y < L$ the shifted entry is a negative zero. Thus the calculation retains every genuine creation and every original cofactor; empty slots in a resulting weak board do not change the original phase.

The exact indicators for an outer and an inner L-strip at endpoint e are respectively

$$\mathbf{1}_X(L + e)(1 - \mathbf{1}_X(e)), \qquad \mathbf{1}_Y(e)(1 - \mathbf{1}_Y(L + e)). \tag{B.17}$$

A nonduplicate bead move is an actual weak strip removal: crossing rows uses the original connected inequalities, and the final inserted label has the required weak ordering. The unmatched lists satisfy the matching supplied by (B.9). At a strict endpoint $f < e < g$, (B.17) therefore records exactly one strip when the two unit bits disagree, and none otherwise. At the left extreme exactly one strip occurs at f; when $f = 0$ its sheet is selected by the shared high endpoint, and when $f = 1$ by the shared low endpoint. Rotation gives exactly one strip at g. There are no records outside $[f, g]$. By (B.16), these are the only two strips on both sheets together.

The jump identity now has nonzero terms only at f and g. At f there is exactly one strip, so write its signed actual remainder as

$$P_{\mathrm{rem}} = (-1)^{s-1}\big(\mathcal{C}_f - \mathcal{R}_{L+f}\big).$$

Equality of the two complete Hall sums makes the jump at g equal to $-P_{\mathrm{rem}}$. Starting from $N_0 = 0$, telescoping gives $N_t = P_{\mathrm{rem}}$ for $f < t \le g$, and $N_t = 0$ outside that interval. This applies whether the two strips lie on the same sheet or on different sheets; no strip is separately observed. Using (B.3), the original weighted-current identity, and $g - f = m$ from (B.8), one obtains

$$mQ_0 = V_L Q_C = \sum_{t=1}^{R} N_t = (g - f)P_{\mathrm{rem}} = mP_{\mathrm{rem}}.$$

Since $m \neq 0$, the entire signed actual remainder is exactly Q_0.

Finally this is the function of an actual, possibly disconnected, strip remainder. Its nonzero affine h_R top permits exactly one connected component of span R. Every other component factor divides h_1^c and is a cell. The exact valuation r removes exactly r such factors, leaving an actual connected J with $Q_0 = h_1^r Q_J$ and $\partial_{h_R} Q_J = h_1^a$. Its area is $R+a$, and $(S+c)-(R+a) = L+r > 0$. This completes the proof of Theorem B.1. □

B.6. **Application to the original aligned comparison.** If C^- is cell-only, write $\kappa(C) = h_1^c$. The full image and Lemma A.2 give $\Gamma = \partial_{h_R} P = h_1^a$ with $0 \le a \le c$. Put $r = c - a$ and

$$L = (M-1)k, \qquad R = k + b, \qquad N_C = Mk + b = L + R.$$

To see the two current coefficients directly, let $\mathcal{C}_D$ be the proper prefix-cut set of the row composition of the actual ribbon D, and put $\epsilon_D = 1 - \mathbf{1}_{1 \in \mathcal{C}_D} - \mathbf{1}_{M-1 \in \mathcal{C}_D}$. The complete ribbon coarsening expansion gives

$$V_{M-1} Q_D = h_1, \qquad \delta_{M-1} Q_D = \epsilon_D h_1. \tag{B.18}$$

Only the one-block term and the cuts at $1, M-1$ can contribute. At $M = 2$ these are two occurrences of the same cut, and differentiation of h_1^2 counts both.

Write $\sigma_T = (-1)^{r(T)-1}$, and let A_s, B_s, C_s and A_f, B_f, C_f be the mark, hatted-mark and hatted-middle factors of the source and fine charts. Their complete canonical normalizations are

$$\Phi_\omega(Q_D) = \Lambda_s Q_F, \quad \Lambda_s = \sigma_D \sigma_F \epsilon_\omega(D) A_s^{M-r(D)} B_s^{M-c(D)} C_s^{M-N_D},$$
$$\Phi_\eta(Q_C) = \Lambda_f Q_F, \quad \Lambda_f = \sigma_C \sigma_F \epsilon_\eta(C) A_f^{|C|-r(C)} B_f^{|C|-c(C)} C_f^{|C|-N_C}.$$

The prescribed outer involution is carried on each coarse polynomial and its normalization together. The source factors have span below p, and the fine factors have span below q; negative exponents are read in the corresponding fraction fields. The source corner formula gives

$$\frac{\kappa(F)}{\kappa(E)} = A_s^{r(D)-1} B_s^{c(D)-1} C_s^{M-1}.$$

In the horizontal ordinary or positive mixed source convention put

$$\zeta_s = \begin{cases} \xi_s/(A_s B_s C_s), & \text{ordinary}, \\ \chi_s/(A_s B_s C_s), & \text{mixed}, \end{cases} \qquad \varepsilon_s^{\mathrm{ray}} = \frac{\sigma_E \zeta_s^{M-1}}{\sigma_D \sigma_F \epsilon_\omega(D)} \in \{1, -1\}.$$

The source covariance and (B.18) then give

$$V_{(M-1)p} Q_F = a_s \frac{\kappa(F)}{\kappa(E)} Q_E, \qquad (a_s, b_s) = \begin{cases} \varepsilon_s^{\mathrm{ray}}(p + \theta_s \epsilon_D, \epsilon_D), & \text{ordinary}, \\ \varepsilon_s^{\mathrm{ray}}(p + \rho_s \epsilon_D, 0), & \text{mixed}. \end{cases}$$
$$\delta_{(M-1)p} Q_F = b_s \frac{\kappa(F)}{\kappa(E)} Q_E,$$

In particular $a_s \neq 0$, since $|\theta_s| < p$, $0 < \rho_s < p$, and $\epsilon_D \in \{-1, 0, 1\}$. Whole antitransposition leaves a_s unchanged and negates b_s, by the physical involution identities in [10, Composition covariance]. An outer-only gauge instead carries its coarse polynomial and normalizer together.

Use the zero-offset fine convention and put $K = N_C$. Write L_E for the low factor called L in (A.1), since here $L = (M-1)k$. For the actual fine row probes put $r_j = r(\mathrm{Row}_j \circ_\eta U)$. Their fresh coefficients satisfy

$$c_j = (-1)^{r_j - 1} A_f^{j-1} \Theta_\eta C_f^j B_f^{j-1} = c_1 \xi_f^{j-1}.$$

The last equality follows from the constant row increment and its sorting phase, as in the row-probe calculation following (A.8); it also holds in the horizontal empty reading. Differentiating the parent and inner-source images at their largest fine pivots gives

$$c_K s_U^c = \Lambda_f \kappa(F), \qquad c_R s_U^a = L_E \sigma_E \kappa(E).$$

Thus

$$\frac{\kappa(F)}{\kappa(E)} = \frac{c_K L_E \sigma_E}{\Lambda_f c_R} s_U^{c-a}.$$

Every factor outside this power of s_U is below q. The exponents c, a are therefore its actual high-prime multiplicities in the two proper corners. Since $K - R = L$ and $\Phi_\eta(h_1) = s_U$, fine covariance gives

$$q\xi_f^L \Phi_\eta(V_L Q_C) = a_s \frac{c_K}{c_R} \Phi_\eta(h_1^{c-a} P),$$
$$\xi_f^L \Phi_\eta(\delta_L Q_C) = b_s \frac{c_K}{c_R} \Phi_\eta(h_1^{c-a} P).$$

The sign in $Q_E = \sigma_E s_E$ cancels the sign in the top-derivative identity. Finally $c_K/c_R = \xi_f^L$, so injectivity in the independent fine pivots gives

$$V_L Q_C = \frac{a_s}{q} h_1^{c-a} P, \qquad \delta_L Q_C = b_s h_1^{c-a} P.$$

Put $m = a_s/q$ and $\alpha = b_s$. Since $\partial_{h_R} P = h_1^a$, the coefficient of $h_R h_1^c$ in $h_1^{c-a} P$ is one. Hence

$$m = [h_R h_1^c]\, V_L Q_C \in \mathbb{Z}, \qquad \alpha = [h_R h_1^c]\, \delta_L Q_C \in \mathbb{Z}.$$

The source weighted output is nonzero, so $m \neq 0$. The ray is primitive, has span R, derivative h_1^c, and exact h_1-valuation $c - a$: P is irreducible of span $R \geq 2$, so it is not associated to h_1. Every hypothesis of Theorem B.1 is therefore supplied by the original comparison. The resulting actual J realizes the whole P. Together with Propositions A.4 and A.6, this exhausts the fine outer interior and completes Proposition 4.1.

Appendix C. Empty overlaps and zero span remainder

C.1. **The whole seam comparison at empty overlap.** An empty chart is always read with its actual horizontal edge convention. For a connected inner W of span d, the full flag and area laws are

$$X_{A\circ W} = dX_A \,\dot\cup \bigcup_{j=0}^{N_A - 1} (jd + I_W),$$
$$Y_{A\circ W} = dY_A \,\dot\cup \bigcup_{j=0}^{N_A - 1} (jd + C_W),$$
$$|A \circ W| = d|A| + N_A(|W| - d).$$

The selected common betas are the empty seams; every other common beta lies inside its actual inner copy. The map has normalization

$$\Phi_W(s_A) = \kappa(W)^{|A| - N_A} s_{A\circ W}.$$

These formulas follow from the actual copy origins, with all original unit flags included.

Lemma C.1. *Suppose both original marks are empty, $p = Ag$, $q = Bg$, $\gcd(A, B) = 1$, $A > B \geq 2$. Then the initially arbitrary C is a ribbon. There are integers $\mu, \ell \geq 1$ such that*

$$|D| = \mu B, \quad |C| = \mu A, \quad N = \mu ABg, \quad |E| = A\ell, \quad |U| = B\ell, \quad n = \mu AB\ell.$$

After a simultaneous actual transposition if needed, there are ribbons T, T' of μ cells with $s_T = s_{T'}$ such that

$$D = T \circ_\varnothing \mathrm{Row}_B, \qquad C = T' \circ_\varnothing \mathrm{Row}_A\,. \tag{C.1}$$

The original g-body seams inside E, U are all near joins. Within either native, all bodies have one literal zero-free core, but retain their own original internal unit sets.

Proof. Take the actual marked common-beta cores. Since D is a ribbon, its core is a row, and the full lowest ordinary component gives an actual equality

$$\mathrm{Row}_{|D|} \circ E_0 \sim \mathrm{Core}(C) \circ U_0.$$

Its zero-free class has full fibers by Proposition 3.7, not by a current-area induction assumption. Apply the fresh-factor descent of Lemma 4.3. In this empty setting every fresh minimum has width zero, since a displayed empty chart is available. The literal empty flag law makes all stripped factors zero-free, and horizontal empty associativity retains the displayed row outer

and the displayed Core(C). If a line factor is encountered, it is a row. Its full line fiber places it on the other parent, and the two actual flags $i, i+p$ inside a proper row power force its period to divide p whenever the native has a proper flag. If it has none, that native is a row already. Thus the descent terminates with two row natives; row-count equality makes Core(C) a row. Common-beta deletion preserves area and span, so $|C| = N_C$ and C is a ribbon.

All nontrivial strips divide the gcd and have strict-area quotient parents. Reattaching the actual empty factors on each side gives

$$E_0 = \mathrm{Row}_A \circ V_E, \qquad U_0 = \mathrm{Row}_B \circ V_U, \qquad N_{V_E} = N_{V_U} = g.$$

The terminal row factor is absorbed into each body; an internal one-cell identity is not used as a proper chart. The span and area laws give the stated integer parameters and $|V_E| = |V_U| = \ell$. Restore the original common betas. In each open g-band the full lists are $(g, I_{V_E} \cup Z_j)/(C_{V_E} \cup Z_j, 0)$, or the corresponding V_U lists. They are actual connected bodies with unchanged height and area. No body has been replaced by an arbitrary Schur mate.

Assign degree $\lfloor j/g \rfloor$ to h_j. In the whole top degree, a matching has no residue inversion. The residue-zero rows exhaust residue-zero columns; every other nonnegative label has nonnegative band displacement, and the total of those displacements is zero. Consequently every nonselected edge stays in its original body band. The complete top component is

$$\mathrm{Top}_g\, Q_S = \left(\prod_i \kappa(B_i) \right) Q_{R_S}(h_g, h_{2g}, \ldots), \tag{C.2}$$

where R_S is the actual ribbon recording the selected seams. The original whole proper equality gives the same entire product $\kappa(E)^{\mu B} = \kappa(U)^{\mu A}$ on the two sides. Cancel it in (C.2). This gives an actual ribbon equality

$$R_F = D \circ_\varnothing A_0 \sim R_H = C \circ_\varnothing B_0, \qquad |A_0| = A, \quad |B_0| = B,$$

of area $\mu AB < n$ unless $\ell = 1$. That exceptional case has one-cell bodies, hence $g = 1$ and original ribbon parents, already handled by Lemma 3.4. No same-area seam induction occurs.

Choose a fresh minimum in the actual seam class. Its width is zero. A nonline minimum would divide both A and B by full smaller-area transport and Lemma 3.5, which is impossible. A line minimum of period d has an actual prime-period line factor, of some prime $\ell_0 \mid d$. The resulting chart is proper and its full line fiber transfers it across the class. The prime is coprime to at least one of A, B; same-parent coprime gluing makes that displayed native a line. Its full line fiber then makes the other displayed native a line as well. Thus we have not assumed that a line minimum divides every period. Opposite orientations cannot occur: a row-A action gives at most μB rows, while a column-B action gives at least $\mu A(B-1)+1 > \mu B$ rows. The reflected inequality handles the reverse.

In the row orientation, the two actual line periods on R_F make its ribbon bits vanish outside both the multiples of A and the multiples of B. Hence only multiples of AB remain. Reading the original rows then gives (C.1); faithfulness gives $s_T = s_{T'}$. The same flags show that the original internal g-seams are near joins. For the column orientation transpose both whole actual charts. If $v_H(W)$ is the horizontal empty displacement, then $v_H(W^{\mathsf{t}}) = e - v_H(W)^{\mathsf{t}}$, and the copy origins verify

$$(A \circ_H W)^{\mathsf{t}} = A^{\mathsf{t}} \circ_H W^{\mathsf{t}}.$$

This sends the grid seams to the row orientation and retains all internal body units. It proves the stated reduction. □

Lemma C.2. *If X is any connected inner of span d and T is a ribbon of $m \geq 2$ cells, with horizontal empty composition, then*

$$V_{(m-1)d} Q_{T \circ X} = d\, \kappa(X)^{m-1} Q_X. \tag{C.3}$$

Proof. Put $\sigma_X = (-1)^{r(X)-1}$. The empty specialization of composition covariance [10, Composition covariance] has $\theta = 0$ and $\Phi_X(h_1) = s_X$. Its Hall identity at h_1 gives $\xi = \delta_d s_X = \sigma_X \kappa(X)$, with $\delta_1 h_1 = 1$. Consequently, for $s \geq 1$, $V_{sd}\Phi_X = d(\sigma_X \kappa(X))^s \Phi_X V_s$. For a ribbon,

$V_{m-1}s_T = (-1)^{r(T)-1}h_1$. Its empty normalizer exponent is zero, and the full row phase is $\sigma_{T\circ X} = (-1)^{r(T)-1}\sigma_X^m$. Substitution cancels both selector signs and proves (C.3), with every native unit and proper factor. □

Proposition C.3. *The original comparison cannot have two empty marks and nondividing periods.*

Proof. Use Lemma C.1; the coprime case has already been excluded. If $\mu \geq 2$, put $X = \mathrm{Row}_B \circ E$ and $Y = \mathrm{Row}_A \circ U$, both of span ABg. The literal reassociations are $F = T \circ X$, $H = T' \circ Y$. The entire proper equality gives $\kappa(X)^\mu = \kappa(Y)^\mu$, hence $\kappa(X) = \kappa(Y)$ by the positive physical associate. Apply (C.3) to the same original whole equality. Cancel the common nonzero factor to obtain $s_X = s_Y$. Now, and only now, use $|X| = |Y| = n/\mu < n$. Their old p- and q-charts are minimal by Lemma 3.5, applied to the original nonline natives. Lower key invariance contradicts $p > q$.

Suppose $\mu = 1$. Both original outers are rows in the chosen physical frame. Write the B successive bodies of U as $B_1, \ldots, B_B$, with near joins and the same literal zero-free core. On the original common function define $\mathcal{L}_s = V_s - (N-s)\delta_s$. Its two row-power identities give $\mathcal{L}_p f = \mathcal{L}_q f = 0$, and

$$[\mathcal{L}_s, \mathcal{L}_t] = (s-t)\mathcal{L}_{s+t}.$$

Choose $i \equiv A^{-1} \pmod B$, $1 \leq i < B$, and $j = (AB - B + 1 - iA)/B$. Then $1 \leq j < A$ and $ip + jq = N - (B-1)g$. Starting with the bracket at p, q and adding p or q successively produces this annihilator; each coefficient is nonzero, since the accumulated positive index is larger than either generator. Every index stays within the original horizon.

The first and last $(B-1)g$ windows of the original fine parent are the actual near words

$$P_- = B_1 \odot \cdots \odot B_{B-1}, \qquad P_+ = B_2 \odot \cdots \odot B_B.$$

Their cut seams have no selected common beta, so the original endpoint coframes restricted to this range are their entire endpoint coframes. The normalized gradient identifies the annihilator just obtained with minus their full balanced endpoint current. They share a literal core. The contact theorem in Theorem E.3 makes the windows equal literally. Equivalently its leading coefficient is

$$-\sum_{Z_{P_-} \setminus Z_{P_+}} z F_z G_z - \sum_{Z_{P_+} \setminus Z_{P_-}} ((B-1)g - z) F_z G_z,$$

with the shared-unit creation already canceled by the two actual kernel equations. All products have the same endpoint sign. Reading successive windows gives $B_1 = \cdots = B_B =: B_0$. Thus $U = \mathrm{Row}_B \circ B_0$ and $H = \mathrm{Row}_{AB} \circ_\varnothing B_0$ literally. This is a proper period-g chart on the same original H, with $(0, g) < (0, q)$, contradicting its own minimum. The physical transpose used in the column branch is undone on this chart. □

C.2. **The zero-remainder geometry.** The next argument covers two cases: both original outers already known to be lines, with $g \mid N$ and at least one positive mark; and exactly one empty and one positive original mark, where the competing outer is still arbitrary. It does not apply the positive-remainder fold at $c = 0$.

Proposition C.4. *Each of the following original nondividing comparisons is impossible:*

1. *both outers are lines, $g \mid N$, and at least one mark is positive;*
2. *exactly one mark is empty and the other is positive, with the original larger-period outer a ribbon and the competing outer initially arbitrary.*

Proof. In the second case the span equation gives $g \mid N$. A positive coincident mark is impossible by Lemma D.2: it would have width strictly between zero and g, but its width is divisible by g. Thus the positive mark is distinct. The empty-side outer is a ribbon. If it is the original source, this is given. Otherwise Corollary 4.6 excludes a nonline distinct source; on its remaining line branch Proposition 4.4(ii) proves that the empty-side outer is a ribbon. This supplies the required role-sensitive entry before any whole-parent structure is asserted.

Take actual whole cores and fresh factors. Every lower width is zero, since every stripped period divides $g \mid N$. At a zero-width lower factor the covariance has no offset correction, so ordinary/mixed type is retained. A mixed terminal native cannot be a line in a compatible common axis. At the terminal row-native comparison, in case (i) the original outers are still lines up to their full transports. One row parent versus a column parent violates row-count invariance; two column parents force equal widths and then equal periods. Thus both whole terminal parents are rows. In case (ii) the empty-side core outer is a row. Its terminal whole parent is a ribbon. A positive-width row action on a multirow outer has positive adjacent gaps and cannot be that ribbon; therefore the positive-side terminal outer is a row and both whole terminal parents are rows.

If the product of stripped periods is a proper divisor D_0 of g, strip the remaining actual row factor of length g/D_0 at this known terminal row. Its outer remains proper because the original copy counts are at least two. This is a supplied factor of a known row, not a new minimum argument. Propagating back up the actual empty-factor chains gives whole $\mathcal{P}(g,0)$, private singleton pins at $jg, (j+1)g-1$, and proper component spans below g. Restore the original common betas. Heights and proper-component support intervals remain, so the original whole proper products are low below g.

At every positive grid position the two full flag bits are both present or both absent, since the two neighboring diagonal heights are one. All positive proper components lie strictly in the private bands. Taking the whole floor-g component and dividing by the entire private proper product gives the actual selected-grid ribbon $(L,Z)/(Z,0)$, where $L = N/g$. No positive across-band entry is suppressed on this residue-zero block. Equality gives an actual Schur-equal pair of such ribbons.

On a positive original side its width is $kg \geq g$. A positive component of its outer interior would give a proper prime of span at least its period, contradicting the whole low corner. Its outer is a ribbon. If that outer had at least two rows in the genuine horizontal convention, its connected mark function of span kg would occur in the proper corner, again impossible. Thus that outer is a row. Its actual singleton pins force $t = 0, s = w$: a tail crossing either private endpoint would increase that singleton's height. The complete row coding therefore makes its selected-grid ribbon an actual positive coprime line-inner composition. The empty side's grid composition retains its original outer.

The selected ribbon class has area $L < n$, since a positive mark supplies additional cells. Its coprime comparison and the full smaller-area/line fiber argument of Lemma C.1 force the two selected natives into the same line orientation. The positive-side outer is a row, so the whole selected ribbons are both rows or both columns. In case (ii) this also forces the originally empty-side outer to be a line. Thus the arbitrary competing outer has been resolved, rather than assumed.

All grid bits are now uniformly absent or uniformly present. The actual attachment bits at w, p make positive charts ordinary, with the corresponding same-axis selector; a mixed attachment is incompatible. In the absent-grid frame, both genuine ordinary clocks are $N-p, N-q$. For the present-grid frame take the full physical antitransposition and the genuine empty-edge gauge to reach this same frame. All cap and operator data are rebuilt.

Put $m = N - q = (L-B)g$. The original first and last m windows are actual connected diagrams with identical cores. The cuts are singleton seams with no selected unit; their endpoint coframes are exactly the restricted source coframes. Consequently the entire observed equation $\mathscr{D}_q(N-q)f_S = 0$ is minus their balanced endpoint current. Theorem E.3 makes the windows literally equal. This supplies all private q comparisons. The original period supplies the p cycle relation, and $L \geq 2\max(A,B) \geq A+B$ gives the complete finite range. All g-bodies agree literally. Hence the same original parent is $\mathrm{Row}_L \circ_{\varnothing} T$ for an actual span-g inner T. The chart is legal and proper, and its score $(0,g)$ improves the original score. Undoing the full transports proves the contradiction in both cases. □

Appendix D. Positive coincident attachment phases

The original phases, rather than phases of a subsequently chosen factor, define coincidence. A positive coincident mark is ordinary and has an internal common phase. The competing original mark can be ordinary or mixed. All results below retain the original two minima and $\mathsf{J}_{<n}$.

D.1. **The full-hat gauge and zero-offset descent.**

Lemma D.1. *If the full hatted mark of an ordinary chart is connected and has the same span as its mark, it is the other legal ordinary overlap on the same inner, up to the prescribed translation. The two gauges interchange the mark and hatted-mark functions and antitranspose the outer. They preserve the whole parent and its score. A connected full-span hat is not Schur-equal to its own mark.*

Proof. In case (a), write the original mark interval at residue r as $[\ell_r, h_r]$. The lower and upper intervals of the inner, with the upper one translated back, are

$$[\ell_r, h_r + \mathbf{1}_{r\geq s}], \qquad [\ell_r - \mathbf{1}_{r<t}, h_r].$$

For $v_b = v_a - e$, their full intersection is

$$W'_- = E \cap (E - v_b) = [\ell_r + 1 - \mathbf{1}_{r<t}, h_r + \mathbf{1}_{r\geq s}] = \widehat{W}_- + e.$$

The full-span hypothesis makes this a connected width-w end mark and forces $0 < t < w$, $0 < s < w$. The new middle has the northwest lower tail $[t, w-1]$, the unchanged private band, and the southeast upper tail $[q, q+s-1]$. The two inherited endpoint-step bounds make both tail/private joins singleton joins. Thus the middle and both one-mark complements are connected skew. The actual steps at t and s give vertical attachments. The hatted middle is still confined to $[w+1, q-2]$, two contents away from all periodic marked hats, proving IV.

Maximality is also checked across the possible displacements. The extreme cells permit only v_a, v_b. The old intersection at v_a does not contain the new last marked singleton; at v_b the displayed full intersection forbids enlargement. Separation is unchanged. This proves all I–IV. The inverse case-(b) construction uses $v_a = v_b + e$ and returns the old mark as the new hat. The copy origins give the literal identity $D \circ_a E = \mathrm{AT}(D) \circ_b E$, with every supplemental convention carried. Consequently the full normalization factors exchange their A, B roles and their row/column exponents, preserving the original whole identity.

For completeness, the case-(a) full hat has southwest cell translated by $-e$ and the same northeast cell, so it has one fewer row than W. The case-(b) full hat has one more row. A full-span component exhausts the entire hat and has multiplicity one, since distinct proper components have disjoint content intervals. Row count is a Schur invariant. Thus the connected mark prime never equals the full-hat prime. □

Lemma D.2. *At an original nondividing comparison with a distinguished positive coincident mark of width u, one has $2 \leq u < g$. The other reconstructed core mark is ordinary; if its actual offset is θ and its original width is w, then*

$$w \geq u, \qquad |\theta| = w - u. \tag{D.1}$$

Both original native proper corners have all component spans below g. If the distinguished side's actual outer is a ribbon, both original parents have $\mathcal{P}(g, u)$ and their entire proper corners are low below g.

Proof. The coincident native remains nonline after the actual marked-core construction. The core class has full fibers. A line alternative would make both displayed natives lines, impossible for a positive coincident mark. Choose a fresh common minimum of period d dividing both current periods. Its lower factor is ordinary: a mixed lower factor would make the upper-period Hall response zero, contradicting that of the ordinary coincident native. The true covariance gives

$$0 = \theta_{\mathrm{lower}} + \epsilon d \theta_{\mathrm{upper}}, \qquad |\theta_{\mathrm{lower}}| < d.$$

Integrality makes both offsets zero. If its lower width is v, the new distinguished width is $(u-v)/d = \lfloor u/d \rfloor$. If initially $u \geq g$, after any product D_0 of stripped periods the width remains at least $g/D_0 \geq 1$. The positive coincident nonline alternative would persist while the gcd strictly decreases to one, where no proper common period exists. Thus $u < g$. The actual phase is internal, so $u \geq 2$.

An ordinary and a mixed displayed core cannot persist together in a full class. Their fresh lower factor would be ordinary, and its nonzero Hall covariance and injectivity would preserve the nonzero and zero Hall responses of the two quotient natives, respectively. The actual quotient marks therefore remain opposite types. Neither could reach a line alternative, again contradicting descent of the gcd. This proves that the other reconstructed core mark is ordinary; it does not relabel an original mixed mark as ordinary.

Continue stripping zero-offset lower factors until the native line alternative. The distinguished terminal line has width zero. If D_0 is the product stripped, then $D_0 \mid g$, $u < D_0$, and the other terminal width is $(w-u)/D_0$. All lower offsets vanish, so the actual ordinary covariance gives $\theta = \pm D_0 \theta_{\text{terminal}}$. An actual marked line has absolute offset equal to its width, proving (D.1).

The corner span estimate propagates up each actual chain separately. A component of span $< D_{\text{rest}}$ maps through a period-d, width-v factor to span at most $d(D_{\text{rest}} - 1) + v < dD_{\text{rest}}$; the remaining local factors have span $< d$. Starting with a terminal line gives native proper-component spans $< D_0 \leq g$. For heights, a zero-offset ordinary lower chart has

$$H_S(dj + r) = \begin{cases} \Omega_r + H_A(j-1), & 0 \leq r < \sigma, \\ \Omega_r + H_A(j), & \sigma \leq r < v, \\ H_W(r) + H_A(j) - 1, & v \leq r < d. \end{cases}$$

Use the actual full-hat dual to read the horizontal convention; an empty factor has only the third line. These identities carry $\mathcal{P}(D_1, v_1)$ to $\mathcal{P}(dD_1, dv_1 + v)$. The dangerous preceding index at the lower marked band is outside the claimed middle, and the private endpoints give exactly the new singleton families. The width digit sum is u. Rotation and antitransposition preserve the corresponding height property.

When the distinguished original outer is a ribbon, its core outer is a row. At the terminal empty-line native its whole parent is a ribbon; the equal terminal parent is also a ribbon by area equal to span. Both have height identically one. The preceding height propagation gives $\mathcal{P}(g, u)$ for both whole cores. Restoring original units preserves height and all proper positive support intervals, so it preserves the component span bounds. The distinguished ribbon corner has only its native proper stock and its width-u local factors. It is therefore low below g; the whole equality gives the same corner on the other side. □

If both marks are coincident, the actual core source outer is a row and terminal row-count comparison also makes Core(C) a line. Area equal to span then makes the original C a ribbon. The two widths lie in $[2, g-1]$ and are congruent modulo g, hence equal. Write

(D.2)

$$p = Ag, \quad q = Bg, \quad \gcd(A, B) = 1, \quad A > B \geq 2, \quad M = \mu B, \quad J = \mu A, \quad N = \mu ABg + u.$$

This includes all possible original copy counts; the apparent below-first-common-multiple case is excluded by the width equation.

D.2. **The whole marked-band determinant.** An original coincident ribbon seam has whole height $\Omega_r + 1$ on its marked band. The g-periodic height propagates this to every internal marked band. Each such band is consequently one connected actual proper component B_i of span u. Private proper components lie in the disjoint private intervals. All original units stay in their own categories.

Assign h_j degree $\lfloor j/g \rfloor$. For a whole matching the exact degree is

$$L - \#\{x : x \bmod g < \pi(x) \bmod g\}, \qquad N = Lg + u.$$

The complete degree-L determinant is therefore obtained by zeroing just the residue-inversion entries. Group marked residues $\leq u$ and private residues $> u$. The marked-to-private block is

zero, and the private block is band triangular with its actual diagonal blocks. Thus

$$\mathrm{Top}_g\, Q_S = G_{\mathrm{private}} Q_{\mathcal{M}}, \qquad \kappa(S) = G_{\mathrm{private}} G_{\mathcal{M}}, \qquad \frac{\mathrm{Top}_g\, Q_S}{\kappa(S)} = \frac{Q_{\mathcal{M}}}{G_{\mathcal{M}}}. \tag{D.3}$$

The phase is fixed by the unique highest raw entry. The remaining matrix $\mathcal{M}$ still has a residue mask: some positive across-band entries were zeroed, so it is not declared an actual Schur matrix. No induction is used on it.

Keep only $x_j = h_{jg+u}$ among the high variables, while keeping every low variable physical. Across different bands the only remaining port has row residue u and column residue 0. Its actual signed inverse ratio is

$$q_i = (M(B_i)^{-1})_{0,u} = \frac{\partial_{h_u} s_{B_i}}{s_{B_i}}.$$

All other low rows and columns are eliminated by their complete inverses. The two endpoint factors cancel in the full proper normalization. Expansion of the ordered block inverse gives the entire polynomial

$$\mathcal{W}_L(x;q) = \sum_{T\subseteq[1,L-1]} (-1)^{|T|} \left(\prod_{i\in T} q_i\right) \prod_{\text{runs cut by } T} x_{\text{run length}}. \tag{D.4}$$

The two parents have the same (D.4), by (D.3). This is an entire specialization of an observed identity, not an assertion of independent end data.

Lemma D.3. *For the two-coincident comparison, the two unordered complete mark/hat pairs agree. After genuine full-hat gauges both charts are ordinary (a) and their functions are the same distinct primes $A = s_W$, $B = s_{\widehat{W}}$. Every internal marked component has class A or B. The corresponding indicator ribbons have actual coprime charts of periods $A_0 = p/g$ and $B_0 = q/g$, whose natives are lines in the same orientation.*

Proof. Use the original common terminal Hall operator δ_{N-u}. On each actual ribbon outer it reads the unique highest generator. Divide its whole output by the common proper corner. The complete local products give

$$\frac{A_s B_s}{\Theta_s} = \frac{A_f B_f}{\Theta_f}.$$

The two mark and full-hat primes have span u, whereas every factor of Θ has smaller span. Unique factorization and positive primitive associates give equality of the unordered pairs and of Θ. Their row counts differ by one, by Lemma D.1; the larger-row member is the genuine case-(a) mark on both sides. The lawful full-hat gauge aligns the pair without making the two diagrams literal.

At an original source seam, the horizontal step contributes class B and the vertical step class A. For a prime f outside this pair, let $n_E(f), n_U(f)$ be its counts among the internal native g bands. The coefficient of x_1^L in (D.4) is $(-1)^{L-1}\prod q_i$. Its valuation at f counts its occurrences, since every proper numerator has span below u. Therefore

$$Mn_E(f) = Jn_U(f), \qquad B_0 n_E(f) = A_0 n_U(f), \qquad 0 \le n_E(f) < A_0, \quad 0 \le n_U(f) < B_0.$$

Coprimality forces both counts to vanish. Every internal class is A or B. Their port ratio q_A/q_B is nonconstant, since its valuation at one of these distinct span-u primes is nonzero. It is transcendental over $\mathbb{Q}$.

The coefficient of an x monomial of degree d in (D.4) is a homogeneous integer polynomial of degree $d-1$ in q_A, q_B. Divide by q_B^{d-1} and use that transcendence. The equality lifts to the formal homogeneous polynomials in two symbols; algebraic independence of the two actual ports is not asserted. Setting those symbols to $1, 0$ gives the entire canonical polynomial of the actual L-cell ribbon cut precisely at class-A positions. The original seam assignment gives literal charts through A_0- and B_0-cell ribbon natives. These are proper, of coprime periods, and of strict area $L < n$. The fresh-minimum/line-fiber argument in Lemma C.1 makes both natives lines in the same orientation. □

For $\mu \geq 2$, this supplies an actual line factor V_{B_0} of the original D: the row orientation gives B_0-divisibility of every actual row length, and the column orientation gives its genuine column version. The remaining outer has μ cells. Regroup through the original mark as in Lemma 4.5, and apply aligned reconstruction at the new period $B_0 p = A_0 q$ before its minimum corollary. The real inner $V_{B_0} \circ E$ retains its old minimal period p and has area below n; the constructed original-U reference is proper, has old minimal period q, and has the same whole function. Lower key invariance gives a contradiction. At $\mu = 1$ the remaining outer would be a cell, so this argument is not used. Instead it supplies actual same-selector line outers of B_0, A_0 cells, respectively. Full transports put both in the ordinary-(a) row frame, where all internal marked components have the single class B. The scalar argument below treats this primitive case.

D.3. **Exactly one coincidence: a prime indicator gives line entry.**

Lemma D.4. *In an original nondividing comparison with exactly one positive coincident mark, let its width be u and let the other width be w. Then $2 \leq u < g$, $w = u + kg$ with $k \geq 1$. The wider side has an original row outer and actual phases $s = t + kg$, $0 \leq t \leq u$. The coincident-side outer is a line. All internal marked components on both parents have one common entire Schur class.*

Proof. If coincidence is on the larger-period side its ribbon outer is given. If it is on the finer side, Corollary 4.6 removes the nonline distinct source, and Proposition 4.4(ii) makes the coincident-side outer a ribbon on the remaining line branch. Only now apply Lemma D.2 to obtain whole $\mathcal{P}(g, u)$ and a whole low proper corner. Core construction retains the two phase positions up to interchange, so a distinct pair cannot become coincident. Equation (D.1) gives $w = u + kg$, $k \geq 1$.

Write the wide period as ag and the coincident period as bg, without ordering a, b. A positive outer-interior component on the wide side would give a prime of span at least $ag + w > g$ in the whole corner. Its outer is therefore a ribbon. If it had two rows, its mark prime of span $w > g$ would occur in that corner. Hence the outer is the original Row_M. The core phase balance gives $|s - t| = kg$. For an ordinary mark, $t > s$ would make its full hat positive throughout width w, by the interval formula $\Omega(r) - 1 + \mathbf{1}_{r<t} + \mathbf{1}_{r \geq s}$. That connected prime occurs in a row-outer corner, contradicting the low bound. Mixed marks already have $t < s$. Thus $s = t + kg$ and $0 \leq t \leq u$. The row coding has only the initial t correction and terminal $Mag + s = Lg + t$ correction. Both are outside the internal marked bands, so their complete component words have period a.

Each marked band is one actual connected span-u component on both sides. On the coincident side a true original seam gives height $\Omega + 1$, propagated by $\mathcal{P}$. On the wide side strengthen the zero-offset height induction in Lemma D.2 by requiring height at least two on internal marked bands. It holds at the terminal width-zero base vacuously. Under a lower (d, v_0) factor, private positions in the new marked band read an outer marked height at least two and add a native height at least one minus one; marked positions read Ω plus a positive outer height. The final short marked portion has the latter form as well. These exhaust the band, including its preceding-private endpoint. Rotation, grouping and unit restoration preserve the assertion. Thus the full proper height is positive throughout the width-u band, with zero proper height at its adjacent singleton cuts.

Fix any connected span-u component prime $f = c_f h_u + d_f$. Localize $c_f \neq 0$ and set $f = \epsilon$, so $h_u = (\epsilon - d_f)/c_f$. A port of class f is c_f/ϵ; other ports are regular units in ϵ. In the entire identity (D.4), put $x_j = (\epsilon/c_f) y_j$ for all j, multiply by c_f/ϵ, and take the constant coefficient. Exactly the cut sets contained in the class-f positions survive. The result is the entire canonical polynomial of their actual indicator ribbon. Equal canonical functions have the same positive Schur associate.

On the wide row side its indicator word has period a. The first $a + k$ cells give an actual ribbon native with a width-k end mark and period a, and a proper row outer. The mark is the full intersection; its complements are connected intervals, $k < a$ gives separation, and its

periodic hatted sets are empty. On the coincident side the indicator native has b cells and empty overlap. In a genuine case-(a) gauge its outer is D_J for $f = s_W$, $\mathrm{AT}(D_J)$ for $f = s_{\widehat{W}}$, and Row_J for any other class. These descriptions are literal consequences of the seam assignment, not formal polynomial realizations.

The indicator class has strict area $L < n$, and its two actual periods a, b are coprime. Full smaller-area transport and the line alternative make both indicator natives lines in the same orientation. The positive wide chart with a row outer consequently gives a whole line indicator. A class outside $\{s_W, s_{\widehat{W}}\}$ would have coincident-side outer Row_J: it cannot yield a column indicator, and a row indicator has no cuts, contradicting the occurrence of the class. Choose any class that occurs. Its indicator has a cut, so it is the whole column. Therefore that class occupies every internal position on both sides. The actual empty line action on the coincident side now makes its original outer a row or column, as claimed. □

D.4. **One whole highest-pole coefficient synchronizes the mark bands.**

Lemma D.5. *Consider the original nondividing comparison in either the primitive two-positive-coincidence regime $\mu = 1$ of* (D.2)*, or the exactly-one-positive-coincidence regime of Lemma D.4. In its permitted ordinary-(a) row frame, let the coincident source S have period Ag, width $2 \le u < g$, phase $0 < \tau < u$, and total span $Lg + u$; the other original period is Bg, where $A, B \ge 2$ and $\gcd(A, B) = 1$. Retain the supplied whole $\mathcal{P}(g, u)$ height data, whole proper corner with all component spans below g, single marked-component Schur class, original A-periodic full slot word, and the complete comparison range $0 \le i \le r = L-B$, together with the held minimum on S. In the first regime $L = AB$ and $r = B(A-1)$. In the second, $L = mA = jB+k$, with the proper original copy counts $m, j \ge 2$ and $k \ge 1$; no ordering of A, B is imposed in this regime.*

The original Hall and negative-clock alternatives in (D.5) *are impossible. The positive-clock alternative constructs an actual proper legal period-g, width-u chart on the same original S, retaining its original marked end windows and attachments. Its score $(u, g+u) < (u, Ag+u)$ contradicts the held source minimum.*

Proof. We treat the primitive two-coincident and the one-coincident line entries together. Normalize the actual coincident side to an ordinary-(a) row composition, applying simultaneous whole antitransposition and then its lawful full-hat dual when needed. Let

$$S = \mathrm{Row}_m \circ_\omega E, \quad p_0 = Ag, \quad N = Lg + u,$$

with true coincident phase $0 < \tau < u$. The other side has period Bg. Write $r = L - B$. In the primitive two-coincident case $L = AB$ and $r = B(A-1) \ge A - 1$. In the one-coincident case $L = mA = jB + k$, $k \ge 1$, and $r \ge A$: if $A \ge B$, use $r \ge 2A - B$; otherwise use $r = (j-1)B + k > A$. All component slots $B_1, \dots, B_{L-1}$ have the class $\phi = s_{B_*}$, where B_* is the actual mark with its common beta τ deleted from both lists. Put $B_0 = B_L = B_*$. The original row chart gives the full A-periodicity of this slot word, and per-side height periodicity gives a common literal zero-free core.

The other original chart supplies one of exactly three whole equations:

$$\mathscr{D}_{Bg}(rg)Q_S = 0, \qquad \mathscr{D}_{Bg}(-rg)Q_S = 0, \qquad \delta_{Bg}Q_S = 0. \tag{D.5}$$

For two coincident row charts only the first occurs. For the one-coincident entry the wide row chart has width $u+kg$ and phase difference kg; its actual ordinary-(a) clock is $Bg(j-1)+kg = rg$. The actual ordinary-(b) transported ratio is its negative. An original mixed chart supplies the Hall zero. These are original clocks, not choices made in the residue calculation.

Apply the appropriate equation to the complete normalized marked-band quotient in (D.3) before specializing any high variable. The operator lowers floor degree by B and fixes the whole low proper product. Retain only $y_j = h_{g+j}$, $0 \le j \le u$, among the high variables. Each actual internal block has one simple ϕ pole, whose residue is its endpoint rank-one adjugate with the same signed proper coefficient. The actual cap cofactors have smaller span and are ϕ-units. At this prime the cap contact coefficient at τ is ϕ divided by a nonzero cap minor, hence zero; the remaining cap coframe is exactly the endpoint coframe of B_*.

The full finite inverse is the sum over increasing band paths. A surviving differentiated edge originally has length B or $B+1$, and every other edge has length one. A B-edge becomes low and uses r inverse component blocks; a $B+1$-edge becomes first-grade and uses only $r-1$. Hence the whole highest pole ϕ^{-r} comes from precisely the $r+1$ paths with a B-edge from i to $i+B$, $0 \le i \le r$. Each has r short edges. The coefficient of y_u in each short-edge contraction is one, from the actual endpoint normalizations. The phases of all paths agree, since they have the same number of actual equal-row-count inverse blocks.

Let $K_u(A,B)$ be the endpoint cross kernel, $f_A = K_u(A,A)$ its actual self-complement, and let $J_u(A,B)$ be the balanced endpoint current of span u. Define the top-free shifted current

$$J_u^\theta(A,B) = J_u(A,B) + \theta(K_u(A,B) - f_A).$$

The long edge for an ordinary clock λ contracts to

$$\sum_{x,y}(x-y-\lambda)h_{x-y}l_{i+B,x}r_{i,y} = -[J_u(B_i,B_{i+B}) + (\lambda-u)K_u(B_i,B_{i+B})].$$

The equal-slot coefficient is $-\lambda$ and is retained. On ϕ the common self-complement is zero. Taking the whole coefficient $\phi^{-r}y_u^r$ consequently gives, respectively,

$$\sum_{i=0}^{r} J_u^{rg-u}(B_i,B_{i+B}) = 0, \quad \sum_{i=0}^{r} J_u^{-rg-u}(B_i,B_{i+B}) = 0, \quad \sum_{i=0}^{r}(K_u(B_i,B_{i+B}) - f_{B_i}) = 0.$$

The common nonzero cap/proper factor has been canceled. Every summand and its denominator omits h_u, whereas ϕ is affine with nonzero coefficient in h_u. Thus the residue congruences are whole identities in the lower raw field. No individual edge was observed separately.

The contact calculation of Theorem E.3, with coherent $h_0 = \nu$, gives endpoint products $P_z > 0$ at one separated negative-lex specialization and

$$\begin{aligned}[\nu^{-1}]_\infty J_u^\theta(E,F) &= -\sum_{Z_E\setminus Z_F} zP_z - \sum_{Z_F\setminus Z_E}(u-z+\theta)P_z,\\ [\nu^{-1}]_\infty(K_u(E,F)-f_E) &= -\sum_{Z_F\setminus Z_E} P_z.\end{aligned} \tag{D.6}$$

All differing positions satisfy $1 \le z \le u-1$. Whole component-class equality supplies the additional balance

$$\sum_{Z_E\setminus Z_F} P_z = \sum_{Z_F\setminus Z_E} P_z =: S_{EF}. \tag{D.7}$$

Indeed the coherent self-complement has expansion $f_E = f_{\mathrm{core}} - \nu^{-1}\sum_{Z_E} P_z + O(\nu^{-2})$, and the two whole Schur functions and their complete proper coefficients agree. Equal row counts make their coherent homogenizations agree. In particular $S_{EF} > 0$ for a nonliteral pair.

For $\theta = rg - u > 0$, every nonliteral edge in (D.6) is strictly negative. For $\theta = -rg-u$, use (D.7); the coefficient is at least $(rg-u+2)S_{EF} > 0$. For the Hall graph it is strictly negative for a nonliteral edge, since equal row counts give equal restored-set sizes. Thus each of the three complete graph sums has one sign at its own first coherent coefficient. Every edge is literal: $B_i = B_{i+B}$. Original A-periodicity and the complete range $0 \le i \le r$ cover all A residue representatives. Coprimality synchronizes the full word:

$$B_0 = B_1 = \cdots = B_L = B_*. \tag{D.8}$$

Only marked slots have been synchronized at this stage.

The unmasked scalar and the original private records.

Return now to the full original equation (D.5). Write $J = X_W \setminus \{\tau\}$, $K = Y_W \setminus \{\tau\}$, and let I_i^P, C_i^P be all private records in band i, including units. Equation (D.8) makes the full lists

$$\begin{aligned} X_S &= \{Lg+\tau\} \cup \bigcup_{i=0}^{L-1} (ig + (J \cup I_i^P)) \cup (Lg + J), \\ Y_S &= \{\tau\} \cup \bigcup_{i=0}^{L-1} (ig + (K \cup C_i^P)) \cup (Lg + K). \end{aligned} \tag{D.9}$$

No positive wrap entry is masked here. Form the actual splices at $(rg-1, Lg-1)$ and at $(u, u+Bg)$. Their common span is $n_0 = rg+u$ and their full $u+1$ end windows coincide. The width-u overall end mark is the original row-regrouping mark. Both splices retain it, its full hat and its phase τ. Their private height words agree, so their zero-free cores are identical. Every differing unit is strictly between the actual contacts $R_0 = rg+\tau$ and $C_0 = \tau$.

The original right coframe through n_0 is exactly the row-R_0 contact coframe of the first splice, normalized at global 0: deleting that contact leaves the identical square source-prefix system. The original left coframe is the column-C_0 contact system of the other splice, normalized at its global top. All later columns against the relevant prefix have negative indices. Literal marked-slot equality was required for these full list identifications. Let K_c and B_c be the resulting entire cross kernel and balanced contact current. The full gradient gives

$$\delta_{Bg} Q_S/\kappa(S) = K_c, \qquad \mathscr{D}_{Bg}(\lambda) Q_S/\kappa(S) = -B_c + (rg-\lambda)K_c.$$

The possible common cap-normalization factor is nonzero and identical on both cofaces, by their closed identical end-window equations.

Both contacts survive in the common coface core: the upper contact is absent from the cap columns and the lower contact from the cap rows, because τ was deleted from both full hat lists. At coherent unit infinity,

$$K_c = K_{\mathrm{core}} + O(\nu^{-1}), \qquad B_c = O(\nu^{-1}), \qquad K_{\mathrm{core}} \neq 0.$$

The nonzero contact proper minor is weakly feasible, and the self pairing is the nonzero full core function times actual nonzero cofactor ratios. The balanced current at identical core data is exactly zero. Consequently a mixed $K_c = 0$ and a negative-clock $-B_c + 2rgK_c = 0$ are impossible. In the positive-clock case the supplied whole zero is $B_c = 0$, and Theorem E.3 gives literal coface equality. All private records therefore have period B in band indices; the original A period and its full finite range synchronize them as well.

For clarity, this last synchronization produces a legal chart with the same original mark, not merely periodic words. Shorten at private singletons, retaining its full first and last $u+1$ windows, to keep one private g band. If the singleton cuts are $a < b$, the full intersection and every possible containing end mark are read in those retained end windows. The new span is $g+u > 2u$, proving separation. Removing either mark changes only its retained end window; the new interior is the connected singleton splice. In the periodic hatted test only the two neighboring copies meet a marked $u+1$ neighborhood, and these are the unchanged windows. Thus I–IV, attachments and the actual hat are preserved even though the span has changed. The resulting actual native T_0 has span $g+u$, and (D.9) proves $S = \mathrm{Row}_L \circ_{\text{original mark}} T_0$ literally. Its score $(u, g+u)$ improves the same original coincident-side minimum. All initial physical transports are undone on this chart. □

Proposition D.6. *The original nondividing comparison cannot have two positive coincident marks.*

Proof. Lemma D.2 gives (D.2) with actual ribbon outers. Lemma D.3 and the strict old-minimum regrouping exclude $\mu \geq 2$. For $\mu = 1$ they give the actual same-selector line input and the single marked-component class. Choose the original larger-period source as S, in its permitted ordinary-(a) row frame. In Lemma D.5 its parameters are $A = p/g$, $B = q/g$, $m = B$, $L = AB$, and $r = B(A-1)$. The whole height and corner data, original slot period and full comparison range are those just established. The other coincident row chart supplies

the positive-clock equation in (D.5). Lemma D.5 gives the forbidden finer chart on that same source. No strict power with a one-cell outer is used in this primitive case. □

Proposition D.7. *The original nondividing comparison cannot have exactly one positive coincident mark.*

Proof. Lemma D.4 supplies actual line outers, the one entire marked-component class, and the real wide-phase order. Normalize the coincident source S in the row frame of Lemma D.5. Here A is its period divided by g, B is the other period divided by g, and $k = (w-u)/g \geq 1$. With the actual copy counts $m, j \geq 2$, $L = mA = jB + k$ and $r = L - B \geq A$, in either ordering of A, B. Lemmas D.2 and D.4 supply the whole height and corner bounds and the single marked-component class; the original row chart supplies the full A-periodic slot word. The original counterpart supplies exactly one of (D.5), over the complete range $0 \leq i \leq r$. Lemma D.5 excludes the Hall and negative-clock alternatives and constructs the forbidden same-source chart in the positive case. The minimum contradicted is the coincident side's own, regardless of which original period was larger. The range bound follows from the proper original counts, not from a least-common-multiple horizon. □

Appendix E. Same-core contact currents

The empty-overlap and coincident-phase arguments use contact currents on actual diagrams. We give the complete rigidity statement, including the case where either contact is itself a restored common beta. No equality of the two diagram functions is assumed in the basic theorem.

E.1. **The exact contact identities.** Let V be a connected zero-free diagram of span m, with full beta lists X, Y. Let E, F be two connected diagrams obtained from V by restoring sets Z_E, Z_F of common betas. Thus their full lists are $(X \cup Z_E, Y \cup Z_E)$ and $(X \cup Z_F, Y \cup Z_F)$, in decreasing order. Suppose that r is a row label shared by E, F, that t is a column label shared by them, and that $0 \leq t < r \leq m$. A right row-r contact vector of E solves every row equation except the one at r; normalize it by $R_E(t) = 1$. A left column-t contact vector of F solves every column equation except the one at t; normalize it by $L_F(r) = 1$. These normalizations are nonzero: deleting a row and a column from a connected full Jacobi–Trudi matrix gives a feasible weak minor. For example, along the interval between the two deleted positions use adjacent shifted matches, and elsewhere use the diagonal matches; connectedness supplies every needed weak inequality. The unique least-square matching then gives nonvanishing.

Pad the vectors by zero on the union of their full label sets, and write $M = (h_{x-y})$ on that union. All occurrences of h_0 have the same value, first 1 and later ν. Let X_{diag} and Y_{diag} be the diagonal matrices of the union row and column labels, respectively, in the orders used by M. Define

$$\begin{aligned} K(E,F) &= L_F M R_E, \qquad f_E = K(E,E), \quad f_F = K(F,F), \\ B_{r,t}(E,F) &= L_F\big((r-t)M - X_{\mathrm{diag}}M + MY_{\mathrm{diag}}\big)R_E. \end{aligned} \tag{E.1}$$

Both self terms use the *same* contacts r, t and their contact normalizations. They are not replaced by global-endpoint self-complements. Put $w_E = MR_E$ and $v_F = L_F M$.

Lemma E.1 (Contact Lagrange identities)**.** *With the preceding data,*

$$K(E,F) - f_E = \sum_{z \in Z_F \setminus Z_E} L_F(z) w_E(z),$$

$$K(E,F) - f_F = \sum_{z \in Z_E \setminus Z_F} v_F(z) R_E(z).$$

$$\begin{aligned} B_{r,t}(E,F) = &\sum_{z \in Z_F \setminus Z_E} (r-z) L_F(z) w_E(z) \\ &+ \sum_{z \in Z_E \setminus Z_F} (z-t) v_F(z) R_E(z). \end{aligned} \tag{E.2}$$

These are identities of entire rational functions.

Proof. On the rows of E, w_E is zero except at r, where its value is f_E because the normalized left contact coefficient is one. Pairing with L_F leaves only that row and the F-only rows, giving the first identity. The column argument gives the second. Expanding the last expression in (E.1) and using both identities gives (E.2). At every shared noncontact unit the own row or column equation is exactly zero. Such units disappear from these sums by those equations, but remain in the complete coframes. □

E.2. **The correct anchor matrix.** For the rest of this appendix assume

$$Z_E \triangle Z_F \subset (t, r). \tag{E.3}$$

Contacts and other units may be shared. Let $\epsilon_r = 1$ if $r \notin X$ and 0 otherwise, and let $\epsilon_t = 1$ if $t \notin Y$ and 0 otherwise. An indicated contact is a shared restored unit and is absent from both core lists. Restore precisely these anchor units to V. After removing contact row r and contact column t, its square matrix has lists

$$I_0 = (X \setminus \{r\}) \cup \big(\{t\} \text{ if } \epsilon_t = 1\big), \qquad C_0 = (Y \setminus \{t\}) \cup \big(\{r\} \text{ if } \epsilon_r = 1\big). \tag{E.4}$$

Write $D = \mathrm{JT}(I_0, C_0)$. It has no common beta and is independent of ν. Its height is the height of V lowered by one on $[t, r)$, so it is a nonzero weak determinant. It can be empty or disconnected.

The anchor input and output are $b = (h_{i-t})_{i \in I_0}$ and $a = (h_{r-j})_{j \in C_0}$. The vector b contains the term νe_t precisely when $\epsilon_t = 1$; a contains νe_r^t precisely when $\epsilon_r = 1$. Their remaining entries and h_{r-t} are independent of ν. Eliminating D from the complete system leaves, on every other restored set Z, a matrix

$$\nu I_Z + E_0[Z, Z], \tag{E.5}$$

where E_0 is independent of ν. Thus the true diagonal includes its cap feedback; no diagonal Schur complement is replaced by ν.

For a differing vacancy z define

$$\begin{aligned} A_z &= [\nu^{\epsilon_t}]\big(h_{z-t} - M(z, C_0)D^{-1}b\big), \\ G_z &= [\nu^{\epsilon_r}]\big(h_{r-z} - aD^{-1}M(I_0, z)\big). \end{aligned} \tag{E.6}$$

Here the brackets select the highest indicated power of ν in these anchor-only expressions. They are independent of the nonanchor sets.

Lemma E.2. *Every A_z, G_z in* (E.6) *is nonzero. On the absent side of a differing record,*

$$\begin{aligned} w_E(z) &= \nu^{\epsilon_t}\big(A_z + O(\nu^{-1})\big) \quad (z \in Z_F \setminus Z_E), \\ v_F(z) &= \nu^{\epsilon_r}\big(G_z + O(\nu^{-1})\big) \quad (z \in Z_E \setminus Z_F). \end{aligned} \tag{E.7}$$

On its own inserted side,

$$\begin{aligned} R_E(z) &= -\nu^{\epsilon_t - 1}\big(A_z + O(\nu^{-1})\big) \quad (z \in Z_E \setminus Z_F), \\ L_F(z) &= -\nu^{\epsilon_r - 1}\big(G_z + O(\nu^{-1})\big) \quad (z \in Z_F \setminus Z_E). \end{aligned} \tag{E.8}$$

All products $A_z G_z$ have one nonzero sign at a single sufficiently separated negative-lex specialization.

Proof. Elimination of D followed by $(\nu I + E_0)^{-1} = \nu^{-1} I + O(\nu^{-2})$ gives (E.7). To obtain the own-side formulas, use the actual zero equations, not the absent residuals. For example, put $T_E(z) = \sum_{y \neq z} h_{z-y} R_E(y)$. Then $0 = \nu R_E(z) + T_E(z)$. After elimination of D, the forcing has leading term $\nu^{\epsilon_t} A_z$, while feedback from another nonanchor unit is one order lower. The equation gives the first formula in (E.8); the left equation gives the other.

The two Cramer numerator boards, in decreasing order, are

$$F_z^{\mathrm{num}} = \frac{(X \setminus \{r\}) \cup \{z\}}{Y \cup (\{r\} \text{ if } \epsilon_r = 1)}, \qquad G_z^{\mathrm{num}} = \frac{X \cup (\{t\} \text{ if } \epsilon_t = 1)}{(Y \setminus \{t\}) \cup \{z\}}. \tag{E.9}$$

The notation denotes row and column lists, not division of diagrams. Their heights are the height of V lowered by one on $[z, r)$ and $[t, z)$, respectively. Connectedness gives weak feasibility and the weak-determinant lemma gives nonvanishing. Their denominators are the same $\det D$.

For the sign, put $a_x = \#\{i \in X : i > x\}$, $b_x = \#\{j \in Y : j < x\}$, and let P count all positive entries of the core matrix. If $d = |X| = |Y|$, the positive-entry counts of the two boards are

$$P_F(z) = P + b_z - (1 - \epsilon_r) b_r + \epsilon_r a_r,$$
$$P_G(z) = P + a_z - (1 - \epsilon_t) a_t + \epsilon_t b_t.$$

Their sizes are $d + \epsilon_r$ and $d + \epsilon_t$. Since every entry strictly above the diagonal of a weak feasible matrix is positive, its negative-lex exponent is its positive-entry count minus the triangular number of its size. The z-dependent part of the sum of these two exponents is $a_z + b_z$ modulo two. The Cramer insertion parity has z-dependent part

$$[a_z - (1 - \epsilon_r)] + [d - b_z].$$

The two dependencies cancel. The denominator is squared, so the product sign is independent of z.

More explicitly, with $A_I(x) = \#\{i \in I_0 : i > x\}$ and $A_C(x) = \#\{j \in C_0 : j > x\}$, the signed numerator multipliers are

ϵ_t	multiplier for A_z	ϵ_r	multiplier for G_z
0	$(-1)^{A_I(z)+A_C(t)}$	0	$(-1)^{A_I(r)+A_C(z)}$
1	$-(-1)^{A_I(t)-A_I(z)}$	1	$-(-1)^{A_C(z)-1-A_C(r)}$.

(E.10)

The minus signs in the second row come from the unit-anchor forcing terms. Apply the weak-determinant sign lemma simultaneously to the finite collection of numerator and denominator boards. This proves the final assertion, with no sign claim at arbitrary moment values. □

E.3. **Rigidity, inclusion, and compensation.**

Theorem E.3 (Same-core contact rigidity)**.** *Under* (E.3)*,*

$$\begin{aligned} B_{r,t}(E,F) = 0 \quad &\Longleftrightarrow \quad E = F, \\ K(E,F) = f_E \quad &\Longleftrightarrow \quad Z_F \subseteq Z_E, \\ K(E,F) = f_F \quad &\Longleftrightarrow \quad Z_E \subseteq Z_F. \end{aligned} \tag{E.11}$$

In addition, with $d = r - t$*,*

$$B_{r,t}(E,F) - 2dK(E,F) + d(f_E + f_F) = 0 \quad \Longleftrightarrow \quad E = F. \tag{E.12}$$

Proof. Put $e = \epsilon_r + \epsilon_t$ and $P_z = A_z G_z$. The exact Lagrange identities and Lemma E.2 give

$$\begin{aligned} [\nu^{e-1}]_\infty B_{r,t}(E,F) &= -\sum_{z \in Z_E \setminus Z_F} (z - t) P_z - \sum_{z \in Z_F \setminus Z_E} (r - z) P_z, \\ [\nu^{e-1}]_\infty (K - f_E) &= -\sum_{z \in Z_F \setminus Z_E} P_z, \\ [\nu^{e-1}]_\infty (K - f_F) &= -\sum_{z \in Z_E \setminus Z_F} P_z. \end{aligned} \tag{E.13}$$

All distance weights are strictly positive. At the one specialization provided by the preceding lemma, a nonempty sum cannot vanish. This proves necessity in (E.11). Sufficiency follows from

the empty exact sums in Lemma E.1. The compensated expression in (E.12) has coefficient

$$\sum_{z\in Z_E\setminus Z_F}(r-z)P_z+\sum_{z\in Z_F\setminus Z_E}(z-t)P_z$$

at the same order, proving its assertion as well.

The leading order is ν^{-1}, ν^0, or ν^1 according as zero, one, or two contacts are shared unit anchors. Thus the proof includes all anchor configurations rather than imposing one universal inverse-unit order. An observed rational identity at $h_0=1$ has the coherent extension used here: the kernels and currents have ordinary degree one, and the extension is ν times their expression at h_j/ν. □

Corollary E.4. *A positive rational constant-weight sum of balanced contact currents on one fixed literal core and the same contacts vanishes only if every edge joins identical diagrams. A corresponding sum of $K-f_E$ or $K-f_F$ gives the indicated edgewise inclusions. The same assertion holds for the compensated current in* (E.12).

Proof. All edges use the same anchor matrix, coherent order, and residual products. At one simultaneous negative-lex specialization their first nonzero coefficients have one sign by (E.13). Positive constant weights preserve that sign. The whole sum, rather than an independently observed edge, therefore gives the conclusion. □

Lemma E.5. *Put $e=\epsilon_r+\epsilon_t$. The self kernel has a nonzero leading coefficient*

$$f_E=\nu^e\Phi+O(\nu^{e-1}),\qquad \Phi\neq 0,$$

independent of all restored units except the shared anchors. The cross kernel $K(E,F)$ has the same leading term.

Proof. In the anchor-only system, the self kernel is the Schur complement $h_{r-t}-aD^{-1}b$. Its coefficient of ν^e, with the actual Cramer sign, is the nonzero full core determinant divided by $\det D$. Eliminating additional nonanchor units changes it only at the next order by (E.5). The first Lagrange identity gives the assertion for the cross kernel. □

E.4. **Global and contact normalizations.** The source coframes used in the scalar identities are often normalized at global column 0 and global row m, rather than at t,r. For cap-preserving cofaces whose contacts lie in identical lower and upper cap windows, both conversions are the same nonzero cap ratios on both diagrams. Indeed the proper system of the lower cap is a closed square block: rows in it have negative entries against all higher columns. Solving this block determines the coefficient at t independently of the private word and the upper contact. The reversed upper-cap system determines the coefficient at r. Weak cofactors make both ratios nonzero, including when an anchor is a shared unit. Consequently B, K, and both corresponding self kernels acquire one common nonzero factor on changing normalization.

Under coherent homogenization the global right and left vectors are bounded. The coefficient at a shared unit contact is of order ν^{-1} with nonzero leading term; at a core contact it has order one. Thus the conversion factors from contact-normalized to globally normalized vectors have respective orders $\nu^{-\epsilon_t}$ and $\nu^{-\epsilon_r}$. These orders also follow directly by solving (E.4). They account for the universal ν^{-1} order of the globally normalized difference sums used in the extension argument below. No assertion that an arbitrary paired cofactor is a low-generator polynomial is required.

Appendix F. An alternative geometric realization of the scalar

The proof in Section 8 uses shifted determinants of the original source. There is another exact realization of the same scalar: modify the two cap cofaces so that the source readers become contact or extension vectors on actual diagrams. This argument uses the same structural, operator, and end-pole inputs as the main proof. We give the additional geometric current lemmas and the construction explicitly.

F.1. **Directional contact rigidity.**

Lemma F.1. *Use the same-core notation and contact normalization of Appendix E, but replace* (E.3) *by*

$$Z_E \setminus Z_F \subset (0, r), \qquad Z_F \setminus Z_E \subset (t, m). \tag{F.1}$$

Then $B_{r,t}(E, F) = 0$ if and only if $E = F$ literally.

Proof. The exact identity (E.2) holds for differing vacancies outside (t, r) too. Use the same anchor matrix (E.4) and numerator lists (E.9). For $z < r$ the first numerator lowers core height on $[z, r)$; for $z > r$ it raises core height on $[r, z)$. The second lowers height on $[t, z)$ when $z > t$ and raises it on $[z, t)$ when $z < t$. All these weak boards are feasible and nonzero because the core is connected. No smaller-area theorem is applied to an exterior replacement. The absent and own-forcing expansions (E.7)–(E.8) therefore remain valid.

There is an additional sorting sign when z crosses an anchor. In the notation of Lemma E.2, the positive-entry counts become

$$P_F(z) = P + b_z - (1 - \epsilon_r)b_r + \epsilon_r a_r + \epsilon_r \mathbf{1}_{z>r},$$
$$P_G(z) = P + a_z - (1 - \epsilon_t)a_t + \epsilon_t b_t + \epsilon_t \mathbf{1}_{z<t}.$$

For a unit lower anchor the Cramer multiplier is now $-(-1)^{A_I(t)+A_I(z)-\mathbf{1}_{z<t}}$, and for a unit upper anchor it is $-(-1)^{A_C(r)+A_C(z)-\mathbf{1}_{z<r}}$; the core-anchor multipliers are as in (E.10). Combining these with the two weak-JT exponents leaves, up to a z-independent parity, $\epsilon_r - \mathbf{1}_{z<r} - \mathbf{1}_{z<t}$. Thus

$$\operatorname{sgn}(A_z G_z) = \eta_0 (-1)^{\mathbf{1}_{z<t}+\mathbf{1}_{z>r}},$$

where η_0 is fixed and nonzero at one simultaneous negative-lex specialization.

The coefficient of the balanced current at $\nu^{\epsilon_r+\epsilon_t-1}$ is still

$$-\sum_{z \in Z_E \setminus Z_F} (z - t) A_z G_z - \sum_{z \in Z_F \setminus Z_E} (r - z) A_z G_z.$$

An E-only unit below t changes the sign of both its product and its distance weight; an F-only unit above r does the same. Condition (F.1) forbids the other two exterior directions. Consequently every nonempty summand has the same strict sign. Vanishing forces both difference sets empty. The converse is the exact empty Lagrange sum. □

F.2. **A column-extension Hall lemma.**

Lemma F.2. *Let E, F be connected diagrams of span m with the same literal zero-free core and a shared column t. Let $t < v \le m$. Suppose that v is either a common core row, absent from both full column sets, or absent from all full row and column sets of E, F. Let R_{ext} be the vector on columns $Y_E \cup \{v\}$ which annihilates every row in X_E and is normalized at global column 0. Assume its square normalization is the actual nonzero return normalization. Let L_F^t be the column-t contact vector of F, normalized at global row m. If*

$$Z_F \setminus Z_E \subset (t, v), \tag{F.2}$$

then

$$L_F^t M R_{\mathrm{ext}} = 0 \quad \Longleftrightarrow \quad Z_F \subseteq Z_E. \tag{F.3}$$

There is a row-extension mirror.

Proof. The right vector annihilates every row of E. Since the core rows are shared, the entire contraction is exactly

$$O = \sum_{z \in Z_F \setminus Z_E} L_F^t(z)(M R_{\mathrm{ext}})(z). \tag{F.4}$$

There is no self term.

All globally normalized contact vectors are bounded at coherent unit infinity, and their nonanchor inserted-unit coefficients are $O(\nu^{-1})$, by the last part of Appendix E. If v is a core row,

compare R_{ext} with the row-v contact vector on E. Completing its omitted equation by adding column v gives the true Schur complement

$$\delta_v = \nu - \text{cap feedback} = \nu + O(1).$$

Its forcing is $O(1)$. Thus the complete two vectors differ by $O(\nu^{-1})$, and their residuals on an absent differing row have the same leading bounded term. This conclusion is obtained from the exact inverse solve, not from the false equality $\delta_v = \nu$.

If v is absent from all flags, restore it as a common unit on both E, F, giving actual diagrams E^+, F^+. The vector R_{ext} is exactly the row-v contact vector on E^+ normalized at 0. In the left system, adding the noncontact unit v changes the coefficient at a different inserted unit z only at $O(\nu^{-2})$: eliminate the common unit-free matrix and expand the full $(\nu I + E_0)^{-1}$. That coefficient was already $O(\nu^{-1})$ and the extra-unit feedback enters one order later.

In both cases $[\nu^{-1}]_\infty O$ is the same as the corresponding globally normalized contact-difference coefficient for the common anchor pair (v, t). Conversion to contact normalization makes it one nonzero common factor times

$$-\sum_{z \in Z_F \setminus Z_E} A_z G_z.$$

Indeed the contact coefficient has order $\nu^{\epsilon_v + \epsilon_t - 1}$ and the two conversion factors have total order $\nu^{-\epsilon_v - \epsilon_t}$. By (F.2), every z lies strictly between the anchors, so Lemma E.2 gives one nonzero sign. A nonempty difference therefore makes O nonzero. The converse follows directly from (F.4).

In the absent-v case the augmented pair is used only to calculate this coefficient; no scalar zero is assigned to it. Reflecting labels by $x \mapsto m - x$, transposing, and interchanging the vectors proves the row-extension statement. This combined transformation preserves the moment difference $x - y$. □

F.3. **The actual cap modifications.** Return to the data of Sections 5 to 7. The source span is N, its observed period is q, and $m = N - q = a + c$. The actual upper cap has rectangular lists (U_{cap}, C_+) and the lower one (I_-, V_{cap}). The return frames are (I_R, C_R) and (I_L, C_L); their common disjoint core is (I_0, C_0). Use the active indices σ, τ of the positive-clock ordinary or the mixed reference deck.

Starting from $S^-_{\mathrm{cap}}, S^+_{\mathrm{cap}}$, replace only the upper cap of S^-_{cap} and the lower cap of S^+_{cap} by the following full lists. All upper lists are translated by a.

(F.5)

	upper rows	upper columns	lower rows	lower columns
ordinary 00	$I_R \cup \{\sigma\}$	C_R	I_L	$C_L \cup \{\tau\}$
mixed 10	I_R	$C_R \setminus \{\sigma\}$	I_L	$C_L \cup \{\tau\}$
mixed 01	$I_R \cup \{\sigma\}$	C_R	$I_L \setminus \{\tau\}$	C_L

Write $\widehat{E}, \widehat{F}$ for the new diagrams. The two mixed rows refer to the actual Boolean pair (e_R, e_L); the ordinary pair is 00. The Boolean endpoint equations license every displayed insertion and deletion. For example the ordinary zero limits make $\sigma \notin I_R$ and $\tau \notin C_L$, and the upper mixed one limit makes $\sigma \in C_R$.

Put $\Delta = \mathbf{1}_{I_0} - \mathbf{1}_{C_0}$. Each new upper difference is $\Delta + \mathbf{1}_\sigma$, and each new lower difference is $\Delta - \mathbf{1}_\tau$, exactly the original cap differences. The rectangular cardinalities and the genuine endpoint memberships are unchanged. Thus every changed record is a paired vacancy restoration or deletion. It preserves the positive height word and singleton ends, while deliberately changing the two separate boundary words. The diagonal criterion gives connected skew diagrams with the same literal zero-free core and the same area as the old cofaces. They remain strict in area relative to the source, and agree with $S^-_{\mathrm{cap}}, S^+_{\mathrm{cap}}$ at every position $c < k < a$. No mark, minimum, or Schur equality is assigned to the new cofaces.

The complete block equations identify the source readers exactly. In the ordinary row of (F.5), the right source vector is the row-r contact on $\widehat{E}$, $r = a + \sigma$, normalized at 0, and the left source vector is the column-t contact on $\widehat{F}$, $t = \tau$, normalized at m. In mixed 10, the right source vector annihilates every row of $\widehat{E}$ and uses its columns plus $v = a + \sigma$, while the left

vector is the column-t contact on $\widehat{F}$. In mixed 01 the right vector is the row-r contact and the left vector is its row-extension counterpart. These identifications retain any returned column m, returned row 0, and all inverse feedback.

Proposition F.3. *In the positive distinct-phase line-outer setting, the original whole ordinary scalar gives $\widehat{E} = \widehat{F}$ literally. The whole mixed scalar gives $Z_{\widehat{F}} \subseteq Z_{\widehat{E}}$ in Boolean state* 10 *and the reverse inclusion in state* 01*. In all cases the full private q-comparison and the original minimum contradiction follow.*

Proof. In the ordinary case the aligned genuine clock is $\lambda = a + \sigma - \tau = r - t$. The entire source scalar is therefore

$$L_{\mathrm{source}}(X_{\mathrm{diag}}M - MY_{\mathrm{diag}} - \lambda M)R_{\mathrm{source}} = -B_{r,t}(\widehat{E}, \widehat{F}),$$

up to the common nonzero global/contact normalization. An $\widehat{E}$-only unit above r can only be an added upper return record in $C_R \setminus C_+$ above σ. An $\widehat{F}$-only unit below t can only be an added lower return record in $I_L \setminus I_-$ below τ. Both are excluded by Theorem 7.4. The remaining changes satisfy (F.1), including all private changes. The supplied whole zero and Lemma F.1 give literal equality.

For mixed 10 put $v = a + \sigma$ and $t = \tau$. The extra column v is absent from both coface column sets. If σ is a core vacancy, Boolean value one restores it in the return, and removing its column in (F.5) makes v an unpaired common core row. If σ is an unpaired weak-core column, its removal makes v absent from all coface lists. These are the two alternatives of Lemma F.2. An $\widehat{F}$-only unit above v is a missing upper cap unit from $U_{\mathrm{cap}} \setminus I_R$ above σ; an $\widehat{F}$-only unit below t is an added lower unit from $I_L \setminus I_-$ below τ. The end-pole filter excludes both. Thus $Z_{\widehat{F}} \setminus Z_{\widehat{E}} \subset (t, v)$ and the whole Hall scalar is exactly the contraction in (F.3). It gives $Z_{\widehat{F}} \subseteq Z_{\widehat{E}}$. Mixed 01 follows from the row-extension mirror and the same outward exclusions.

The cap modifications preserve every position $c < k < a$. Hence these conclusions give all the original private comparisons used in Lemma 9.1. That lemma and Theorem 9.2 construct the legal same-parent gcd chart before its improved score is compared with the original prescribed minimum. The proof consumes the same complete scalar as Section 8, without requiring constant cap-response coefficients or separately vanishing response modes. □

Declarations

Funding. This research received no external funding.

Competing interests. The author has no relevant interests to disclose.

Data availability. No datasets were generated or analyzed.

Computational verification. The proofs do not depend on numerical computation. Scripts used for verification and figures are available at `https://github.com/johnnyshields/papers`.

Generative AI use. During the preparation of this work, the author used OpenAI's ChatGPT (GPT-6 Astra, GPT-6.1 Sol) and Anthropic's Claude (Opus 5.5, Sonnet 5.5) to assist with mathematical research, manuscript drafting, and the development of auxiliary verification code. After using these tools, the author reviewed and edited as needed. The author takes full responsibility for the content of the work.

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